THIRSTY'S CODEX

STANDARDS EDITION

VOLUME VI / 06 OF 20

DATA, INTELLIGENCE, AND HUMAN OVERSIGHT

Lifecycle Sections 151-188

NO PROOF, NO CLAIM

CONTINUOUS MASTER FOLIO / SOURCE-PRESERVING EDITION

Edition Statement

THIRSTY'S CODEX / STANDARDS EDITION / SOURCE-PRESERVING SET

This volume is one part of a continuous 20-volume edition. Canonical lifecycle section numbers remain stable across the set; no volume restarts or renumbers them.

The visual treatment has been rebuilt as a unified standards library: coordinated covers, volume architecture, source dividers, running headers, continuous master folios, local page numbers, and source-identity marks.

The source corpus remains authoritative over the presentation layer. Status terms are preserved exactly. CONTENT-COMPLETE does not silently become source-reconciled, hostile-reviewed, owner-accepted, ratified, or published.

Every registered source witness is identified by path, byte count, and SHA-256 digest. The complete source archive is preserved as an embedded attachment in Volume XX, while the two supplied foundational PDFs are preserved as attachments in Volume I.

CONTINUITY RULE

Read the master folio and canonical section sequence as one uninterrupted work. Each succeeding volume begins where the preceding volume ends.

Standards Edition Set Map

VOLUME I
Foundations and Governance
Doctrine, prompts, front matter, shared governance, roles, and output contracts

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Research, Strategy, and Risk
Lifecycle Sections 001-018

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Requirements through Hardware Architecture
Lifecycle Sections 019-060

VOLUME IV
Electrical Engineering through User Validation
Lifecycle Sections 061-106

VOLUME V
Software Systems through Smart Contracts
Lifecycle Sections 107-150

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Authority, constitutions, policies, schemas, provenance, evaluation, and verifica...

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Applied Corpus and V3Q
Agent-output corpus, V3Q implementation, fixtures, tests, and governed examples

VOLUME XVII
Training Views and Indexes
Continuous-pretraining, evaluation, retrieval, and training views with canonical ...

VOLUME XVIII
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151

SECTION 151 - DATABASE ENGINEERING

CODEX/03_SECTIONS/151_DATABASE_ENGINEERING.txt

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7,272 BYTES / PRESERVED SOURCE IDENTITY

SECTION 151 — Database Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Design, build, test, operate, and retire database capabilities that preserve authorized data meaning, integrity, availability, security, performance, recoverability, and evidence.

REQUIRED OUTCOME

An approved database design with schemas, constraints, indexes, transactions, access paths, backup and restore behavior, operational ownership, lifecycle state, and bounded evidence.

WHY THIS MATTERS

Database behavior can silently change truth, authority, privacy, service continuity, and recovery. A working query or healthy instance does not prove sound integrity, isolation, access control, or recoverability.

APPLICABILITY

Apply when a database stores, transforms, serves, replicates, audits, or governs material data, or when schema, query, transaction, availability, or recovery behavior affects outcomes.

NON-APPLICABILITY CONDITIONS

No material database behavior, data meaning, access, performance, evidence, lifecycle, or recovery is changed, and an exact approved baseline covers the work. Record scope, baseline, owner, evidence date, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requirements, domain and data models, information classification, application contracts, workload, users, operators, recovery objectives, security controls, legal duties, and acceptance authority.

REQUIRED INPUTS

Authorized objective and scope; entities and relationships; workload and concurrency; data classes; queries; retention; migrations; backup, restore, failover, monitoring, and support constraints; assumptions and unknowns.

HUMAN RESPONSIBILITIES AND AUTHORITY

The human owner approves data purpose, schema meaning, access, risk, service objectives, migrations, exceptions, recovery tradeoffs, and acceptance. The owner assigns database, data, security, privacy, operations, and retirement authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized database definitions, migration history, query plans, workload evidence, operational records, and recovery test results. It may identify integrity, concurrency, access, performance, compatibility, and recovery risks; propose schemas, constraints, indexes, queries, migrations, tests, runbooks, and evidence plans; and explain expected impacts and unresolved assumptions.

AI must preserve the distinction between observed engine behavior, inferred behavior, and recommendation. It must not choose data meaning, grant access, expose credentials or protected data, execute production changes, approve migrations or risk, declare recovery or correctness from partial evidence, or treat a benchmark, checksum, backup, replica, or successful query as proof beyond its stated scope. Any mutation-capable proposal must identify the authorized actor, target environment, change scope, dependencies, validation, audit record, and forward recovery or compensation path; unresolved authority, evidence, or safety gaps require stop and owner review.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Researcher, Implementer, Tester, Verifier, Security Auditor, Governance Reviewer, Observer, Critic, Optimizer, Documentation Writer, Memory Curator, and Red Team. Existing role contracts remain controlling.

REQUIRED ACTIVITIES

1. Establish scope, workload, data ownership, criticality, environments, and lifecycle.
2. Define schemas, keys, constraints, transactions, isolation, indexes, queries, access paths, and compatibility.
3. Design access control, encryption, auditing, backup, restore, replication, failover, migration, and deletion.
4. Test normal, concurrent, malformed, unauthorized, degraded, recovery, and upgrade behavior; baseline evidence and open gaps.

HUMAN-AI INTERACTION REQUIREMENTS

AI may inspect authorized schemas and plans, propose queries, indexes, tests, migrations, and documentation. It must label observation, recommendation, execution, and verification; it must not alter production, infer permission, expose secrets, approve risk, or declare correctness from a sample or benchmark.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent schemas, migrations, transactions, credentials, roles, replicas, backups, logs, evidence, and lifecycle states as versioned artifacts. Separate logical design from physical engine behavior and preserve idempotency, locking, timeout, retry, compensation, and audit semantics.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show baseline, engine and version, environment, owner, data scope, active changes, locks, replication, backup freshness, alerts, evidence, and available pause, revoke, restore, failover, rollback, and retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Database design and data dictionary; schema and migration records; query and workload catalog; access and audit model; backup, restore, failover, performance, test, risk, decision, evidence, and handoff records.

EVIDENCE REQUIREMENTS

Claims identify database, version, configuration, dataset, workload, environment, time, method, actor, result, and limitation. A successful migration, query plan, backup, or checksum proves only its bounded claim.

VERIFICATION AND ACCEPTANCE TESTS

Verify constraints, transaction behavior, isolation, concurrency, authorization, injection resistance, performance under stated load, backup restoration, failover, migration compatibility, deletion, audit, and independent reproducibility.

FAILURE STATES AND ANTI-PATTERNS

Unowned schema changes, shared privileged accounts, unsafe dynamic SQL, missing constraints, unbounded queries, hidden copies, untested restore, false replica assumptions, destructive rollback, and treating uptime as integrity.

RECOVERY AND RECONSIDERATION

Stop or isolate affected writes on corruption, unauthorized access, migration failure, capacity exhaustion, or evidence loss. Preserve logs and state; use approved restore, reconciliation, controlled forward migration, failover, or compensation, then reassess downstream consumers.

COMPLETION CRITERIA

Scope, ownership, schema, controls, workload, compatibility, operations, recovery, tests, evidence, unresolved gaps, and next review point are recorded. Status remains CONTENT-COMPLETE until reconciliation, hostile review, and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off schemas, contracts, migrations, access, monitoring, backup, restore, performance, incidents, and retirement obligations to Storage, Data, Application, Security, Testing, Operations, Migration, and Retirement sections. Record reopening triggers and exact baseline.

RELATED CODEX SECTIONS

019 Requirements Engineering; 020 Systems Engineering; 030 Information Architecture; 031 Data Architecture; 032 Application Architecture; 203 Cryptography; 285 Database Administration; 307 Data Operations; 328 Backup Engineering; 329 Restore Engineering; 382 Data Migration Operations; 391-395 Data Retention through Data Deletion.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Sections 019-032, Owner Cockpit Requirements, Status and Evidence Lexicon, and the integration roadmap. Engine-specific locking, licensing, jurisdiction, retention, workload, and recovery standards remain unresolved until supplied and verified.

152

SECTION 152 - STORAGE ENGINEERING

CODEX/03_SECTIONS/152_STORAGE_ENGINEERING.txt

SHA-256 8b9f03f73696581fd7e4961aa782e45049121c1d5e7eecf57e5aafb4dd89d8ea

6,765 BYTES / PRESERVED SOURCE IDENTITY

SECTION 152 — Storage Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Engineer storage systems and media so data is placed, protected, retrieved, replicated, retained, restored, migrated, and retired within declared authority and service boundaries.

REQUIRED OUTCOME

A storage design covering media, namespaces, capacity, performance, durability, availability, access, encryption, lifecycle, backup, restore, portability, and disposal with owners and evidence.

WHY THIS MATTERS

Storage loss, leakage, corruption, or exhaustion can invalidate every downstream function. Capacity and redundancy diagrams do not prove recoverability or lawful deletion.

APPLICABILITY

Apply when files, objects, blocks, volumes, caches, archives, backups, or physical media carry material data or service state.

NON-APPLICABILITY CONDITIONS

No material storage location, data class, access, lifecycle, capacity, recovery, or disposal behavior changes and an exact approved baseline covers the work. Record the determination and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Data architecture, classifications, workload, access, durability and recovery objectives, regions, suppliers, media constraints, retention, deletion, cost, environmental, security, and operational requirements.

REQUIRED INPUTS

Authorized scope; data volumes and growth; read/write patterns; latency; durability; availability; replication; backup; restore; encryption; users; regions; retention; disposal; assumptions and unknowns.

HUMAN RESPONSIBILITIES AND AUTHORITY

The human owner approves storage classes, locations, risk, cost, residency, retention, deletion, recovery objectives, exceptions, and acceptance, and assigns data, security, operations, supplier, and disposal authorities.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized storage inventories, classifications, configurations, capacity and performance telemetry, replication state, backup and restore evidence, provider terms, and disposal records. It may compare storage options, identify capacity, durability, exposure, residency, portability, cost, lifecycle, and recovery risks, and propose designs, tests, controls, runbooks, and evidence requirements.

AI must distinguish observed media or provider behavior from inference and recommendation. It must not select a region or provider as an authority decision, expose data or keys, change storage configuration, claim durability, residency, deletion, isolation, or restoration from incomplete evidence, or approve cost, legal, security, or recovery risk. A mutation-capable proposal must identify the authorized actor, exact scope, environment, copies and dependencies affected, validation, append-only audit evidence, and forward recovery or compensation path; missing authority, evidence, or disposal coverage requires stop and owner review.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Researcher, Implementer, Tester, Verifier, Security Auditor, Governance Reviewer, Observer, Critic, Optimizer, Documentation Writer, Memory Curator, and Red Team. Existing role contracts remain controlling.

REQUIRED ACTIVITIES

1. Classify data and define storage, access, lifecycle, durability, recovery, and disposal objectives.
2. Compare media and provider options against workload, portability, residency, security, capacity, and cost.
3. Design namespaces, quotas, replication, encryption, backup, restore, monitoring, migration, and secure disposal.
4. Exercise loss, corruption, provider outage, quota exhaustion, stale replica, restore, and deletion paths.

HUMAN-AI INTERACTION REQUIREMENTS

AI may compare authorized options and analyze telemetry, but must not choose regions, expose data, change storage, or claim durability. Recommendations must state workload, assumptions, provider, evidence, and owner decision.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Model storage classes, locations, copies, replicas, caches, backups, keys, quotas, lifecycle states, and evidence separately. Enforce identity, access, retention, deletion, integrity, portability, and recovery at storage boundaries.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show storage class, location, owner, capacity, utilization, copies, freshness, encryption, retention, incidents, cost, evidence, and authorized quota, isolate, restore, migrate, delete, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Storage landscape and classification; capacity and performance model; access and protection design; replication, backup, restore, migration, retention, disposal, test, risk, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify media, provider, region, configuration, dataset, time, workload, test method, result, and limitation. A redundant copy or vendor statement is not proof of restoration, isolation, or deletion.

VERIFICATION AND ACCEPTANCE TESTS

Verify authorized access, encryption, integrity, capacity thresholds, performance, replication lag, backup restoration, provider loss, portability, retention expiry, deletion, and residual-copy behavior.

FAILURE STATES AND ANTI-PATTERNS

Unbounded growth, hidden copies, public buckets, shared keys, untested backups, region assumptions, cache leakage, quota blindness, provider lock-in without decision, and deletion claims that omit replicas or archives.

RECOVERY AND RECONSIDERATION

On loss, corruption, exposure, exhaustion, or provider failure, restrict affected access, preserve evidence, and use approved restore, failover, substitution, quota, migration, or compensation paths. Reassess data and service impact before resuming.

COMPLETION CRITERIA

Storage scope, ownership, classification, capacity, access, durability, recovery, lifecycle, disposal, tests, evidence, and gaps are recorded for the exact environment. Status remains CONTENT-COMPLETE pending later gates.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off storage contracts, locations, quotas, keys, monitoring, backup, restore, migration, retention, disposal, and provider obligations to Data, Database, Infrastructure, Cloud, Security, Operations, Testing, and Retirement work.

RELATED CODEX SECTIONS

030 Information Architecture; 031 Data Architecture; 032 Application Architecture; 193 Infrastructure Security; 194 Cloud Security; 275 Infrastructure Engineering; 277 Cloud Engineering; 286 Storage Administration; 328-330 Backup, Restore, and Failover Engineering; 391-401 retention and disposal sections.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Sections 019-032, Owner Cockpit Requirements, Status and Evidence Lexicon, and roadmap. Provider-specific durability, residency, environmental, legal, pricing, and disposal requirements remain unresolved.

153

SECTION 153 - DATA ENGINEERING

CODEX/03_SECTIONS/153_DATA_ENGINEERING.txt

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6,545 BYTES / PRESERVED SOURCE IDENTITY

SECTION 153 — Data Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Engineer reliable data collection, movement, transformation, publication, and observability so authorized data reaches intended consumers with preserved meaning, lineage, quality, security, and lifecycle control.

REQUIRED OUTCOME

A versioned data-flow design with source and destination contracts, pipelines, schedules, state, lineage, quality gates, idempotency, recovery, ownership, and evidence.

WHY THIS MATTERS

Pipelines can produce plausible but wrong results through loss, duplication, late data, schema drift, or silent transformation. Successful execution is not proof of valid data.

APPLICABILITY

Apply when data is ingested, transformed, enriched, joined, streamed, batched, exported, synchronized, or published across a material boundary.

NON-APPLICABILITY CONDITIONS

No material flow, transformation, schema, lineage, quality, access, timing, cost, or recovery behavior changes and an exact baseline covers the work. Record the determination.

PREREQUISITES AND DEPENDENCIES

Data and information architecture, source authority, contracts, workload, latency, quality, privacy, security, retention, consumers, recovery, and operational ownership.

REQUIRED INPUTS

Sources, destinations, schemas, events, schedules, volumes, ordering, lateness, transformations, dependencies, credentials, quality rules, SLAs, assumptions, and known gaps.

HUMAN RESPONSIBILITIES AND AUTHORITY

The human owner approves purpose, sources, transformations, sharing, quality thresholds, cost, risk, exceptions, and acceptance, and assigns data, pipeline, security, privacy, operations, and retirement authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized source and destination contracts, pipeline definitions, lineage, run records, schemas, quality results, offsets, logs, quarantines, and replay evidence. It may propose mappings, transformations, quality gates, orchestration, tests, observability, capacity controls, and recovery procedures, while identifying assumptions, uncertainty, data loss, duplication, lateness, drift, access, and cost risks.

AI must preserve source provenance and distinguish observed records and run outcomes from inferred meaning or simulated results. It must not approve data sharing, alter a pipeline, replay or backfill material data, bypass quality or access controls, expose credentials or protected records, or declare data correctness from pipeline completion, samples, checksums, or green status alone. Any mutation-capable proposal must identify the authorized actor, exact flow and data window, idempotency and dependency effects, validation, audit and lineage records, and forward recovery or compensation path; unresolved authority, provenance, or quality gaps require quarantine or owner review.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Researcher, Implementer, Tester, Verifier, Observer, Security Auditor, Governance Reviewer, Critic, Optimizer, Documentation Writer, Memory Curator, Coordinator, and Red Team.

REQUIRED ACTIVITIES

1. Map sources, consumers, contracts, lineage, ownership, and data states.
2. Design batch and streaming paths with validation, ordering, deduplication, replay, idempotency, quarantine, and reconciliation.
3. Define orchestration, secrets, observability, capacity, cost, access, retention, and recovery.
4. Test late, duplicate, malformed, partial, out-of-order, unavailable, and replayed data; baseline evidence.

HUMAN-AI INTERACTION REQUIREMENTS

AI may inspect authorized pipeline metadata and propose mappings or tests. It must label inference and simulated results, preserve provenance, and never alter flows, approve sharing, or treat pipeline completion as data correctness.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent data products, stages, checkpoints, lineage, schemas, quality results, offsets, retries, quarantines, replays, evidence, and lifecycle states separately. Design for idempotency, backpressure, ordering, replay, compensation, and safe stop.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show pipeline version, source freshness, offsets, lag, quality gates, failures, quarantines, active runs, owners, dependencies, authority, and controls for pause, replay, quarantine, compensate, and retire.

REQUIRED OUTPUTS AND ARTIFACTS

Pipeline catalog; data contracts; lineage and mapping records; orchestration and runbook design; quality and reconciliation rules; observability, security, recovery, test, decision, evidence, and handoff artifacts.

EVIDENCE REQUIREMENTS

Identify source, run, code, configuration, schema, data window, environment, actor, method, result, and limitation. A green run, checksum, or sample supports only its bounded claim.

VERIFICATION AND ACCEPTANCE TESTS

Verify completeness, schema compatibility, ordering, deduplication, replay, idempotency, quality gates, access, lineage, latency, backpressure, failure recovery, and independent result reproduction.

FAILURE STATES AND ANTI-PATTERNS

Silent drops, duplicate loads, unbounded retries, hidden manual fixes, hard-coded credentials, schema drift, unbounded backlogs, missing lineage, destructive replay, and optimistic freshness claims.

RECOVERY AND RECONSIDERATION

Pause or quarantine affected stages on corruption, drift, access failure, or quality breach. Preserve offsets and evidence; use approved replay, reconciliation, corrected source, backfill, compensation, or safe degradation.

COMPLETION CRITERIA

Contracts, owners, lineage, transformations, quality, security, operations, recovery, tests, evidence, gaps, and next action are recorded. Status remains CONTENT-COMPLETE pending later review gates.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off contracts, mappings, offsets, runbooks, alerts, quality rules, recovery, retention, and migration obligations to Data Modeling, Integration, Storage, Database, Platform, Security, Testing, Data Operations, and Retirement.

RELATED CODEX SECTIONS

030 Information Architecture; 031 Data Architecture; 041 Integration Architecture; 043 Event-Driven Architecture; 151 Database Engineering; 154 Data Modeling; 156 Data Quality Engineering; 157 Data Integration; 159-163 Warehousing, Lake, Analytics, BI, and Analysis; 307 Data Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the canonical contract, shared governance, Sections 019-032, and roadmap. Source-specific standards for streaming, orchestration, residency, quality, cost, retention, and sector controls remain unresolved.

154

SECTION 154 - DATA MODELING

CODEX/03_SECTIONS/154_DATA_MODELING.txt

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6,313 BYTES / PRESERVED SOURCE IDENTITY

SECTION 154 — Data Modeling

STATUS: CONTENT-COMPLETE

PURPOSE

Define stable data meaning, structure, relationships, identifiers, constraints, and representations that support authorized processes, functions, systems, analysis, and lifecycle decisions.

REQUIRED OUTCOME

Conceptual, logical, and physical models with controlled mappings, definitions, ownership, constraints, variants, lineage, quality rules, change impact, and acceptance boundaries.

WHY THIS MATTERS

Ambiguous entities and identifiers create duplicate identity, invalid joins, privacy mistakes, and incompatible systems. A diagram is not proof that meanings are shared or enforced.

APPLICABILITY

Apply when data entities, events, relationships, schemas, identifiers, classifications, or transformations materially affect behavior or decisions.

NON-APPLICABILITY CONDITIONS

No material meaning, structure, relationship, identifier, constraint, interface, quality, or lifecycle behavior changes and an exact model baseline covers the work. Record scope and trigger.

PREREQUISITES AND DEPENDENCIES

Domain and information architecture, requirements, processes, functions, data subjects, source systems, contracts, jurisdictions, quality, privacy, security, records, and lifecycle rules.

REQUIRED INPUTS

Business terms, entities, events, identifiers, relationships, cardinality, constraints, states, classifications, sample records, consumers, source authority, variants, assumptions, and disputes.

HUMAN RESPONSIBILITIES AND AUTHORITY

The human owner approves canonical meaning, keys, classifications, relationships, exceptions, quality thresholds, compatibility, and acceptance; assigns domain, data, privacy, security, and records authorities.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized glossaries, requirements, domain evidence, source schemas, contracts, representative records, lineage, mappings, and model versions. It may propose terms, entities, relationships, identifiers, constraints, model mappings, compatibility views, test cases, and impact analyses, and may surface ambiguity, disagreement, privacy risk, and unresolved assumptions for human adjudication.

AI must distinguish source-defined meaning from inference and must not declare a canonical term, resolve identity, approve classifications, expose records, alter a model or schema, or claim semantic agreement from a diagram, sample, or schema inspection. Any mutation-capable proposal must identify the authorized owner, model level and version, affected consumers and mappings, validation evidence, immutable decision and audit records, and forward migration, compatibility, or compensation path; unresolved meaning, identity, authority, or evidence conflicts require preservation of the prior version and owner review.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Researcher, Observer, Governance Reviewer, Security Auditor, User Experience Reviewer, Implementer, Tester, Verifier, Critic, Documentation Writer, Memory Curator, and Red Team.

REQUIRED ACTIVITIES

1. Establish vocabulary, scope, subjects, contexts, and model levels.
2. Define entities, events, identifiers, relationships, states, constraints, classifications, and ownership.
3. Map conceptual to logical and physical structures, contracts, lineage, variants, and migration impact.
4. Validate with domain owners and test representative, ambiguous, missing, duplicate, and incompatible records.

HUMAN-AI INTERACTION REQUIREMENTS

AI may propose models and mappings from authorized evidence, but must distinguish source meaning from inference. It must not declare a canonical term, infer identity, expose records, or alter schemas without authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Keep meaning, model level, schema, storage, interface, and implementation distinct. Version identifiers, constraints, mappings, lineage, ownership, compatibility, and lifecycle states; make critical constraints enforceable.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show model version, domain owner, scope, mappings, terms, identifiers, constraints, consumers, conflicts, evidence, and authorized approve, supersede, restrict, migrate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Model charter; glossary and data dictionary; conceptual, logical, physical, event, state, and lineage models; contracts; mapping and compatibility records; decisions, tests, evidence, dissent, and handoff.

EVIDENCE REQUIREMENTS

Claims identify source, definition, model level, version, scope, population, method, owner, and limitation. A sample, diagram, or schema inspection proves only its bounded observation.

VERIFICATION AND ACCEPTANCE TESTS

Verify term consistency, key uniqueness, cardinality, constraints, null and unknown handling, temporal behavior, mappings, schema compatibility, access implications, and independent reproduction.

FAILURE STATES AND ANTI-PATTERNS

Entity soup, overloaded identifiers, hidden many-to-many relations, accidental null semantics, physical design masquerading as meaning, copied definitions, and unowned model changes.

RECOVERY AND RECONSIDERATION

Freeze affected changes when meaning or identity conflicts arise; preserve prior versions and evidence. Use owner adjudication, mapping, quarantine, controlled migration, compatibility views, or compensation, then reassess consumers.

COMPLETION CRITERIA

Terms, entities, relationships, constraints, owners, mappings, consumers, tests, evidence, gaps, and reopening conditions are explicit. Status remains CONTENT-COMPLETE until later gates.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off models and contracts to Database, Data, Integration, API, Application, Analytics, ML, Privacy, Security, Records, Testing, Migration, Operations, and Retirement work.

RELATED CODEX SECTIONS

016 Domain Analysis; 030 Information Architecture; 031 Data Architecture; 041 Integration Architecture; 042 API Architecture; 151 Database Engineering; 155 Data Governance; 159 Data Warehousing; 175 Knowledge Graph Engineering; 391-395 data lifecycle sections.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the canonical contract, shared governance, Sections 016, 019-032, and roadmap. Domain-specific vocabularies, modeling notation, legal identity rules, and sector standards remain unresolved.

155

SECTION 155 - DATA GOVERNANCE

CODEX/03_SECTIONS/155_DATA_GOVERNANCE.txt

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5,898 BYTES / PRESERVED SOURCE IDENTITY

SECTION 155 — Data Governance

STATUS: CONTENT-COMPLETE

PURPOSE

Govern the purpose, ownership, authority, access, quality, sharing, retention, correction, and retirement of data across its lifecycle.

REQUIRED OUTCOME

A data-governance operating model with accountable owners, classifications, policies, decision rights, exceptions, evidence, review cadence, and enforcement paths.

WHY THIS MATTERS

Data governance determines who may use data and why. Unowned classification, undocumented purpose, or unenforced policy converts technical convenience into unauthorized consequence.

APPLICABILITY

Apply whenever data affects rights, decisions, safety, privacy, security, compliance, service, finance, models, records, or material operations.

NON-APPLICABILITY CONDITIONS

A bounded non-material change has no effect on purpose, ownership, access, classification, quality, retention, sharing, or lifecycle; record the exact baseline and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Authority, purpose, data inventory, owners, subjects, classifications, jurisdictions, policies, contracts, retention, access, quality, risk, and acceptance rules.

REQUIRED INPUTS

Data domains, purposes, lawful or organizational basis, owners, custodians, consumers, classifications, access roles, decisions, exceptions, incidents, evidence, and unknowns.

HUMAN RESPONSIBILITIES AND AUTHORITY

The human owner appoints data owners and stewards, resolves conflicting rights and purposes, approves policy, exceptions, sharing, retention, deletion, risk, and acceptance. Authority is not inferred from technical access.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory authorized data, compare governance sources, identify missing ownership or control evidence, and draft classifications, policies, registers, decision records, and review questions. It must preserve provenance and uncertainty, distinguish recommendation from authorization, and use only approved sources. It must not infer lawful basis, ownership, consent, access rights, retention authority, or policy exceptions; approve sharing or access; alter authoritative records or controls; or conceal conflicts, dissent, or unresolved gaps. Any governance mutation or consequential recommendation requires the applicable policy evaluation, capability scope, immutable decision record, human authority, and append-only audit linkage before execution or acceptance.

APPLICABLE EXISTING AGENT ROLES

Governance Reviewer, Architect, Researcher, Analyst, Security Auditor, Verifier, Tester, Documentation Writer, Memory Curator, Critic, User Experience Reviewer, Implementer, and Red Team.

REQUIRED ACTIVITIES

1. Inventory data and assign ownership, purpose, classification, and lifecycle.
2. Define policy, access, quality, sharing, retention, correction, deletion, exception, and escalation rules.
3. Bind rules to systems, contracts, evidence, monitoring, reviews, and enforcement points.
4. Review changes and incidents; record decisions, dissent, evidence, and reopening conditions.

HUMAN-AI INTERACTION REQUIREMENTS

AI may inventory, compare, and draft governance artifacts from authorized sources. It must not infer lawful basis, approve access, classify sensitive data conclusively, waive policy, or conceal uncertainty.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent policies, owners, classifications, permissions, decisions, exceptions, evidence, expiries, and supersession as immutable attributable records with enforceable controls and append-only audit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show purpose, owner, classification, authority, policy status, expiry, consumers, exceptions, evidence, and controls to approve, deny, restrict, revoke, quarantine, correct, retain, or delete.

REQUIRED OUTPUTS AND ARTIFACTS

Governance charter; data inventory; ownership and classification register; policy and decision records; access and exception register; retention and deletion records; audit, evidence, review, and handoff package.

EVIDENCE REQUIREMENTS

Each decision identifies authority, scope, version, purpose, subjects, policy, evidence, date, expiry, dissent, and limitation. A policy document alone does not prove enforcement.

VERIFICATION AND ACCEPTANCE TESTS

Verify ownership, purpose, classification, access, policy enforcement, exception expiry, audit integrity, subject rights, retention, deletion, and independent decision reproduction.

FAILURE STATES AND ANTI-PATTERNS

Implied consent, ownerless data, blanket access, policy without enforcement, permanent exceptions, hidden secondary use, mutable audit, and treating a dashboard or attestation as compliance proof.

RECOVERY AND RECONSIDERATION

Suspend affected use on authority, classification, policy, access, or evidence conflict. Preserve records; use restriction, quarantine, human adjudication, correction, revocation, or compensation, then review downstream effects.

COMPLETION CRITERIA

Scope, authority, owners, policies, controls, evidence, decisions, exceptions, review, gaps, and next action are recorded. Status remains CONTENT-COMPLETE pending reconciliation and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off enforceable rules and records to Data, Security, Privacy, Legal, Records, Application, Testing, Operations, AI Governance, and Retirement sections with reopening triggers.

RELATED CODEX SECTIONS

016 Domain Analysis; 019 Requirements Engineering; 030-031 Information and Data Architecture; 045-049 Security through Compliance Architecture; 155-158 data governance and migration; 183 AI Governance; 391-406 lifecycle governance and retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on shared governance, the canonical contract, Sections 016 and 019-032, Owner Cockpit Requirements, and roadmap. Jurisdictional law, sector rules, roles, lawful bases, and enforcement technology remain unresolved.

156

SECTION 156 - DATA QUALITY ENGINEERING

CODEX/03_SECTIONS/156_DATA_QUALITY_ENGINEERING.txt

SHA-256 10c60c40df11aa570081ad91c2e7d43f662ea87842ef3970ed7bad847f4b1a31

5,604 BYTES / PRESERVED SOURCE IDENTITY

SECTION 156 — Data Quality Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Engineer measurable, observable, explainable, and recoverable data quality across source, transformation, storage, use, and lifecycle boundaries.

REQUIRED OUTCOME

Quality dimensions, rules, thresholds, sampling, monitoring, ownership, quarantine, correction, reconciliation, evidence, and impact handling for each material data product.

WHY THIS MATTERS

Plausible data can be incomplete, biased, stale, duplicated, or misclassified. A score without population, rule, threshold, and consequence is not a quality claim.

APPLICABILITY

Apply when data informs decisions, automation, reporting, models, records, compliance, safety, service, or material operations.

NON-APPLICABILITY CONDITIONS

No material quality rule, threshold, source, transformation, consumer, or lifecycle behavior changes and an exact approved baseline covers the work; record the decision.

PREREQUISITES AND DEPENDENCIES

Purpose, data contract, population, dimensions, business rules, tolerances, source lineage, consumers, sampling, incidents, risk, and correction authority.

REQUIRED INPUTS

Datasets, profiles, rules, thresholds, samples, missingness, drift, duplicates, reconciliations, alerts, defects, impacts, evidence, and assumptions.

HUMAN RESPONSIBILITIES AND AUTHORITY

The owner approves quality dimensions, thresholds, exceptions, risk acceptance, correction, quarantine, consumer notification, and acceptance; assigns domain, steward, and verification authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may profile authorized data, propose quality dimensions and rules, detect anomalies or drift, explain observed defects, and draft monitoring, quarantine, correction, reconciliation, and evidence artifacts. It must state population, sampling, method, uncertainty, limitations, and possible bias, protect sensitive values, and keep recommendations traceable to source and rule versions. It must not declare data fit for consequential use, silently impute or correct records, release quarantined data, change thresholds, suppress alerts, or notify consumers without authorized review. Any quality-control mutation or acceptance decision requires the applicable policy evaluation, capability scope, immutable decision record, human authority, and append-only audit linkage.

APPLICABLE EXISTING AGENT ROLES

Analyst, Researcher, Observer, Architect, Implementer, Tester, Verifier, Governance Reviewer, Security Auditor, Critic, Optimizer, Documentation Writer, Memory Curator, and Red Team.

REQUIRED ACTIVITIES

1. Define quality dimensions, rules, populations, thresholds, validity, and owners.
2. Profile and monitor data; detect missingness, drift, duplication, bias, corruption, and schema change.
3. Route failures to quarantine, correction, reconciliation, consumer notification, and evidence preservation.
4. Test representative and adversarial cases and review threshold fitness over time.

HUMAN-AI INTERACTION REQUIREMENTS

AI may profile authorized data and propose rules or anomaly explanations. It must disclose sampling and uncertainty, protect values, and never auto-correct consequential data or declare fitness without authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent quality rules, results, populations, samples, thresholds, lineage, alerts, quarantines, corrections, and invalidation as versioned records. Separate data quality from model confidence and service uptime.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show dataset, rule version, population, freshness, score, threshold, uncertainty, impact, owner, alert, quarantine, correction, and evidence actions.

REQUIRED OUTPUTS AND ARTIFACTS

Quality charter; rule catalog; profiles; monitoring and alert definitions; defect, quarantine, correction, reconciliation, impact, test, decision, evidence, and handoff records.

EVIDENCE REQUIREMENTS

Identify population, sampling, dimensions, rule, threshold, source, version, method, time, result, uncertainty, and limitation. A sample never proves all records.

VERIFICATION AND ACCEPTANCE TESTS

Verify rule correctness, representative coverage, threshold behavior, false positives and negatives, drift, access, correction, quarantine, consumer notification, and reproduction.

FAILURE STATES AND ANTI-PATTERNS

Single metric quality, biased samples, threshold gaming, silent imputation, destructive correction, alert fatigue, stale profiles, and quality gates that bypass owner decisions.

RECOVERY AND RECONSIDERATION

Quarantine affected data, preserve prior versions and evidence, notify consumers, and use approved correction, reconciliation, reprocessing, or restricted-use paths. Reassess decisions and models already using it.

COMPLETION CRITERIA

Rules, thresholds, populations, owners, monitoring, response, tests, evidence, limitations, and review cadence are explicit. Status remains CONTENT-COMPLETE pending later gates.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off quality contracts and alerts to Data Engineering, Data Governance, Integration, Warehousing, Analytics, ML, Testing, Operations, Incident, and Migration work.

RELATED CODEX SECTIONS

031 Data Architecture; 153 Data Engineering; 154 Data Modeling; 157 Data Integration; 158 Data Migration; 161-167 Analytics through ML Engineering; 254 Data Testing; 307 Data Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the canonical contract, shared governance, data and information architecture, and roadmap. Sector quality definitions, sampling standards, tolerances, and legal consequences remain unresolved.

157

SECTION 157 - DATA INTEGRATION

CODEX/03_SECTIONS/157_DATA_INTEGRATION.txt

SHA-256 3803974bf5090e6f781aa5eec10308d50052cce0b62c597d6e0a29b0e26cf661

5,259 BYTES / PRESERVED SOURCE IDENTITY

SECTION 157 — Data Integration

STATUS: CONTENT-COMPLETE

PURPOSE

Define controlled exchange of data across systems, domains, organizations, and providers while preserving meaning, identity, authority, quality, security, and lifecycle.

REQUIRED OUTCOME

Integration contracts, mappings, transport, timing, ordering, error, retry, reconciliation, lineage, ownership, compatibility, and recovery are explicit.

WHY THIS MATTERS

Integration can create duplicate truth, unauthorized disclosure, silent loss, or incompatible change. Connectivity alone is not semantic or operational success.

APPLICABILITY

Apply when data crosses a system, trust, domain, supplier, jurisdiction, protocol, or lifecycle boundary.

NON-APPLICABILITY CONDITIONS

A bounded change has no material effect on data meaning, contract, access, timing, lineage, quality, or recovery and an exact baseline covers it; record why.

PREREQUISITES AND DEPENDENCIES

Source authority, schemas, mappings, identity, access, protocol, workload, ordering, latency, errors, quality, privacy, security, retention, and owners.

REQUIRED INPUTS

Endpoints, events, files, schemas, mappings, transformations, credentials, volumes, schedules, dependencies, consumers, failure cases, and evidence limits.

HUMAN RESPONSIBILITIES AND AUTHORITY

The owner approves sources, consumers, contracts, sharing, exceptions, risk, compatibility, and acceptance; assigns integration, data, security, privacy, and operations authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized schemas and exchange evidence, propose mappings, transformations, contract tests, compatibility findings, and failure-handling options, and draft integration records and runbooks. It must preserve source and destination meaning, identity, lineage, access boundaries, uncertainty, and version differences; distinguish delivery from semantic success; and surface conflicts or missing evidence. It must not infer permission or semantic equivalence, publish or transmit data, change contracts, credentials, routing, retries, or replay state, or declare an exchange accepted without human authority. Any integration mutation or consequential replay requires the applicable policy evaluation, capability scope, immutable decision record, human approval, and append-only audit linkage.

APPLICABLE EXISTING AGENT ROLES

Architect, Implementer, Analyst, Researcher, Tester, Verifier, Security Auditor, Governance Reviewer, Observer, Critic, Coordinator, Documentation Writer, Memory Curator, and Red Team.

REQUIRED ACTIVITIES

Define boundaries and contracts; map and transform data; implement validation, identity, ordering, retry, idempotency, reconciliation, monitoring, quarantine, and recovery; test compatibility and failure.

HUMAN-AI INTERACTION REQUIREMENTS

AI may propose mappings and analyze authorized exchanges. It must not infer permission, publish data, modify integration, or claim semantic equivalence without human review and evidence.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate transport, contract, mapping, source, destination, identity, authority, lineage, and lifecycle. Design explicit timeouts, retries, deduplication, compensation, and version negotiation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show integration version, endpoints, owners, data classes, lag, failures, retries, reconciliation, authority, evidence, and controls for pause, replay, quarantine, revoke, and retire.

REQUIRED OUTPUTS AND ARTIFACTS

Integration catalog, data contracts, mappings, lineage, security model, runbooks, compatibility matrix, test results, evidence, decisions, exceptions, and handoff.

EVIDENCE REQUIREMENTS

Identify producer, consumer, contract, version, data window, environment, method, result, actor, and limitation. A successful transfer proves delivery only.

VERIFICATION AND ACCEPTANCE TESTS

Verify schema, semantics, identity, authorization, ordering, duplicates, retries, partial failure, reconciliation, latency, compatibility, audit, and recovery.

FAILURE STATES AND ANTI-PATTERNS

Point-to-point sprawl, shared credentials, silent coercion, unbounded retry, missing ownership, contract drift, duplicate delivery, and integration that bypasses governance.

RECOVERY AND RECONSIDERATION

Pause affected paths, preserve messages and evidence, quarantine unsafe data, and use approved replay, reconciliation, version fallback, manual exchange, or compensation.

COMPLETION CRITERIA

Contracts, owners, mappings, controls, tests, evidence, gaps, and reopening triggers are recorded; status remains CONTENT-COMPLETE pending later review.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off contracts to API, Event, Data, Application, Security, Privacy, Testing, Operations, Migration, and Retirement sections with exact versions and recovery notes.

RELATED CODEX SECTIONS

031 Data Architecture; 041 Integration Architecture; 042 API Architecture; 043 Event-Driven Architecture; 153 Data Engineering; 156 Data Quality; 158 Data Migration; 307 Data Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on shared governance, the canonical contract, Sections 019-032, and roadmap. Protocol, interoperability, residency, supplier, and sector-specific obligations remain unresolved.

158

SECTION 158 - DATA MIGRATION

CODEX/03_SECTIONS/158_DATA_MIGRATION.txt

SHA-256 119588bb92559f17c25197a7b215e09f76848611bc0a472e3c687afb37223145

5,672 BYTES / PRESERVED SOURCE IDENTITY

SECTION 158 — Data Migration

STATUS: CONTENT-COMPLETE

PURPOSE

Move or transform data between versions, systems, stores, regions, or ownership boundaries without losing meaning, authority, integrity, continuity, or evidence.

REQUIRED OUTCOME

A migration plan with inventory, mapping, reconciliation, controls, rehearsal, cutover, coexistence, rollback or compensation, validation, communications, and retirement.

WHY THIS MATTERS

Migration changes historical truth and future behavior. A completed transfer can still lose records, alter meaning, duplicate identity, or leave unauthorized residual copies.

APPLICABILITY

Apply to material movement, conversion, consolidation, separation, modernization, provider change, schema change, region change, or retention transition.

NON-APPLICABILITY CONDITIONS

No material data, meaning, authority, continuity, lifecycle, or evidence changes; exact baseline covers it and the decision records why migration analysis is unnecessary.

PREREQUISITES AND DEPENDENCIES

Source and target models, inventories, mappings, owners, quality, access, retention, volume, downtime, recovery, legal, security, privacy, and acceptance authority.

REQUIRED INPUTS

Migration scope, records, schemas, mappings, transformations, counts, checksums, exceptions, cutover, coexistence, reconciliation, rollback, communications, and unknowns.

HUMAN RESPONSIBILITIES AND AUTHORITY

The owner approves scope, timing, risk, downtime, mapping, cutover, residual copies, rollback or compensation, communications, and acceptance; assigns source, target, security, privacy, and operations authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized migration inventories, draft mappings, transformations, rehearsal plans, test cases, reconciliation logic, cutover criteria, communications, and evidence packages. It must preserve provenance, source and target versions, exceptions, uncertainty, residual-copy implications, and the distinction between count, checksum, and semantic equivalence. It must not infer authority or equivalence, expose records, execute transfer, cutover, deletion, destructive cleanup, or compensation, or declare migration success without authorized validation. Any migration mutation or cutover requires the applicable policy evaluation, capability scope, immutable decision record, human approval, and append-only audit linkage; rollback must use an approved compensation path rather than destructive reversal.

APPLICABLE EXISTING AGENT ROLES

Architect, Planner, Implementer, Analyst, Researcher, Tester, Verifier, Observer, Governance Reviewer, Security Auditor, Critic, Coordinator, Documentation Writer, Memory Curator, and Red Team.

REQUIRED ACTIVITIES

Inventory and classify; map and rehearse; validate quality; secure and transfer; reconcile; execute controlled cutover; preserve evidence; verify consumers; retire or quarantine the source.

HUMAN-AI INTERACTION REQUIREMENTS

AI may draft mappings, scripts, test cases, and reconciliation queries from authorized data. It must not execute destructive migration, infer equivalence, expose records, or declare success from counts alone.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Preserve source, target, versions, mappings, lineage, checkpoints, idempotency, audit, compatibility, reconciliation, compensation, and residual-copy semantics. Prefer reversible stages and explicit cutover gates.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show migration baseline, phase, counts, exceptions, quality, owners, approvals, source and target state, cutover readiness, residual copies, and authorized pause, abort, reconcile, compensate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Migration charter, inventory, mapping, transformation, rehearsal, security, runbook, reconciliation, cutover, validation, rollback or compensation, communication, evidence, and handoff records.

EVIDENCE REQUIREMENTS

Identify source, target, scope, version, population, method, counts, exceptions, environment, authority, result, and limitation. A checksum does not prove semantic equivalence.

VERIFICATION AND ACCEPTANCE TESTS

Verify inventory completeness, mapping, transformation, authorization, counts, checksums, semantics, referential integrity, performance, cutover, coexistence, recovery, and deletion of residual copies.

FAILURE STATES AND ANTI-PATTERNS

Big-bang cutover, untested rollback, hidden source writes, count-only validation, lost exceptions, duplicate identities, unbounded downtime, and destructive cleanup before acceptance.

RECOVERY AND RECONSIDERATION

Pause writes or isolate phases on mismatch; preserve both states and evidence. Use reconciliation, replay, prior version, coexistence, controlled forward repair, or compensation; never silently erase the source.

COMPLETION CRITERIA

Scope, authority, mappings, tests, cutover, reconciliation, residual copies, evidence, gaps, and next decision are recorded. Status remains CONTENT-COMPLETE pending later gates.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off to Database, Storage, Data Engineering, Integration, Application, Security, Privacy, Testing, Operations, Data Migration Operations, Retention, and Retirement.

RELATED CODEX SECTIONS

031 Data Architecture; 151-157 Database through Data Integration; 255-300 delivery and deployment; 382 Data Migration Operations; 391-398 retention, archival, deletion, and decommissioning.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the canonical contract, shared governance, data architecture, and roadmap. Actual data volume, legal basis, downtime, reconciliation tolerance, and owner-approved recovery path remain unresolved.

159

SECTION 159 - DATA WAREHOUSING

CODEX/03_SECTIONS/159_DATA_WAREHOUSING.txt

SHA-256 242433dc44cbc148421ea7f66c7a32b2287f943fe3c22125f9ba19ce19dfa1ea

5,536 BYTES / PRESERVED SOURCE IDENTITY

SECTION 159 — Data Warehousing

STATUS: CONTENT-COMPLETE

PURPOSE

Design governed analytical stores that preserve historical meaning, dimensional context, lineage, quality, access, performance, retention, and decision evidence.

REQUIRED OUTCOME

A warehouse model with source contracts, grain, dimensions, facts, history, refresh, quality, lineage, security, ownership, recovery, and lifecycle controls.

WHY THIS MATTERS

Warehouses can make stale or transformed data appear authoritative. A report that runs successfully does not prove its grain, lineage, freshness, or interpretation.

APPLICABILITY

Apply when historical, integrated, subject-oriented, or decision-support data is consolidated for reporting, analysis, compliance, or operations.

NON-APPLICABILITY CONDITIONS

No material analytical meaning, source, grain, refresh, access, quality, retention, or consumer behavior changes and an exact baseline covers the work; record the reason.

PREREQUISITES AND DEPENDENCIES

Business definitions, source systems, grain, history, dimensions, measures, refresh, quality, lineage, users, access, retention, performance, and acceptance authority.

REQUIRED INPUTS

Source inventory, model, mappings, snapshots, schedules, late data, corrections, consumers, reports, controls, privacy, security, recovery, assumptions, and gaps.

HUMAN RESPONSIBILITIES AND AUTHORITY

The owner approves definitions, metrics, sources, refresh, access, quality, retention, exceptions, and acceptance; assigns data, reporting, privacy, security, and operations authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized schemas, lineage, load histories, quality results, query plans, metric definitions, and warehouse metadata; identify inconsistencies; draft mappings, tests, documentation, and bounded analysis; and propose model, refresh, quality, access, cost, or recovery improvements.

AI must distinguish source facts, transformed values, inferred relationships, metric definitions, recommendations, and verified results. It must preserve grain, time window, source identity, transformation version, freshness, privacy classification, and known limitations in every material output.

AI must not silently redefine a metric, merge incompatible grains, infer sensitive attributes, publish consequential reports, change production schemas or loads, widen access, delete history, approve retention exceptions, or declare data quality or business truth without delegated authority and human acceptance.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Researcher, Data Engineer, Implementer, Tester, Verifier, Governance Reviewer, Security Auditor, Observer, Critic, Documentation Writer, Memory Curator, and Red Team.

REQUIRED ACTIVITIES

Define grain and history; map sources; build conformed dimensions and facts; establish refresh, quality, lineage, access, reconciliation, observability, recovery, and consumer validation.

HUMAN-AI INTERACTION REQUIREMENTS

AI may analyze authorized warehouse metadata and propose models or queries. It must preserve metric definitions and provenance and must not publish consequential reports or infer authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate source, staging, warehouse, mart, cache, report, and evidence. Preserve slowly changing history, late-arriving data, correction, reconciliation, access, retention, and reproducible query semantics.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show warehouse version, grain, freshness, source status, quality, metric owner, consumers, access, costs, incidents, evidence, and pause, correct, restrict, restore, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Warehouse charter; dimensional and lineage models; metric catalog; source and refresh contracts; quality, access, retention, recovery, report, test, decision, evidence, and handoff records.

EVIDENCE REQUIREMENTS

Identify source, grain, transformation, window, refresh, model, query, environment, actor, result, and limitation. A report or row count proves only its declared slice.

VERIFICATION AND ACCEPTANCE TESTS

Verify grain, joins, history, late data, refresh, reconciliation, access, performance, report definitions, correction, recovery, retention, and independent reproduction.

FAILURE STATES AND ANTI-PATTERNS

Metric drift, double counting, stale dimensions, hidden filters, untracked corrections, unrestricted marts, snapshot confusion, and dashboards presented as decisions or truth.

RECOVERY AND RECONSIDERATION

Quarantine affected loads or reports, preserve versions and evidence, notify consumers, and use approved correction, backfill, prior snapshot, restricted mode, or compensation.

COMPLETION CRITERIA

Definitions, grain, sources, owners, refresh, quality, access, recovery, tests, evidence, gaps, and review cadence are recorded. Status remains CONTENT-COMPLETE.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off models and metric contracts to Data Engineering, Data Quality, BI, Analytics, Database, Storage, Governance, Security, Testing, Operations, and Migration.

RELATED CODEX SECTIONS

031 Data Architecture; 153-156 Data Engineering through Data Quality; 160 Data Lake; 161 Analytics Engineering; 162 Business Intelligence; 163 Data Analysis; 254 Data Testing; 307 Data Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the canonical contract, shared governance, Sections 019-032, and roadmap. Industry warehouse patterns, metric authority, refresh tolerance, residency, and retention rules remain unresolved.

160

SECTION 160 - DATA LAKE ENGINEERING

CODEX/03_SECTIONS/160_DATA_LAKE_ENGINEERING.txt

SHA-256 91066ccab0056bd170ccf8a741b49f28bec5a6a526c35836b8b87ce8d26d838d

5,546 BYTES / PRESERVED SOURCE IDENTITY

SECTION 160 — Data Lake Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Engineer governed lake environments that retain diverse data with discoverability, provenance, security, quality, lifecycle, cost, and controlled analytical use.

REQUIRED OUTCOME

A lake-zone and catalog design covering ingestion, formats, schemas, metadata, access, isolation, quality, lineage, retention, recovery, and consumer contracts.

WHY THIS MATTERS

Cheap scale can become an unsearchable, overexposed, stale copy of everything. Flexibility does not remove ownership, quality, retention, or deletion duties.

APPLICABILITY

Apply when varied structured, semi-structured, unstructured, streaming, experimental, or historical data is collected for authorized reuse.

NON-APPLICABILITY CONDITIONS

No material source, zone, classification, access, retention, lineage, quality, cost, or consumer behavior changes and an exact baseline covers the work.

PREREQUISITES AND DEPENDENCIES

Purpose, sources, formats, zones, catalog, subjects, classification, access, quality, lineage, workloads, retention, deletion, cost, and recovery objectives.

REQUIRED INPUTS

Sources, schemas, formats, partitions, volumes, zones, metadata, consumers, identities, policies, quality, lifecycle, incidents, and unknowns.

HUMAN RESPONSIBILITIES AND AUTHORITY

The owner approves collection, reuse, zones, access, retention, deletion, cost, risk, and acceptance; assigns data, platform, privacy, security, and operations authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized catalogs, schemas, object metadata, lineage, quality results, partition statistics, access records, and lifecycle signals; identify drift or duplicate data; draft classifications, contracts, tests, mappings, documentation, and bounded queries; and propose quarantine, correction, cost, or recovery actions.

AI must preserve source identity, zone, classification, purpose, schema version, time window, lineage, freshness, and limitations. It must keep exploratory observations separate from validated or authoritative data and must surface uncertainty when metadata is incomplete or conflicting.

AI must not read, copy, expose, reclassify, retain, delete, publish, grant access to, or move data across zones or jurisdictions without explicit authorization. It must not declare a lake zone trusted, complete, compliant, or safe merely because ingestion succeeded, catalog metadata exists, or a query returned results.

APPLICABLE EXISTING AGENT ROLES

Architect, Data Engineer, Researcher, Analyst, Implementer, Tester, Verifier, Governance Reviewer, Security Auditor, Observer, Critic, Optimizer, Documentation Writer, Memory Curator, and Red Team.

REQUIRED ACTIVITIES

Classify and catalog sources; design zones, formats, partitions, schema evolution, quality gates, lineage, access, lifecycle, observability, cost, recovery, and deletion; validate consumers.

HUMAN-AI INTERACTION REQUIREMENTS

AI may classify authorized metadata and propose transformations or queries. It must not infer permission, expose raw data, publish unverified findings, or turn exploratory data into authoritative evidence.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate raw, validated, curated, sandbox, quarantine, archive, backup, and evidence zones. Enforce catalog, identity, purpose, lineage, retention, deletion, and isolation at boundaries.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show zone, source, owner, format, freshness, quality, access, classification, cost, retention, incidents, evidence, and controls for quarantine, restrict, restore, delete, and retire.

REQUIRED OUTPUTS AND ARTIFACTS

Lake charter; zone and catalog model; format and schema contracts; lineage, quality, access, retention, cost, recovery, test, decision, evidence, and handoff records.

EVIDENCE REQUIREMENTS

Identify source, zone, format, schema, partition, population, environment, method, actor, result, and limitation. Catalog presence or object count proves neither quality nor authorization.

VERIFICATION AND ACCEPTANCE TESTS

Verify isolation, access, encryption, catalogability, schema evolution, partition behavior, quality, lineage, cost controls, recovery, retention, deletion, and consumer interpretation.

FAILURE STATES AND ANTI-PATTERNS

Data swamp, public raw zone, orphaned objects, missing catalog, schema drift, copied sensitive data, unbounded retention, and exploratory output treated as source truth.

RECOVERY AND RECONSIDERATION

Quarantine affected zones, restrict access, preserve evidence, and use approved prior data, reprocessing, reconciliation, deletion, restore, or restricted operation. Reassess downstream uses.

COMPLETION CRITERIA

Zones, sources, ownership, catalog, quality, access, lineage, lifecycle, recovery, tests, evidence, gaps, and next action are recorded; status remains CONTENT-COMPLETE.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off zone and contract definitions to Storage, Data Engineering, Data Governance, Warehousing, Analytics, ML, Security, Privacy, Testing, Operations, and Retention work.

RELATED CODEX SECTIONS

031 Data Architecture; 152 Storage Engineering; 153 Data Engineering; 155-156 Governance and Quality; 159 Data Warehousing; 160-164 Analytics and Data Science; 391-395 data lifecycle sections.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the canonical contract, shared governance, Sections 019-032, and roadmap. Provider formats, residency, catalog standards, deletion propagation, cost limits, and sector rules remain unresolved.

161

SECTION 161 - ANALYTICS ENGINEERING

CODEX/03_SECTIONS/161_ANALYTICS_ENGINEERING.txt

SHA-256 c619584d5ecf90d4d813617ed27b251cf25db057d19002ba469ff586c994fe43

5,174 BYTES / PRESERVED SOURCE IDENTITY

SECTION 161 — Analytics Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Engineer trustworthy analytical models, metrics, transformations, tests, documentation, and delivery practices that turn governed data into reproducible decision support.

REQUIRED OUTCOME

Versioned analytical models with metric definitions, lineage, tests, owners, freshness, access, documentation, deployment, monitoring, and recovery.

WHY THIS MATTERS

Analytics can encode business rules as hidden software. A polished metric may be stale, duplicated, biased, or semantically wrong.

APPLICABILITY

Apply when reusable transformations, metrics, analytical models, semantic layers, or decision datasets are built or changed.

NON-APPLICABILITY CONDITIONS

No material metric, transformation, lineage, consumer, access, quality, or lifecycle behavior changes and an exact analytical baseline covers the work.

PREREQUISITES AND DEPENDENCIES

Data contracts, business definitions, source lineage, consumers, tests, freshness, quality, access, deployment, ownership, and acceptance rules.

REQUIRED INPUTS

Models, metrics, SQL or code, dependencies, tests, samples, thresholds, reports, owners, environments, evidence, assumptions, and known gaps.

HUMAN RESPONSIBILITIES AND AUTHORITY

The owner approves metric meaning, thresholds, access, publication, exceptions, risk, and acceptance; assigns data, domain, analytics, privacy, security, and verification authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized model code, schemas, lineage, tests, metric catalogs, query plans, freshness signals, and bounded samples; draft transformations, tests, documentation, impact analyses, and explanations; and propose corrections or optimization experiments.

AI must preserve metric definitions, grain, population, filters, time window, source and model versions, transformation lineage, freshness, uncertainty, and known exclusions. It must distinguish a syntactically valid query from a semantically valid measure and a useful analysis from a decision.

AI must not redefine metrics, change business meaning, publish consequential results, modify production models, bypass review gates, infer authorization, or present correlation, forecast, or generated narrative as causation or verified fact without delegated authority and human acceptance.

APPLICABLE EXISTING AGENT ROLES

Analyst, Data Engineer, Researcher, Architect, Implementer, Tester, Verifier, Governance Reviewer, Security Auditor, Critic, Optimizer, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Define metrics and grain; build tested transformations; document lineage; validate freshness and quality; deploy controlled changes; monitor drift; notify consumers; preserve recovery.

HUMAN-AI INTERACTION REQUIREMENTS

AI may draft transformations and tests from authorized sources. It must preserve metric definitions, disclose uncertainty, and never publish consequential analysis or infer business authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Treat analytical models as versioned software with contracts, tests, lineage, environments, dependencies, access, observability, rollback, and retirement semantics.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show model version, metric owner, source freshness, tests, consumers, deployment, incidents, evidence, and controls to approve, pause, revert, correct, restrict, and retire.

REQUIRED OUTPUTS AND ARTIFACTS

Metric catalog; analytical models; lineage; tests; deployment and monitoring design; quality and access records; decisions, evidence, limitations, and handoff.

EVIDENCE REQUIREMENTS

Identify source, model, metric, version, population, query, environment, method, result, and limitation. A chart or passing test proves only its bounded claim.

VERIFICATION AND ACCEPTANCE TESTS

Verify metric definitions, grain, joins, freshness, tests, access, reproducibility, change impact, deployment, failure, correction, and consumer understanding.

FAILURE STATES AND ANTI-PATTERNS

Metric duplication, hidden filters, untested transformations, stale dashboards, circular validation, manual edits, and treating engagement or correlation as causation.

RECOVERY AND RECONSIDERATION

Withdraw affected outputs, preserve versions and evidence, notify consumers, and use prior model, corrected data, restricted publication, or controlled reprocessing.

COMPLETION CRITERIA

Models, metrics, owners, tests, lineage, evidence, limitations, gaps, and review cadence are explicit. Status remains CONTENT-COMPLETE pending later gates.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off to BI, Data Analysis, Data Science, Data Quality, Data Engineering, Testing, Governance, Security, Operations, and Retirement.

RELATED CODEX SECTIONS

031 Data Architecture; 156 Data Quality; 159-160 Warehousing and Lake; 162 Business Intelligence; 163 Data Analysis; 164 Data Science; 254 Data Testing; 307 Data Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the canonical contract, shared governance, data architecture, and roadmap. Domain metric authority, statistical standards, access rules, and sector reporting obligations remain unresolved.

162

SECTION 162 - BUSINESS INTELLIGENCE

CODEX/03_SECTIONS/162_BUSINESS_INTELLIGENCE.txt

SHA-256 16b3d053badd50f4b6cc51437a1158bb9f3de08ea4008aeb0810c09475f0c21b

5,333 BYTES / PRESERVED SOURCE IDENTITY

SECTION 162 — Business Intelligence

STATUS: CONTENT-COMPLETE

PURPOSE

Design governed reporting and decision-support experiences that present timely, understandable, accessible, and traceable information without overstating certainty.

REQUIRED OUTCOME

A BI model with audience, decision purpose, metrics, lineage, refresh, permissions, accessibility, explanation, alerts, owner, evidence, and correction paths.

WHY THIS MATTERS

Dashboards influence action while hiding definitions and uncertainty. Visual clarity and a green indicator are not proof of valid decisions.

APPLICABILITY

Apply when reports, dashboards, scorecards, alerts, self-service analytics, or executive views influence material decisions or operations.

NON-APPLICABILITY CONDITIONS

No material metric, audience, decision, data, access, presentation, or alert behavior changes and an exact BI baseline covers the work.

PREREQUISITES AND DEPENDENCIES

Decision purpose, audiences, data, metrics, definitions, refresh, lineage, access, accessibility, thresholds, risk, and acceptance authority.

REQUIRED INPUTS

Reports, queries, metric definitions, filters, audiences, source windows, permissions, refresh results, user research, incidents, evidence, and limitations.

HUMAN RESPONSIBILITIES AND AUTHORITY

The owner approves decisions, metrics, publication, access, thresholds, exceptions, risk, and acceptance; assigns business, data, privacy, security, accessibility, and reporting authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized report definitions, semantic models, lineage, refresh status, accessibility metadata, bounded report data, and alert history; draft summaries, visual alternatives, test cases, and explanations; and identify stale, contradictory, or anomalous presentation signals.

AI must identify the report version, source window, filters, audience, freshness, metric definition, uncertainty, and limitations for every material summary. It must distinguish displayed values from interpretation and interpretation from a recommended decision.

AI must not change metric meaning, hide contrary evidence, suppress or trigger consequential alerts, infer a user's authority, publish or withdraw a report, expose restricted data, or make a business decision without explicit delegation, accessible human review, and recorded acceptance.

APPLICABLE EXISTING AGENT ROLES

Analyst, Researcher, Architect, Data Engineer, User Experience Reviewer, Tester, Verifier, Governance Reviewer, Security Auditor, Critic, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Define decisions and audiences; establish semantic models; build reports and alerts; validate definitions, accessibility, refresh, permissions, uncertainty, failure, and correction; monitor use and drift.

HUMAN-AI INTERACTION REQUIREMENTS

AI may summarize authorized reports and propose views, but must cite sources, disclose freshness and uncertainty, and never make the decision, hide contrary evidence, or publish without authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate source, metric, semantic model, visualization, alert, interpretation, and decision. Preserve lineage, filters, permissions, accessibility, snapshots, corrections, and audit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show report version, audience, metric definitions, freshness, sources, permissions, alerts, uncertainty, incidents, evidence, and controls to restrict, correct, withdraw, and retire.

REQUIRED OUTPUTS AND ARTIFACTS

BI charter; semantic model; metric and report catalog; audience and accessibility evidence; refresh and alert design; lineage, access, test, decision, evidence, and handoff records.

EVIDENCE REQUIREMENTS

Identify source, metric, filter, window, report version, audience, environment, method, result, and limitation. A screenshot proves only what it displayed at that time.

VERIFICATION AND ACCEPTANCE TESTS

Verify metric meaning, filters, freshness, permissions, accessibility, alert thresholds, uncertainty, contradictory data, reproducibility, and correction or withdrawal.

FAILURE STATES AND ANTI-PATTERNS

Dashboard theater, hidden filters, stale refresh, inaccessible charts, vanity metrics, alert overload, false precision, and conclusions presented without source or limitations.

RECOVERY AND RECONSIDERATION

Withdraw or label affected reports, preserve snapshots and evidence, notify decision makers, and use prior validated views, manual review, corrected data, or restricted publication.

COMPLETION CRITERIA

Purpose, audience, metrics, sources, access, accessibility, refresh, tests, evidence, gaps, and review authority are explicit. Status remains CONTENT-COMPLETE.

HANDOFF AND CONTINUITY REQUIREMENTS

Hand off to Analytics, Data Quality, Data Governance, Data Engineering, UX, Accessibility, Security, Testing, Operations, and Records.

RELATED CODEX SECTIONS

030 Information Architecture; 031 Data Architecture; 156 Data Quality; 159-161 Warehouse, Lake, and Analytics Engineering; 163 Data Analysis; 247 Accessibility Testing; 307 Data Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the canonical contract, shared governance, data and information architecture, and roadmap. Decision authority, reporting law, accessibility standard, and metric definitions remain unresolved.

163

SECTION 163 - DATA ANALYSIS

CODEX/03_SECTIONS/163_DATA_ANALYSIS.txt

SHA-256 43d060ad40bcbc631aa9d36274fe5785b0fb67d575f49400e4ec8b2a9a715a0b

7,451 BYTES / PRESERVED SOURCE IDENTITY

SECTION 163 — Data Analysis

STATUS: CONTENT-COMPLETE

PURPOSE

Define a governed method for turning authorized data into bounded descriptive, diagnostic, exploratory, and decision-support findings. Data Analysis preserves the difference between observation, calculation, interpretation, recommendation, and decision.

REQUIRED OUTCOME

- A question, population, scope, time window, unit of analysis, and decision context.
- Reproducible data preparation, measures, methods, assumptions, limitations, and findings.
- Traceability from source data through transformations, calculations, visualizations, conclusions, and owner action.
- Explicit uncertainty, quality, bias, privacy, security, accessibility, and non-applicability determinations.

WHY THIS MATTERS

Analysis can make incomplete or biased data appear authoritative. A clear analytical chain lets owners distinguish what the data shows from what an analyst infers and what a decision maker chooses.

APPLICABILITY

Applies when data is used to describe conditions, explain variation, compare groups, diagnose causes, monitor outcomes, support decisions, or assess a change. It applies more deeply when results affect rights, safety, finance, service, policy, models, or public claims.

NON-APPLICABILITY CONDITIONS

May be recorded as non-applicable for formatting, transcription, or reversible inspection with no analytical claim or material decision effect. The record must name scope, owner, evidence date, and reopening trigger; a chart or query is not automatically exempt.

PREREQUISITES AND DEPENDENCIES

Requires Sections 019, 030, 031, 155, and 156 for the question, information meaning, data lineage, governance, and quality context. Privacy, security, accessibility, records, and domain authorities must be identified or marked unresolved.

REQUIRED INPUTS

Authorized question; source and version; population and sampling frame; definitions; measures; time and geography; inclusion rules; transformations; quality results; lawful access; hypotheses; decision criteria; and known confounders and limitations.

HUMAN RESPONSIBILITIES AND AUTHORITY

The human owner approves the question, population, sensitive uses, interpretation, decision threshold, publication, and acceptance. Humans resolve ambiguous meaning, competing interests, residual risk, and action; analysts must not turn findings into policy without authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may profile, summarize, calculate, visualize, detect patterns, propose analyses, and identify gaps within authorized data. It must show provenance and uncertainty and must not invent observations, infer permission, hide adverse results, or present correlation as causation.

APPLICABLE EXISTING AGENT ROLES

Analyst, Researcher, Observer, Architect, Data Governance Reviewer, Security Auditor, Privacy Reviewer, Critic, Tester, Verifier, Documentation Writer, Memory Curator, and Decision Arbiter Advisory. Existing role contracts remain controlling.

REQUIRED ACTIVITIES

1. Frame the question, decision, scope, population, unit, and non-goals.
2. Inspect source authority, lineage, quality, missingness, bias, and access.
3. Define measures, transformations, comparison groups, and uncertainty.
4. Run reproducible analysis and sensitivity checks.
5. Separate results, interpretation, recommendation, and owner decision.
6. Review privacy, security, accessibility, fairness, and misuse risks.
7. Record limitations, dissent, failed analyses, and reopen conditions.
8. Validate findings with domain owners and independently reproduce material results.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, method, filters, code or equivalent procedure, confidence limits, and data freshness. AI suggestions remain proposed until reviewed. Owners must correct definitions and assumptions without erasing prior findings, and non-AI evidence and appeal paths must remain available.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Store versioned questions, datasets, transformations, notebooks or procedures, measures, outputs, provenance, permissions, decisions, and evidence. Separate raw, prepared, derived, cached, and published data. Preserve repeatability, access controls, deletion obligations, and invalidation when sources change.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show question, owner, dataset version, scope, freshness, quality, method, status, uncertainty, findings, dissent, and decision. Provide authorized controls to pause, restrict, rerun, correct, quarantine, withdraw, publish, or reopen. Do not equate a green chart with truth or accepted action.

REQUIRED OUTPUTS AND ARTIFACTS

Analysis charter; data and lineage record; preparation procedure; measure definitions; analysis artifact; visualizations; findings and limitations; privacy and bias review; decision record; reproducibility package; and continuity handoff.

EVIDENCE REQUIREMENTS

Each claim identifies source, version, population, period, method, transformation, analyst or tool, result, uncertainty, and limitation. Samples, dashboards, correlations, and model-generated summaries support only bounded claims. Missing or contradictory evidence is marked UNKNOWN or NOT VERIFIED.

VERIFICATION AND ACCEPTANCE TESTS

Reproduce key results from the archived inputs and procedure; alter filters and definitions to test sensitivity; inject missing, duplicate, stale, and biased records; verify access and privacy controls; compare independent calculations; and confirm conclusions do not exceed evidence.

FAILURE STATES AND ANTI-PATTERNS

Cherry-picked periods; p-hacking; silent filtering; denominator changes; proxy measures treated as outcomes; causal language from correlation; data leakage; hidden missingness; unauthorized disclosure; automated recommendations presented as decisions; and untraceable spreadsheet edits.

RECOVERY AND RECONSIDERATION

Quarantine affected outputs when lineage, quality, permission, or method fails. Preserve prior artifacts, identify safe uses, notify owners, correct or rerun from a governed source, and reassess decisions that relied on invalid findings. Reopen after population, definition, policy, or source change.

COMPLETION CRITERIA

The question, scope, source, method, transformations, results, uncertainty, limitations, controls, verification, owner decision, and reopening condition are recorded. Status remains CONTENT-COMPLETE until source reconciliation and owner acceptance are separately completed.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve identifiers, inputs, code or procedure, environment, dependencies, decisions, dissent, evidence, and open gaps. Hand off to Statistics, Data Quality, Business Intelligence, Operations, Model Evaluation, Governance, and Records work with the next safe action and blocked actions stated.

RELATED CODEX SECTIONS

019 Requirements Engineering; 030 Information Architecture; 031 Data Architecture; 155 Data Governance; 156 Data Quality Engineering; 161 Analytics Engineering; 162 Business Intelligence; 180 Model Evaluation; 307 Data Operations; 313 Observability Engineering.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance requirements, Sections 030-032, and the Status and Evidence Lexicon. Domain-specific statistical standards, lawful purposes, population definitions, quality thresholds, accessibility needs, and decision authority remain unresolved until supplied and verified.

164

SECTION 164 - DATA SCIENCE

CODEX/03_SECTIONS/164_DATA_SCIENCE.txt

SHA-256 40c4ebf4a06b700fc5d119f9e486666dff3b0dbad0e184dcbe47f0c8da7a03ea

6,806 BYTES / PRESERVED SOURCE IDENTITY

SECTION 164 — Data Science

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed scientific investigation of data to answer authorized questions, create explanatory or predictive knowledge, and support decisions without confusing exploratory evidence with validated capability.

REQUIRED OUTCOME

- A research question, hypothesis, population, data basis, method, evaluation plan, and decision boundary.
- Reproducible experiments with traceable features, assumptions, uncertainty, limitations, and alternatives.
- Explicit separation of exploration, selection, validation, recommendation, and deployment authority.

WHY THIS MATTERS

Data science combines analysis, statistics, modeling, domain judgment, and experimentation. Without governance, flexible exploration can become undisclosed selection, overfitting, unfair decision support, or an unsupported claim of operational value.

APPLICABILITY

Applies to predictive, explanatory, causal, simulation, optimization, experimentation, or decision-support work using data and models. Depth increases with human impact, automation, external publication, regulated use, or irreversibility.

NON-APPLICABILITY CONDITIONS

May be non-applicable for descriptive inspection with no new claim, decision, model, or material data use. Record the bounded rationale, owner, evidence date, and reopening trigger; exploratory code remains subject to access and retention controls.

PREREQUISITES AND DEPENDENCIES

Requires authorized objectives, data lineage and quality, domain definitions, statistical method, privacy and security review, and acceptance authority. Depends on Sections 163, 165, 167, 180, and 183 for analysis, statistics, engineering, evaluation, and governance.

REQUIRED INPUTS

Question; hypotheses; data and lawful purpose; labels or outcomes; population; sampling; features; assumptions; baseline; evaluation metric; error costs; subgroup plan; compute; reproducibility environment; and intended use and non-use.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve the research question, sensitive data use, hypotheses, success criteria, subgroup treatment, publication, deployment, and residual risk. Domain and affected-party experts resolve meaning and consequences; the scientist recommends but does not authorize.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate hypotheses, code, features, experiments, summaries, and literature maps when authorized. It must preserve provenance, disclose generated work, flag leakage and uncertainty, and must not fabricate data, citations, experiments, or validation.

APPLICABLE EXISTING AGENT ROLES

Researcher, Analyst, Architect, Statistician, Data Governance Reviewer, Privacy Reviewer, Security Auditor, Critic, Tester, Verifier, Documentation Writer, Memory Curator, and Decision Arbiter Advisory. Existing role contracts remain controlling.

REQUIRED ACTIVITIES

1. Frame the question, hypothesis, population, decision, and exclusions.
2. Audit data provenance, quality, representativeness, consent, and leakage.
3. Establish baselines, design experiments, and predeclare evaluation where possible.
4. Run, compare, and stress-test analyses or models.
5. Assess subgroup, uncertainty, causal, privacy, security, and operational implications.
6. Record negative results, researcher degrees of freedom, dissent, and open gaps.
7. Obtain independent review and owner acceptance before consequential use.

HUMAN-AI INTERACTION REQUIREMENTS

Expose generated code, assumptions, sources, data, evaluation, uncertainty, and human review. AI outputs remain hypotheses or recommendations until accepted. Provide correction, reproducibility, non-AI review, and stop paths; never let fluency substitute for evidence.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version questions, datasets, features, code, environments, experiments, models, metrics, reports, decisions, and artifacts. Isolate exploratory and approved environments. Track lineage, random seeds where relevant, access, compute, dependencies, and invalidation when data or definitions change.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show research state, owner, hypothesis, data, population, method, experiment history, metrics, subgroups, uncertainty, risks, decisions, and evidence freshness. Allow authorized pause, restrict, rerun, reject, accept, publish, withdraw, and reopen actions.

REQUIRED OUTPUTS AND ARTIFACTS

Research charter; data and feature register; experiment plan; reproducible environment; analysis and model artifacts; evaluation report; subgroup and risk review; decision record; negative-results log; and handoff package.

EVIDENCE REQUIREMENTS

Claims identify question, source, version, population, method, environment, metric, uncertainty, reviewer, result, and limitation. A notebook, benchmark, or paper is not proof of generalization or deployment fitness. Missing evidence constrains status.

VERIFICATION AND ACCEPTANCE TESTS

Reproduce experiments; rerun with held-out and shifted data; test leakage, missingness, confounding, subgroup performance, access controls, and dependency changes; compare to baseline; and verify the claim's scope matches the evidence.

FAILURE STATES AND ANTI-PATTERNS

HARKing; unreported experiments; train-test contamination; convenient outcome definitions; benchmark overreach; silent subgroup exclusion; fabricated citations; irreproducible environments; and treating exploratory novelty as operational validity.

RECOVERY AND RECONSIDERATION

Quarantine affected findings when data, code, method, permission, or evaluation fails. Preserve history, identify downstream decisions, correct or rerun, notify owners, and reopen after drift, changed objectives, new populations, or invalid assumptions.

COMPLETION CRITERIA

Question, data, method, experiments, results, uncertainty, limitations, risks, verification, decisions, and handoff are complete and attributable. Status remains CONTENT-COMPLETE until owner acceptance and source reconciliation.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve research IDs, datasets, code, environments, dependencies, results, negative findings, decisions, dissent, and gaps. Hand off to Statistics, Machine Learning Engineering, Model Evaluation, MLOps, Governance, and Operations with next safe action stated.

RELATED CODEX SECTIONS

163 Data Analysis; 165 Statistics; 167 Machine Learning Engineering; 179 Model Training; 180 Model Evaluation; 183 AI Governance; 184 MLOps; 254 Data Testing; 307 Data Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Sections 030-032, and evidence conventions. Specific scientific discipline standards, lawful data bases, domain outcomes, reproducibility infrastructure, and deployment authority remain unresolved until supplied.

165

SECTION 165 - STATISTICS

CODEX/03_SECTIONS/165_STATISTICS.txt

SHA-256 eb0f8bb03e9fecfc063a891ca2ced4356cb36b41aeda1f77692758dbcf1c7fe6

6,542 BYTES / PRESERVED SOURCE IDENTITY

SECTION 165 — Statistics

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed statistical reasoning for describing uncertainty, estimating quantities, testing claims, designing samples and experiments, and supporting decisions without overstating precision or causality.

REQUIRED OUTCOME

- A defined estimand, population, sampling design, assumptions, method, uncertainty, and decision context.
- Reproducible calculations and transparent treatment of missingness, dependence, multiplicity, confounding, and sensitivity.
- An interpretation bounded by the design and accepted by the accountable human owner.

WHY THIS MATTERS

Statistical language can conceal consequential assumptions. Confidence intervals, significance tests, forecasts, and averages are useful only when their population, design, uncertainty, and limits are understood.

APPLICABILITY

Applies to estimation, inference, forecasting, experimentation, sampling, measurement, risk quantification, quality control, and statistical claims. Greater review is required for regulated, rights-affecting, safety, financial, or public decisions.

NON-APPLICABILITY CONDITIONS

May be non-applicable to exact deterministic counts or formatting with no inferential, predictive, comparative, or decision claim. Record the bounded rationale and reopening trigger; a small sample is not a reason to omit statistical qualification.

PREREQUISITES AND DEPENDENCIES

Requires a defined population, data lineage, measurement meaning, sampling frame, domain context, and decision owner. Depends on Sections 154, 156, 163, 164, 180, and 254 for modeling, quality, analysis, evaluation, and tests.

REQUIRED INPUTS

Estimand; population; sampling frame; unit; design; outcome; covariates; missing-data plan; assumptions; model; prior or baseline; error tolerance; power or precision target; multiplicity; subgroup plan; and decision consequences.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve estimands, assumptions, acceptable error, subgroup treatment, stopping rules, publication, and use. A qualified statistician may advise, but only the owner accepts residual uncertainty and resulting action.

AI RESPONSIBILITIES AND BOUNDARIES

AI may calculate, simulate, explain, compare methods, and flag violations. It must expose assumptions and code or procedure, preserve exact inputs, and must not invent p-values, confidence, samples, causal effects, or significance.

APPLICABLE EXISTING AGENT ROLES

Analyst, Researcher, Observer, Critic, Verifier, Tester, Data Governance Reviewer, Privacy Reviewer, Security Auditor, Documentation Writer, Memory Curator, and Decision Arbiter Advisory. Existing role contracts remain controlling.

REQUIRED ACTIVITIES

1. Define estimand, population, unit, outcome, and decision.
2. Audit sampling, measurement, missingness, dependence, and data quality.
3. Select and justify design and method before interpreting results.
4. Calculate uncertainty, sensitivity, subgroup, and multiplicity effects.
5. Test assumptions and compare plausible alternatives.
6. Separate statistical result from causal or policy interpretation.
7. Record limitations, failed checks, dissent, and owner decision.

HUMAN-AI INTERACTION REQUIREMENTS

Show formulas or procedures, assumptions, sample construction, uncertainty, and interpretation limits. AI-generated statistical explanations require review. Users must be able to inspect raw definitions and non-AI calculations; no automated threshold may silently become policy.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version estimands, datasets, sampling frames, code, randomization, models, results, and decisions. Preserve units and denominators. Store uncertainty and invalidation rules with results, and prevent dashboards from hiding revisions or adverse subgroups.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show estimand, population, design, sample, assumptions, uncertainty, method, subgroup results, status, and owner. Provide authorized controls to rerun, challenge, restrict, publish, withdraw, correct, and reopen. Explain what the result does not establish.

REQUIRED OUTPUTS AND ARTIFACTS

Statistical analysis plan; data and sampling record; estimand definition; method and assumptions; code or procedure; results; uncertainty and sensitivity report; subgroup analysis; limitations; decision record; and reproducibility handoff.

EVIDENCE REQUIREMENTS

Each result identifies population, sample, design, period, measure, method, assumptions, software or tool version, uncertainty, reviewer, and limitation. Significance, precision, and forecast accuracy support only bounded claims and never prove causation alone.

VERIFICATION AND ACCEPTANCE TESTS

Reproduce calculations; perturb samples and assumptions; test missingness, dependence, multiplicity, leakage, and subgroup effects; compare independent implementations; verify denominators and units; and confirm interpretation matches the design.

FAILURE STATES AND ANTI-PATTERNS

P-hacking; optional stopping; multiple testing without control; Simpson's paradox ignored; ecological fallacy; confidence mistaken for probability of truth; averages hiding harm; causal claims from association; and fabricated statistical certainty.

RECOVERY AND RECONSIDERATION

Quarantine results when assumptions, data, sampling, or calculations fail. Preserve prior outputs, identify affected decisions, rerun or revise with qualified review, and notify owners. Reopen after population, measure, policy, data, or decision changes.

COMPLETION CRITERIA

Estimand, design, data, method, assumptions, results, uncertainty, limitations, verification, and owner decision are explicit. Status remains CONTENT-COMPLETE until source reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve analysis-plan identity, data, sample, code, environment, results, failed checks, decisions, dissent, and gaps. Hand off to Data Analysis, Data Science, Experimentation, Model Evaluation, Quality, Governance, and Operations.

RELATED CODEX SECTIONS

154 Data Modeling; 156 Data Quality Engineering; 163 Data Analysis; 164 Data Science; 180 Model Evaluation; 239 Performance Testing; 254 Data Testing; 369 Experimentation; 370 A or B Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, data and analysis sections, and evidence lexicon. Sector-specific statistical standards, estimands, lawful purposes, sampling frames, error tolerances, and decision authorities remain unresolved until supplied.

166

SECTION 166 - OPERATIONS RESEARCH

CODEX/03_SECTIONS/166_OPERATIONS_RESEARCH.txt

SHA-256 9f8b2d3bc6147be03411543711a98b1f0c9a0cc212cb9072ec3b0c18e45cce04

6,783 BYTES / PRESERVED SOURCE IDENTITY

SECTION 166 — Operations Research

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed use of mathematical models, optimization, simulation, queuing, scheduling, and decision analysis to improve authorized operations while preserving constraints, human authority, fairness, safety, resilience, and explainability.

REQUIRED OUTCOME

- A decision problem with objective, decision variables, constraints, scenarios, stakeholders, and authority.
- A validated model with data lineage, assumptions, sensitivity, uncertainty, feasibility, and failure behavior.
- A recommendation that is explicitly distinct from an authorized operational decision or execution.

WHY THIS MATTERS

Optimization can produce an efficient answer to the wrong objective or consume hidden safety, equity, accessibility, or resilience margins. Operations Research makes objectives and constraints inspectable before a solver's output becomes action.

APPLICABILITY

Applies to allocation, routing, scheduling, capacity, inventory, queues, reliability, staffing, pricing, logistics, simulation, resource planning, or multi-criteria decisions. Review depth rises with scarcity, human impact, automation, and irreversibility.

NON-APPLICABILITY CONDITIONS

May be non-applicable to descriptive arithmetic or reversible planning with no optimization, simulation claim, external effect, or resource commitment. Record scope, owner, assumptions, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires an authorized objective, domain and process context, data quality, constraints, decision rights, and validation plan. Depends on Sections 163, 165, 301-323, and 369 for analysis, statistics, operations, performance, and experimentation.

REQUIRED INPUTS

Decision statement; stakeholders; objective hierarchy; variables; hard and soft constraints; capacities; costs; scenarios; data; uncertainty; fairness and safety limits; baseline; solver; runtime budget; and execution authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans set objectives, protected constraints, tradeoffs, priorities, exceptions, and acceptance. They decide whether to act, communicate, or override a recommendation. Analysts cannot relax a constraint or commit resources without explicit authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may formulate alternatives, solve bounded problems, simulate scenarios, and explain sensitivity. It must expose objective and constraint choices and cannot hide infeasibility, manipulate weights, claim optimality without proof, or execute without delegation.

APPLICABLE EXISTING AGENT ROLES

Architect, Analyst, Researcher, Planner, Operations Reviewer, Critic, Security Auditor, Privacy Reviewer, Tester, Verifier, Governance Reviewer, Documentation Writer, Memory Curator, Optimizer, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

1. Define decision, authority, stakeholders, objective, and exclusions.
2. Separate hard constraints from preferences and document their precedence.
3. Audit data, scenarios, uncertainty, and common-cause dependencies.
4. Build and validate the model against observed cases and baselines.
5. Test sensitivity, infeasibility, adversarial input, and degraded resources.
6. Compare alternatives and expose distributional effects and residual risk.
7. Obtain owner decision and define monitoring, override, and recovery.

HUMAN-AI INTERACTION REQUIREMENTS

Show objective, constraints, weights, assumptions, alternatives, uncertainty, and affected parties. Recommendations stay non-binding until approved. Provide manual override, appeal, explanation, and safe halt; prevent conversational changes to constraints from bypassing authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version models, data, solver, parameters, scenarios, constraints, results, decisions, and execution records. Enforce hard constraints at execution boundaries. Preserve feasibility, idempotency, override, audit, rollback-by-compensation, and degraded-mode behavior.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show decision state, objective, constraints, scenario, result, sensitivity, uncertainty, affected groups, owner, and authority. Allow authorized approve, reject, constrain, rerun, pause, override, compensate, and reopen actions. Never show optimal as automatically correct.

REQUIRED OUTPUTS AND ARTIFACTS

Decision charter; model and data specification; constraint register; solver and parameter record; scenario and sensitivity report; feasibility and fairness review; recommendation; decision record; execution or non-execution record; and continuity handoff.

EVIDENCE REQUIREMENTS

Claims identify model, data, version, objective, constraints, scenario, solver, run, result, uncertainty, validation, and limitation. A solver status or simulation is not proof of real-world optimality or outcome.

VERIFICATION AND ACCEPTANCE TESTS

Reproduce runs; remove or vary constraints; test infeasible, stale, missing, adversarial, and extreme inputs; compare baseline and alternative objectives; verify hard constraints at execution; and simulate failure, interruption, override, and recovery.

FAILURE STATES AND ANTI-PATTERNS

Objective gaming; hidden constraints; proxy optimization; false optimality; infeasibility masked by relaxation; fairness traded away in a score; solver success treated as outcome; stale capacity; and automated execution without authority.

RECOVERY AND RECONSIDERATION

Stop or constrain execution when data, model, authority, feasibility, or safety fails. Preserve the recommendation and inputs, identify affected decisions, use approved fallback or manual planning, and reassess before resuming. Reopen after objective, capacity, policy, population, or risk changes.

COMPLETION CRITERIA

Decision, objective, constraints, data, model, validation, sensitivity, recommendation, authority, verification, and recovery are explicit. Status remains CONTENT-COMPLETE until source reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve model IDs, inputs, parameters, solver, scenarios, results, decisions, overrides, execution evidence, dissent, and gaps. Hand off to Data Analysis, Statistics, Data Science, Operations, AI, Testing, Governance, and Incident work.

RELATED CODEX SECTIONS

163 Data Analysis; 165 Statistics; 164 Data Science; 018 Risk Management; 027 Process Architecture; 029 Operational Architecture; 301 Site Reliability Engineering; 321 Capacity Management; 369 Experimentation.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Sections 027-032, and evidence conventions. Actual objectives, constraints, fairness criteria, operational authority, solver standards, sector duties, and outcome data remain unresolved until supplied and verified.

167

SECTION 167 - MACHINE LEARNING ENGINEERING

CODEX/03_SECTIONS/167_MACHINE_LEARNING_ENGINEERING.txt

SHA-256 50fd95f9d72795248f01cf0203251fab6fab41dd1ca7b6eb42035b0014dc6f0e

6,539 BYTES / PRESERVED SOURCE IDENTITY

SECTION 167 — Machine Learning Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define the engineering discipline for building, integrating, evaluating, deploying, and maintaining machine-learning systems with explicit data, model, interface, human-control, security, and lifecycle boundaries.

REQUIRED OUTCOME

- A traceable ML system from objective and data to model, service, decision, monitoring, and retirement.
- Reproducible builds, versioned artifacts, evaluation evidence, deployment controls, and rollback-by-compensation paths.
- Clear separation between technical capability, authorized use, and observed outcome.

WHY THIS MATTERS

ML behavior depends on data, training, code, configuration, environment, and context. Engineering discipline prevents a benchmark or successful deployment from being mistaken for reliable, fair, secure, or accepted operation.

APPLICABILITY

Applies when learned parameters or learned representations affect software, hardware, decisions, automation, ranking, prediction, or control. Depth increases with external impact, autonomy, sensitive data, drift, or hard-to-reverse actions.

NON-APPLICABILITY CONDITIONS

May be non-applicable to deterministic rules, documentation-only work, or isolated research with no model artifact or operational claim. Record scope, owner, evidence date, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires Sections 031, 164, 179, 180, 183, and 184 for data, science, training, evaluation, governance, and MLOps. Requirements, domain outcomes, safety, privacy, security, accessibility, and operating authority must be known or open.

REQUIRED INPUTS

Use case; prohibited uses; data and labels; model contract; baseline; metrics; error costs; environments; compute; dependencies; threat model; evaluation plan; deployment target; monitoring; human review; and retirement conditions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve use, data purpose, model class, risk tolerance, evaluation gates, deployment, monitoring thresholds, exceptions, and retirement. Engineers report deviations and do not accept model risk or expand purpose by implementation convenience.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate code, configurations, tests, model comparisons, and diagnostics within scope. It must preserve artifact identity and cannot invent training results, certify security, authorize deployment, or conceal uncertainty and failure.

APPLICABLE EXISTING AGENT ROLES

Architect, Implementer, Refactorer, Researcher, Analyst, Tester, Verifier, Security Auditor, Privacy Reviewer, Governance Reviewer, Critic, Optimizer, Documentation Writer, Memory Curator, and Red Team.

REQUIRED ACTIVITIES

1. Define objective, contract, users, prohibited uses, and failure costs.
2. Establish data, feature, label, lineage, split, and leakage controls.
3. Build reproducibly and record dependencies, seeds, configuration, and hardware.
4. Evaluate accuracy, calibration, robustness, subgroup, privacy, security, and operational behavior.
5. Integrate identity, authorization, observability, fallback, and human review.
6. Validate deployment, rollback, drift, incident, and retirement procedures.
7. Baseline evidence and obtain owner acceptance.

HUMAN-AI INTERACTION REQUIREMENTS

Expose model, data, version, prompt or configuration, tool, authority, output, confidence limits, and review state. AI-generated engineering changes remain proposals until reviewed. Maintain manual fallback, correction, appeal, and stop controls.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version data, features, code, model, weights, tokenizer, configuration, environment, image, endpoint, policy, evaluation, and deployment. Enforce contracts at serving boundaries. Record provenance, drift, access, latency, cost, errors, and compensation actions.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show model lineage, deployment, scope, health, drift, evaluation, incidents, authority, and evidence freshness. Allow authorized pause, traffic reduction, revoke, quarantine, rollback, fallback, retrain, replace, and retire actions. Do not equate service uptime with model validity.

REQUIRED OUTPUTS AND ARTIFACTS

ML system charter; model and data cards; feature and label register; reproducible build; evaluation report; threat and privacy review; serving contract; deployment record; monitoring plan; incident and retirement plan; decision record; and handoff package.

EVIDENCE REQUIREMENTS

Each claim identifies artifact versions, data, split, environment, metric, subgroup, method, result, uncertainty, reviewer, and limitation. A benchmark, training run, or endpoint response proves only its bounded claim. Failed evidence constrains status.

VERIFICATION AND ACCEPTANCE TESTS

Rebuild from declared inputs; reproduce evaluation; test leakage, shift, malformed input, adversarial input, access, latency, resource, fallback, revocation, and rollback; compare baselines; and verify monitoring detects invalidation.

FAILURE STATES AND ANTI-PATTERNS

Training-serving skew; leakage; undocumented features; benchmark overreach; silent retraining; insecure model artifacts; threshold tuning on test data; unmonitored drift; hidden human work; and deployment without approved evaluation.

RECOVERY AND RECONSIDERATION

Stop or narrow affected use on drift, unsafe error, data breach, artifact mismatch, authority loss, or evidence failure. Preserve the exact state, notify owners, use approved fallback or prior version, investigate, and requalify before resuming.

COMPLETION CRITERIA

Objective, contract, data, build, evaluation, deployment, monitoring, controls, recovery, lifecycle, and owner decision are explicit. Status remains CONTENT-COMPLETE until source reconciliation and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve model, data, code, environment, dependencies, metrics, decisions, incidents, dissent, and gaps. Hand off to Deep Learning, AI Engineering, Training, Evaluation, MLOps, Model Operations, Security, and Governance.

RELATED CODEX SECTIONS

031 Data Architecture; 164 Data Science; 168 Deep Learning Engineering; 169 AI Engineering; 179 Model Training; 180 Model Evaluation; 184 MLOps; 185 Model Operations; 253 Model Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Sections 030-032, and the evidence lexicon. Actual model classes, domain thresholds, data rights, threat environment, compute, support, and deployment authority remain unresolved until supplied.

168

SECTION 168 - DEEP LEARNING ENGINEERING

CODEX/03_SECTIONS/168_DEEP_LEARNING_ENGINEERING.txt

SHA-256 23129329138dfc75ebd222269950bc816c7c3bf19ff18d520fd094017516f19f

6,187 BYTES / PRESERVED SOURCE IDENTITY

SECTION 168 — Deep Learning Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed engineering for deep neural systems, including architecture, training computation, data pipelines, optimization, evaluation, serving, resource use, safety controls, and lifecycle.

REQUIRED OUTCOME

- An attributable neural-system design with architecture, data, objective, compute, artifacts, and operational contract.
- Reproducible training and evaluation with robustness, calibration, subgroup, privacy, security, and resource evidence.
- Controlled deployment, monitoring, fallback, and retirement.

WHY THIS MATTERS

Deep models can fail outside benchmarks, encode data defects, consume extraordinary resources, and be difficult to interpret or reproduce. Explicit engineering records make those risks visible and actionable.

APPLICABILITY

Applies to neural networks, representation learners, multimodal models, sequence models, foundation models, and neural components used in consequential systems. It applies to training, fine-tuning, distillation, compression, and serving.

NON-APPLICABILITY CONDITIONS

May be non-applicable to non-neural deterministic components or conceptual research with no model artifact or operational claim. Record exact scope, owner, date, and reopening condition.

PREREQUISITES AND DEPENDENCIES

Requires data and model contracts, compute and supply-chain controls, evaluation, safety, privacy, security, and deployment authority. Depends on Sections 167, 179-184, and 253.

REQUIRED INPUTS

Task; architecture; data; objective; tokenizer or input representation; compute budget; optimizer; checkpoints; reproducibility settings; metrics; threat model; intended and prohibited uses; serving constraints; and human-control plan.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve architecture class, data use, compute, evaluation gates, release, acceptable residual risk, and public claims. Engineers cannot infer safety or authorization from scale, loss, or benchmark improvement.

AI RESPONSIBILITIES AND BOUNDARIES

AI may suggest architectures, code, hyperparameters, diagnostics, and tests. It must label generated work, preserve exact configuration, and must not fabricate runs, claim interpretability, or authorize release.

APPLICABLE EXISTING AGENT ROLES

Architect, Implementer, Researcher, Analyst, Optimizer, Tester, Verifier, Security Auditor, Privacy Reviewer, Safety Auditor, Critic, Red Team, Documentation Writer, Memory Curator, and Governance Reviewer.

REQUIRED ACTIVITIES

1. Define task, contract, threat, protected properties, and prohibited uses.
2. Audit data, preprocessing, splits, contamination, and representativeness.
3. Establish architecture, objective, resource, checkpoint, and reproducibility controls.
4. Train, validate, stress-test, and compare baselines and alternatives.
5. Evaluate robustness, calibration, subgroup, privacy, security, latency, and cost.
6. Verify serving, fallback, monitoring, rollback, and retirement.
7. Record failed runs, uncertainty, dissent, and owner decisions.

HUMAN-AI INTERACTION REQUIREMENTS

Expose architecture, checkpoint, data, configuration, output limits, review, and uncertainty. Generated changes remain proposed. Preserve manual review, stop, fallback, and appeal paths for model outputs and deployment actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version code, data, preprocessing, architecture, weights, tokenizer, hardware, libraries, checkpoints, evaluation, container, endpoint, and policy. Protect artifacts and secrets, enforce input/output contracts, and record resource and failure telemetry.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show checkpoint lineage, training state, compute, evaluations, deployment, drift, incidents, and authority. Provide authorized controls to pause training, revoke artifacts, reduce traffic, fallback, rollback, quarantine, retrain, and retire.

REQUIRED OUTPUTS AND ARTIFACTS

Neural-system charter; architecture and data record; training configuration; checkpoint register; reproducible environment; evaluation and robustness report; security and safety review; serving contract; deployment and retirement plan; decision record.

EVIDENCE REQUIREMENTS

Claims identify architecture, weights, data, compute, configuration, evaluation set, metric, subgroup, method, result, uncertainty, and limitation. Loss and benchmark scores do not prove real-world safety, fairness, or usefulness.

VERIFICATION AND ACCEPTANCE TESTS

Reproduce checkpoints; test preprocessing and serving parity; exercise corruption, shift, adversarial input, resource exhaustion, access, fallback, rollback, and authority revocation; and verify evidence invalidation after configuration changes.

FAILURE STATES AND ANTI-PATTERNS

Checkpoint confusion; undocumented preprocessing; contamination; unstable training; benchmark fixation; hidden unsafe capability; unbounded resource use; artifact tampering; and silent post-release retraining.

RECOVERY AND RECONSIDERATION

Freeze affected checkpoints and deployments on unsafe behavior, mismatch, breach, drift, or evidence loss. Preserve artifacts, use approved prior or manual mode, investigate, re-evaluate, and reopen before resuming.

COMPLETION CRITERIA

Architecture, data, training, evaluation, serving, controls, resources, recovery, evidence, and authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve checkpoints, data, code, hardware, environment, metrics, decisions, incidents, dissent, and gaps. Hand off to ML Engineering, Training, Evaluation, MLOps, Model Operations, Safety, Security, and Governance.

RELATED CODEX SECTIONS

167 Machine Learning Engineering; 169 AI Engineering; 179 Model Training; 180 Model Evaluation; 182 AI Safety; 184 MLOps; 185 Model Operations; 253 Model Testing; 264 Build Engineering.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance. Architecture-specific safety criteria, compute authority, data rights, robustness thresholds, deployment context, and support obligations remain unresolved until supplied.

169

SECTION 169 - AI ENGINEERING

CODEX/03_SECTIONS/169_AI_ENGINEERING.txt

SHA-256 291ef89db6482e04e31f73a974db38cf400416e1e814c1216918bc62d3389a49

6,530 BYTES / PRESERVED SOURCE IDENTITY

SECTION 169 — AI Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define engineering of AI-enabled systems as socio-technical products: models, prompts, tools, data, policies, interfaces, people, controls, evaluation, operations, and lifecycle.

REQUIRED OUTCOME

- A complete AI system boundary and contract, including capability, authority, data, tools, users, providers, and failure behavior.
- Reproducible implementation, evaluation, observability, security, privacy, safety, accessibility, and human-control evidence.
- A governed path from prototype to operation and retirement.

WHY THIS MATTERS

AI behavior emerges from more than a model. Prompts, retrieval, tools, permissions, interfaces, operators, and fallback determine real consequences. Engineering the whole system prevents model-centric claims from hiding socio-technical failure.

APPLICABILITY

Applies to systems that use models, agents, classifiers, generators, recommenders, speech, vision, retrieval, or learned components to inform or act. Depth increases with autonomy, sensitive data, external effects, and long-running operation.

NON-APPLICABILITY CONDITIONS

May be non-applicable to ordinary deterministic software with no AI component or to clearly labeled exploration with no operational claim. Record scope, owner, evidence date, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires requirements, process, data, security, privacy, safety, accessibility, evaluation, governance, and operational contexts. Depends on Sections 167-170, 180-186, and 188.

REQUIRED INPUTS

Use case; users and affected parties; model and provider; prompts and policies; tools; data; identity; permissions; autonomy; interfaces; constraints; evaluation; monitoring; fallback; incident; and retirement requirements.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans define purpose, prohibited use, authority, user notice, review, risk acceptance, deployment, external commitments, and retirement. Engineers cannot grant autonomy, access, or decision authority through wiring or prompt design.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft code, prompts, schemas, tests, evaluations, and architectures. It must preserve provenance and disclose uncertainty. It must not claim alignment, safety, authorization, or production readiness without evidence and human decision.

APPLICABLE EXISTING AGENT ROLES

Architect, Implementer, Researcher, Analyst, Prompt Engineer, Security Auditor, Privacy Reviewer, Safety Auditor, Governance Reviewer, Tester, Verifier, Critic, Red Team, Documentation Writer, Memory Curator, and Human-in-the-Loop Operator.

REQUIRED ACTIVITIES

1. Define system purpose, boundary, users, affected parties, and prohibited use.
2. Map model, prompt, policy, retrieval, tool, data, identity, provider, and human paths.
3. Specify permissions, autonomy, approval, stop, appeal, fallback, and evidence contracts.
4. Build and evaluate normal, adversarial, ambiguous, and failure behavior.
5. Validate privacy, security, safety, accessibility, reliability, cost, and operational readiness.
6. Instrument traces, decisions, tool calls, outputs, incidents, and human review.
7. Baseline, hand off, and define change and retirement gates.

HUMAN-AI INTERACTION REQUIREMENTS

Show actor, model, prompt, policy, tool, data, authority, output, confidence limits, review, and state. Humans must correct intent and stop consequences. No chat path may bypass identity, evaluation, audit, approval, or recovery.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version system, model, provider, prompt, policy, tools, data, retrieval index, configuration, interface, evaluation, and deployment. Enforce capability and authority at tool and state boundaries. Preserve traceability and compensation, not destructive reversal.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show system state, model and provider, active policy, authority, tools, data, traces, evaluations, incidents, costs, and evidence. Provide authorized pause, revoke, restrict, isolate, fallback, compensate, rollback configuration, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

AI system charter; threat and impact assessment; model, prompt, data, tool, policy, and provider registers; architecture; evaluation suite; trace schema; deployment record; runbook; incident and retirement plan; decision and handoff package.

EVIDENCE REQUIREMENTS

Claims identify system version, model, provider, prompt, policy, tools, data, environment, evaluator, scenario, result, uncertainty, and limitation. A response, demo, benchmark, or trace is bounded evidence, not general safety proof.

VERIFICATION AND ACCEPTANCE TESTS

Trace every consequential path; test prompt injection, tool misuse, privilege, data leakage, hallucination, refusal, ambiguity, provider loss, model drift, authority revocation, human review, fallback, and compensation; reproduce accepted claims.

FAILURE STATES AND ANTI-PATTERNS

Model-only testing; hidden tools; implicit authority; prompt as security boundary; unlogged decisions; unsafe defaults; provider lock-in; reward or engagement gaming; human review theater; and rollback that destroys prior evidence.

RECOVERY AND RECONSIDERATION

Stop or narrow affected capability on unsafe output, policy conflict, breach, drift, trace loss, provider failure, or authority expiry. Preserve immutable decisions and evidence, use fallback or compensation, review affected outcomes, and requalify before resuming.

COMPLETION CRITERIA

Purpose, boundary, actors, models, prompts, data, tools, policies, controls, evaluation, operation, recovery, evidence, and authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve system IDs, versions, configurations, traces, decisions, evaluations, incidents, dissent, dependencies, and gaps. Hand off to Generative AI, Agents, MLOps, Model Operations, Safety, Governance, Security, and HITL Operations.

RELATED CODEX SECTIONS

056 AI Architecture; 167 Machine Learning Engineering; 170 Generative AI Engineering; 178 Agent Systems Engineering; 180 Model Evaluation; 182 AI Safety; 183 AI Governance; 184 MLOps; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Sections 030-032, and the evidence lexicon. Actual use cases, impact classes, authority, provider terms, safety thresholds, user notices, and operational context remain unresolved until supplied.

170

SECTION 170 - GENERATIVE AI ENGINEERING

CODEX/03_SECTIONS/170_GENERATIVE_AI_ENGINEERING.txt

SHA-256 ae6b43cca97c900e6c6510ffb80242844e220ccc91a054312fb95abcef2c85fd

6,491 BYTES / PRESERVED SOURCE IDENTITY

SECTION 170 — Generative AI Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define engineering for systems that generate text, code, images, audio, video, structured content, or multimodal outputs, including grounding, safety, provenance, evaluation, tools, user control, and lifecycle.

REQUIRED OUTCOME

- A generation contract specifying purpose, audience, inputs, output limits, sources, authority, review, and prohibited uses.
- Reproducible prompts, model, retrieval, tools, safety controls, evaluations, traces, and release conditions.
- Safe handling of uncertainty, refusal, correction, disclosure, misuse, and recovery.

WHY THIS MATTERS

Plausible output can be wrong, copied, biased, unsafe, private, or action-inducing. Generative engineering must make provenance, uncertainty, control, and human responsibility visible at the point of use.

APPLICABILITY

Applies to generation, transformation, summarization, translation, code synthesis, multimodal creation, retrieval-augmented generation, and tool-using generation. Review depth increases with external publication, rights impact, autonomy, or sensitive data.

NON-APPLICABILITY CONDITIONS

May be non-applicable to deterministic templates or private experimentation with no consequential output. Record scope, owner, evidence date, data handling, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires content and data rights, model and provider records, prompt and tool contracts, evaluation, safety, privacy, security, accessibility, and human-review authority. Depends on Sections 169, 171-173, 180-186, and 188.

REQUIRED INPUTS

Purpose; audience; source rights; model; prompt; context; retrieval; tools; output schema; refusal and disclosure rules; quality and safety metrics; review; logging; cost; retention; and retirement conditions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve use, source rights, output claims, review level, publication, action, exceptions, and risk. They retain responsibility for consequential content and cannot delegate authorship, approval, or accountability merely because a model generated it.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate drafts, alternatives, tests, and bounded transformations. It must label generated content, preserve sources and uncertainty, and must not fabricate citations, claim originality, publish, or execute without authority.

APPLICABLE EXISTING AGENT ROLES

Architect, Prompt Engineer, Researcher, Analyst, Implementer, Content Reviewer, Security Auditor, Privacy Reviewer, Safety Auditor, Governance Reviewer, Tester, Verifier, Critic, Red Team, Documentation Writer, and Human-in-the-Loop Operator.

REQUIRED ACTIVITIES

1. Define use, audience, harms, prohibited output, and action boundary.
2. Establish source, license, consent, privacy, retrieval, and provenance rules.
3. Specify prompt, model, context, tool, output, refusal, and disclosure contracts.
4. Evaluate factuality, grounding, relevance, bias, safety, privacy, robustness, and accessibility.
5. Validate human review, correction, appeal, logging, and safe fallback.
6. Red-team misuse, injection, exfiltration, reproduction, and unsafe action paths.
7. Baseline, release, monitor, and retire with owner acceptance.

HUMAN-AI INTERACTION REQUIREMENTS

Show generated status, sources, timestamp, model, context, uncertainty, review, and actionability. Require human confirmation for consequential output or tool calls. Preserve edit, reject, report, cite, non-AI, and recovery paths.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version model, provider, prompt, system instruction, retrieval index, sources, tools, policies, output schema, filters, evaluator, and deployment. Enforce permissions and output validation outside the model. Preserve trace and content lineage.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show generation request, model, sources, context, policy, tools, output status, reviewer, traces, cost, incidents, and evidence. Allow authorized pause, restrict, regenerate, correct, withdraw, quarantine, revoke, compensate, and retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Generation charter; source-rights and provenance register; model and prompt records; tool and output contracts; evaluation and red-team report; disclosure and review design; trace schema; release record; incident and retirement plan.

EVIDENCE REQUIREMENTS

Claims identify request, model, prompt, context, source, tool, policy, output, reviewer, method, date, uncertainty, and limitation. A fluent response or citation does not prove truth, permission, originality, or safety.

VERIFICATION AND ACCEPTANCE TESTS

Test grounding, citation, hallucination, prompt injection, data leakage, toxic or biased output, copyright and license handling, malformed schema, refusal, ambiguity, provider loss, human review, correction, withdrawal, and tool authorization.

FAILURE STATES AND ANTI-PATTERNS

Uncited generation; source laundering; prompt-only security; hidden retrieval; fabricated confidence; unsafe tool use; model output treated as record; reviewer rubber-stamping; and silent publication or data retention.

RECOVERY AND RECONSIDERATION

Quarantine harmful or unsupported outputs, preserve request and lineage, notify affected owners, withdraw or correct published content, review downstream actions, and use manual or constrained fallback. Reopen after model, source, policy, audience, or risk changes.

COMPLETION CRITERIA

Use, audience, sources, model, prompt, tools, controls, evaluation, review, provenance, recovery, evidence, and authority are explicit. Status remains CONTENT-COMPLETE until reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve prompts, models, sources, contexts, tools, outputs, traces, decisions, incidents, dissent, and gaps. Hand off to NLP, Vision, Speech, Prompt Engineering, Evaluation, Safety, Governance, MLOps, and HITL Operations.

RELATED CODEX SECTIONS

054 Content Architecture; 056 AI Architecture; 089 Conversational Design; 097 Content Design; 169 AI Engineering; 171 Natural Language Processing; 180 Model Evaluation; 182 AI Safety; 186 Prompt Engineering; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, AI and application architecture, and evidence lexicon. Actual content rights, risk class, disclosure rules, model terms, user expectations, and review authority remain unresolved until supplied.

171

SECTION 171 - NATURAL LANGUAGE PROCESSING

CODEX/03_SECTIONS/171_NATURAL_LANGUAGE_PROCESSING.txt

SHA-256 063585b40be24315bffc288154093fbe47e78d7c7aefbee8324b6e14c36e6eb3

6,260 BYTES / PRESERVED SOURCE IDENTITY

SECTION 171 — Natural Language Processing

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed engineering of systems that process, classify, retrieve, translate, summarize, generate, or reason over human language while preserving meaning, provenance, privacy, accessibility, and human authority.

REQUIRED OUTCOME

- A language task and contract with population, languages, context, source, output, error, and review boundaries.
- Evaluation for meaning, robustness, subgroup and language coverage, privacy, security, and usability.
- Traceable deployment, correction, refusal, fallback, and retirement paths.

WHY THIS MATTERS

Language varies by dialect, culture, domain, power, and context. A fluent or high-scoring output can still misrepresent intent, exclude speakers, disclose private content, or cause harm.

APPLICABILITY

Applies to classification, extraction, search, translation, question answering, dialogue, summarization, sentiment, generation, moderation, and language interfaces. Depth increases with rights, safety, employment, health, legal, or public impact.

NON-APPLICABILITY CONDITIONS

May be non-applicable to deterministic string handling with no semantic, user, or decision effect, or exploratory work with no operational claim. Record scope and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires language definitions, source rights, data lineage, domain review, evaluation, privacy, security, accessibility, and human oversight. Depends on Sections 097, 170, 180, 182, 186, and 188.

REQUIRED INPUTS

Task; language and dialect coverage; user population; corpus; labels; context; tokenizer; model; output schema; error costs; bias plan; privacy rules; translation and accessibility needs; evaluation; and fallback.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve language scope, labels, prohibited uses, publication, moderation thresholds, appeals, and residual risk. Native speakers, domain experts, and affected communities must review material language decisions.

AI RESPONSIBILITIES AND BOUNDARIES

AI may annotate, transform, generate, and evaluate language within scope. It must preserve source and language context, disclose uncertainty, and must not invent translation, consent, meaning, or citations.

APPLICABLE EXISTING AGENT ROLES

Researcher, Analyst, Domain Expert, Implementer, Prompt Engineer, UX Reviewer, Accessibility Reviewer, Privacy Reviewer, Security Auditor, Safety Auditor, Tester, Verifier, Critic, Red Team, Documentation Writer, and Governance Reviewer.

REQUIRED ACTIVITIES

1. Define task, language, dialect, audience, context, and prohibited use.
2. Audit corpus rights, representation, labels, annotation quality, and leakage.
3. Define semantic, structural, safety, privacy, and accessibility contracts.
4. Evaluate per-language and subgroup behavior, ambiguity, drift, and adversarial input.
5. Validate review, correction, appeal, refusal, and fallback.
6. Monitor distribution, errors, user impact, and model or provider changes.
7. Baseline and hand off evidence and unresolved language gaps.

HUMAN-AI INTERACTION REQUIREMENTS

Show source language, transformation, model, context, confidence limits, review state, and citation. Users must correct or contest language output and access a safe non-AI path. No fluency score substitutes for meaning review.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version corpora, annotations, tokenizer, model, prompts, translation memory, retrieval, policy, output schema, and evaluator. Preserve character and locale handling, provenance, access, redaction, trace, and fallback across interfaces.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show language coverage, model, source, output, uncertainty, quality, subgroup results, incidents, reviewer, and freshness. Allow authorized restrict, correct, regenerate, withdraw, quarantine, fallback, revoke, and retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Language task charter; corpus and rights register; label guide; model and prompt record; per-language evaluation; bias and accessibility review; output and appeal contract; deployment record; incident plan; and handoff package.

EVIDENCE REQUIREMENTS

Claims identify language, dialect, corpus, model, context, task, metric, subgroup, method, result, reviewer, and limits. Aggregate scores hide gaps unless disaggregated. Missing language or population evidence is UNKNOWN.

VERIFICATION AND ACCEPTANCE TESTS

Test ambiguity, dialect, code-switching, translation, long context, malformed input, prompt injection, privacy, toxicity, accessibility, refusal, correction, appeal, provider loss, and per-language regression. Independently reproduce claims.

FAILURE STATES AND ANTI-PATTERNS

English-only generalization; label leakage; dialect erasure; fluency mistaken for accuracy; translation hallucination; hidden moderation bias; unsupported language claims; and irreversible language decisions without appeal.

RECOVERY AND RECONSIDERATION

Restrict affected language or task on harmful errors, rights impact, drift, source loss, or evidence failure. Preserve inputs and outputs, notify owners, use human review or fallback, correct downstream records, and reopen after corpus or population change.

COMPLETION CRITERIA

Task, language scope, data, model, evaluation, controls, review, accessibility, privacy, recovery, evidence, and authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve corpus, annotations, model, prompts, evaluations, translations, traces, decisions, incidents, dissent, and gaps. Hand off to Generative AI, Content, Localization, Speech, Evaluation, Safety, Governance, and HITL Operations.

RELATED CODEX SECTIONS

089 Conversational Design; 097 Content Design; 101 Localization; 102 Internationalization; 170 Generative AI Engineering; 180 Model Evaluation; 182 AI Safety; 186 Prompt Engineering; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Information and AI Architecture, and evidence lexicon. Actual languages, populations, corpora, rights, evaluation thresholds, cultural authorities, and legal duties remain unresolved.

172

SECTION 172 - COMPUTER VISION

CODEX/03_SECTIONS/172_COMPUTER_VISION.txt

SHA-256 a272557cb63e0744a3f8377c8581b5a9e753582afdd366d36facb08f6a97eaf2

6,299 BYTES / PRESERVED SOURCE IDENTITY

SECTION 172 — Computer Vision

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed engineering for systems that interpret images, video, depth, or spatial signals, including detection, recognition, segmentation, tracking, generation, and visual decision support.

REQUIRED OUTCOME

- A visual task contract covering sensor, scene, population, environment, output, uncertainty, and action boundary.
- Evidence for accuracy, robustness, subgroup and condition coverage, privacy, security, safety, and accessibility.
- Controlled deployment, human review, incident, fallback, and retirement paths.

WHY THIS MATTERS

Vision systems are sensitive to lighting, viewpoint, occlusion, sensor quality, context, and representation. Errors can identify the wrong person, miss danger, expose private imagery, or trigger physical consequence.

APPLICABILITY

Applies to image or video analysis, biometric or object recognition, inspection, medical or industrial vision, autonomous perception, visual search, and image generation. Depth increases with surveillance, physical action, identity, safety, or rights impact.

NON-APPLICABILITY CONDITIONS

May be non-applicable to static non-semantic graphics processing or exploratory imagery with no operational claim. Record exact scope, sensor, population, owner, and reopening condition.

PREREQUISITES AND DEPENDENCIES

Requires sensor and image rights, data lineage, scene definitions, evaluation, privacy, security, safety, accessibility, and human authority. Depends on Sections 059, 060, 170, 180, 182, and 188.

REQUIRED INPUTS

Task; sensor; resolution; scene and environment; subjects; labels; model; thresholds; output schema; occlusion and quality rules; privacy and retention; adversarial plan; evaluation; physical action; and fallback.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve collection, surveillance or identification purpose, thresholds, use, disclosure, physical action, review, retention, and exception. Domain and affected-party experts must review material visual classifications.

AI RESPONSIBILITIES AND BOUNDARIES

AI may annotate, analyze, generate, and diagnose imagery within authorization. It must preserve image provenance and uncertainty and must not infer identity, consent, or physical truth beyond evidence.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Implementer, Safety Auditor, Privacy Reviewer, Security Auditor, Accessibility Reviewer, Tester, Verifier, Critic, Red Team, Governance Reviewer, Documentation Writer, and Human-in-the-Loop Operator.

REQUIRED ACTIVITIES

1. Define visual task, subjects, scene, sensor, environment, and prohibited use.
2. Audit collection rights, representation, labels, quality, and leakage.
3. Define output, confidence, uncertainty, privacy, safety, and action contracts.
4. Evaluate conditions, subgroups, occlusion, shift, adversarial input, and sensor failure.
5. Validate human review, appeal, correction, fallback, and physical stop.
6. Monitor drift, sensor changes, errors, incidents, and retention.
7. Baseline and hand off exact visual evidence.

HUMAN-AI INTERACTION REQUIREMENTS

Show source image or lawful reference, sensor, time, model, region, confidence limits, reviewer, and actionability. Provide human confirmation and non-AI inspection where material. Never equate a bounding box or score with identity or intent.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version sensors, calibration, images, labels, preprocessing, models, thresholds, policies, outputs, traces, and retention. Enforce access, redaction, provenance, latency, and safety interlocks outside the model. Preserve safe physical fallback.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show sensor, scene, model, output, uncertainty, conditions, alerts, reviewer, retention, and evidence. Allow authorized pause sensing, restrict access, correct, quarantine, revoke, fallback, stop physical action, and retire.

REQUIRED OUTPUTS AND ARTIFACTS

Vision task charter; sensor and rights register; dataset and label guide; model card; condition and subgroup evaluation; privacy and safety review; output and action contract; deployment record; incident plan; and handoff.

EVIDENCE REQUIREMENTS

Claims identify sensor, scene, population, conditions, data, model, threshold, metric, subgroup, method, result, reviewer, and limitation. Aggregate accuracy does not prove performance under all conditions or safe action.

VERIFICATION AND ACCEPTANCE TESTS

Test lighting, angle, occlusion, blur, weather, sensor loss, spoofing, adversarial input, privacy, bias, latency, false positives, false negatives, human review, physical stop, fallback, and retention deletion. Reproduce material claims.

FAILURE STATES AND ANTI-PATTERNS

Biometric overreach; dataset bias; benchmark-only validation; identity inferred from appearance; confidence treated as truth; hidden sensor degradation; surveillance without authority; and automated physical action without interlock.

RECOVERY AND RECONSIDERATION

Stop or restrict vision use on unsafe error, sensor drift, privacy event, spoofing, or evidence failure. Preserve records, notify owners, use manual inspection or safe state, correct downstream decisions, and reopen after scene or sensor changes.

COMPLETION CRITERIA

Task, sensor, data, model, conditions, evaluation, privacy, safety, action, review, recovery, evidence, and authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve sensor and calibration records, images or references, labels, models, evaluations, traces, decisions, incidents, dissent, and gaps. Hand off to Embedded, Robotics, Safety, Privacy, Security, AI, MLOps, Evaluation, and HITL Operations.

RELATED CODEX SECTIONS

058 Robotics Architecture; 059 Embedded Systems Architecture; 072 Signal Processing; 170 Generative AI Engineering; 180 Model Evaluation; 182 AI Safety; 198 Embedded Security; 252 Safety Testing; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, AI Architecture, and evidence lexicon. Actual sensors, image rights, populations, environmental limits, safety thresholds, retention, and action authority remain unresolved.

173

SECTION 173 - SPEECH ENGINEERING

CODEX/03_SECTIONS/173_SPEECH_ENGINEERING.txt

SHA-256 379eb02ff0a9b16c05d2637f382d27e277564a4954a9f481a9a0fc5e66b29bea

6,277 BYTES / PRESERVED SOURCE IDENTITY

SECTION 173 — Speech Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed engineering for speech capture, recognition, synthesis, diarization, translation, voice interaction, and audio classification with explicit consent, identity, context, quality, accessibility, and human-control boundaries.

REQUIRED OUTCOME

- A speech task and audio contract covering speakers, languages, environment, consent, retention, output, and action.
- Evidence for intelligibility, word or intent accuracy, subgroup and acoustic coverage, privacy, security, and safety.
- Controlled capture, review, correction, fallback, and retirement.

WHY THIS MATTERS

Speech carries identity, emotion, private content, accent, disability, and environmental context. Errors can mishear intent, impersonate a person, exclude a speaker, or create an irreversible transaction.

APPLICABILITY

Applies to speech-to-text, text-to-speech, voice authentication, diarization, call analysis, spoken dialogue, audio event detection, and voice cloning. Depth increases with biometrics, transactions, surveillance, health, safety, or public communication.

NON-APPLICABILITY CONDITIONS

May be non-applicable to non-semantic audio formatting or clearly isolated research with no identity, content, user, or operational claim. Record scope, consent, owner, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires audio rights and consent, language and accessibility context, data lineage, evaluation, privacy, security, safety, and human review. Depends on Sections 088-089, 170-173, 180, 182, and 188.

REQUIRED INPUTS

Task; speakers; language and accent; audio conditions; consent; microphone; corpus; model; output; confidence; identity boundary; retention; disclosure; evaluation; fallback; and transaction authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve capture, biometric or voice use, retention, disclosure, synthesis, transaction, review, and appeal. Affected speakers and accessibility experts must inform material acceptance decisions.

AI RESPONSIBILITIES AND BOUNDARIES

AI may transcribe, synthesize, classify, translate, and identify bounded audio. It must preserve source and consent, disclose uncertainty and synthetic status, and must not infer identity, emotion, or authorization without evidence.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Implementer, UX Reviewer, Accessibility Reviewer, Privacy Reviewer, Security Auditor, Safety Auditor, Tester, Verifier, Critic, Red Team, Governance Reviewer, Documentation Writer, and HITL Operator.

REQUIRED ACTIVITIES

1. Define speech task, population, languages, environment, consent, and prohibited use.
2. Audit corpus rights, speaker representation, labels, acoustic conditions, and leakage.
3. Specify input, output, identity, synthetic disclosure, retention, and action contracts.
4. Evaluate accents, noise, overlap, impairment, code-switching, spoofing, and drift.
5. Validate human correction, appeal, safe confirmation, and non-AI fallback.
6. Monitor errors, incidents, consent, provider changes, and retention.
7. Baseline and hand off evidence and unresolved coverage.

HUMAN-AI INTERACTION REQUIREMENTS

Show recording state, consent, speaker or synthetic status, language, model, transcript, confidence, reviewer, and action. Require confirmation for consequential intent or identity. Preserve text, visual, manual, and recovery paths.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version microphones, codecs, audio, transcripts, labels, models, voices, prompts, thresholds, policies, and retention. Protect recordings and voice artifacts, preserve timing and provenance, and enforce transaction controls outside recognition.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show capture, consent, speaker, model, output, confidence, retention, reviewer, incidents, and evidence. Allow authorized stop recording, correct, delete, restrict, revoke, fallback, quarantine, and retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Speech charter; consent and rights register; corpus and label guide; model and voice record; acoustic and subgroup evaluation; privacy and security review; output and confirmation contract; deployment and incident plan.

EVIDENCE REQUIREMENTS

Claims identify audio source, speaker population, language, condition, model, output, metric, subgroup, method, result, consent, reviewer, and limitation. Word accuracy or naturalness does not prove intent or identity accuracy.

VERIFICATION AND ACCEPTANCE TESTS

Test accents, noise, overlap, silence, impairment, language switching, replay, spoofing, privacy, synthetic disclosure, confirmation, correction, deletion, provider loss, and emergency fallback. Independently reproduce material results.

FAILURE STATES AND ANTI-PATTERNS

Silent recording; voice cloning without consent; accent exclusion; transcript treated as verbatim truth; confidence treated as authorization; biometric overreach; hidden audio retention; and speech-only access to essential service.

RECOVERY AND RECONSIDERATION

Stop capture or action on consent loss, unsafe misrecognition, spoofing, privacy event, drift, or evidence failure. Preserve required records, correct affected transactions, notify owners, use manual fallback, and reopen after population or provider change.

COMPLETION CRITERIA

Task, audio, consent, population, model, output, evaluation, privacy, accessibility, confirmation, recovery, evidence, and authority are explicit. Status remains CONTENT-COMPLETE pending acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve audio references, consent, transcripts, voices, models, evaluations, traces, decisions, incidents, dissent, and gaps. Hand off to NLP, Generative AI, UX, Accessibility, Security, Privacy, Evaluation, Operations, and HITL.

RELATED CODEX SECTIONS

087 Sound Design; 088 Voice Interface Design; 089 Conversational Design; 101 Localization; 170 Generative AI Engineering; 180 Model Evaluation; 182 AI Safety; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance. Actual consent, speaker populations, languages, biometric law, retention, accessibility, provider terms, and transaction authority remain unresolved.

174

SECTION 174 - REINFORCEMENT LEARNING

CODEX/03_SECTIONS/174_REINFORCEMENT_LEARNING.txt

SHA-256 334332b83b0fc6a20b7e4745ef8c3c9aa52ba0056cd0b98956e256591b484284

6,391 BYTES / PRESERVED SOURCE IDENTITY

SECTION 174 — Reinforcement Learning

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed engineering for agents that learn policies through interaction, reward, feedback, or planning, with explicit environment, objective, constraint, exploration, oversight, and deployment boundaries.

REQUIRED OUTCOME

- A formal task and environment contract with states, actions, observations, rewards, constraints, termination, and authority.
- Evidence that behavior meets protected outcomes beyond reward optimization in simulation or training.
- Controlled exploration, evaluation, deployment, monitoring, interruption, and recovery.

WHY THIS MATTERS

An agent can maximize a reward while exploiting its proxy, harming users, or violating constraints. Interactive learning also creates data, state, and action effects that are difficult to reverse.

APPLICABILITY

Applies to reinforcement learning, preference optimization, online learning, planning agents, adaptive control, bandits, and systems that learn from actions or feedback. Depth increases with physical action, external users, autonomy, or live exploration.

NON-APPLICABILITY CONDITIONS

May be non-applicable to static supervised inference or offline conceptual discussion with no policy, action, or learning claim. Record exact scope and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires environment and domain models, reward authority, safety constraints, evaluation, simulation validity, security, privacy, operations, and human interruption. Depends on Sections 073, 074, 164, 167, 180, 182, 184, and 188.

REQUIRED INPUTS

Task; state; observation; action; reward; hard constraints; environment; simulator; feedback; exploration budget; safety envelope; policy; human approval; monitoring; termination; and compensation.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans define protected outcomes, constraints, reward legitimacy, exploration limits, deployment, intervention, and acceptable residual risk. They may pause, override, revoke, or retire; the learner cannot redefine its objective.

AI RESPONSIBILITIES AND BOUNDARIES

AI may learn policies and propose actions within a sandbox or delegated boundary. It must expose reward and uncertainty and must not explore live without authority, hide reward hacking, or treat success in simulation as safe operation.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Planner, Implementer, Safety Auditor, Security Auditor, Privacy Reviewer, Tester, Verifier, Critic, Red Team, Governance Reviewer, Optimizer, Documentation Writer, and HITL Operator.

REQUIRED ACTIVITIES

1. Define environment, task, state, actions, reward, constraints, and authority.
2. Validate simulator, feedback, reward, data, and safety envelope.
3. Isolate exploration and record seeds, trajectories, policies, and configuration.
4. Evaluate reward hacking, distribution shift, rare events, adversarial behavior, and subgroups.
5. Test human stop, override, fallback, termination, and compensation.
6. Gate promotion from simulation to offline, shadow, limited, and live operation.
7. Monitor, investigate, and retire policy with evidence.

HUMAN-AI INTERACTION REQUIREMENTS

Show current state, action, policy, reward, constraint, authority, uncertainty, and review. Require confirmation for consequential actions and preserve an immediate stop path. Human feedback must be attributable and cannot be silently converted into policy.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version environments, simulators, policies, trajectories, reward functions, constraints, seeds, feedback, checkpoints, actions, and outcomes. Enforce safety constraints outside the policy where possible and preserve idempotent interruption and compensation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show policy, environment, current state, action queue, reward, constraints, exploration mode, reviewer, incidents, and evidence. Provide authorized pause, stop, override, reduce scope, revoke, fallback, compensate, rollback configuration, and retire.

REQUIRED OUTPUTS AND ARTIFACTS

RL task charter; environment and simulator specification; reward and constraint register; trajectory and policy records; evaluation and red-team report; promotion gates; intervention runbook; decision record; and continuity package.

EVIDENCE REQUIREMENTS

Claims identify environment, policy, reward, constraint, scenario, trajectory, seed, evaluator, outcome, uncertainty, and limitation. Reward, episode count, or simulation success does not prove protected outcome or safe deployment.

VERIFICATION AND ACCEPTANCE TESTS

Test reward misspecification, constraint violation, exploration, rare state, adversarial input, simulator gap, feedback poisoning, action interruption, authority revocation, fallback, compensation, and live promotion gates. Reproduce accepted runs.

FAILURE STATES AND ANTI-PATTERNS

Reward hacking; proxy-only success; unrestricted exploration; simulator overconfidence; hidden policy state; unsafe retries; feedback poisoning; no human stop; and destructive rollback of prior policy evidence.

RECOVERY AND RECONSIDERATION

Stop policy actions on constraint violation, reward anomaly, unsafe trajectory, drift, feedback poisoning, or authority loss. Preserve state and immutable decisions, use safe fallback or compensation, review impact, and requalify before resuming.

COMPLETION CRITERIA

Task, environment, reward, constraints, policy, exploration, evaluation, intervention, recovery, evidence, and authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve environments, policies, trajectories, rewards, constraints, seeds, feedback, evaluations, interventions, incidents, decisions, dissent, and gaps. Hand off to Robotics, Control, ML, Safety, Evaluation, MLOps, Operations, and HITL.

RELATED CODEX SECTIONS

073 Control Systems Engineering; 074 Robotics Engineering; 136 Simulation Engineering; 164 Data Science; 167 Machine Learning Engineering; 180 Model Evaluation; 182 AI Safety; 184 MLOps; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance. Actual environments, reward authority, safety envelopes, action consequences, simulation validity, feedback rights, and promotion authority remain unresolved.

175

SECTION 175 - KNOWLEDGE GRAPH ENGINEERING

CODEX/03_SECTIONS/175_KNOWLEDGE_GRAPH_ENGINEERING.txt

SHA-256 13a485745a2e479833b4a2d8beec19e119bfe78eaa3c55269dbb341de26aecc4

6,284 BYTES / PRESERVED SOURCE IDENTITY

SECTION 175 — Knowledge Graph Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed engineering of graph-based knowledge representations that connect entities, concepts, events, claims, sources, provenance, permissions, and temporal context for authorized retrieval and reasoning.

REQUIRED OUTCOME

- A graph scope, vocabulary, identity policy, relationship model, provenance model, and query contract.
- Traceable ingestion, reconciliation, inference, access, quality, update, and retirement behavior.
- Explicit distinction between sourced facts, derived relationships, hypotheses, and generated answers.

WHY THIS MATTERS

Graphs can make an inferred or stale relationship appear canonical. Identity collisions, provenance loss, and unrestricted traversal can create privacy, security, and decision harm while appearing semantically coherent.

APPLICABILITY

Applies to entity graphs, semantic search, lineage graphs, recommendation, dependency maps, retrieval systems, reasoning systems, and graph-based agents. Depth increases with identity, sensitive data, automated inference, or external action.

NON-APPLICABILITY CONDITIONS

May be non-applicable to isolated relational lookup with no graph semantics, inference, or cross-source identity. Record exact scope, owner, evidence date, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires ontology or vocabulary authority, data lineage, identity rules, access policy, quality rules, query purpose, and human acceptance. Depends on Sections 030, 031, 155, 156, 176, 177, 180, and 183.

REQUIRED INPUTS

Graph purpose; entity types; identifiers; relationship semantics; source data; provenance; temporal rules; inference rules; access; sensitivity; quality thresholds; query users; and update and deletion obligations.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve canonical concepts, identity resolution, inference classes, sensitive links, publication, access, correction, retention, and deletion. Domain owners decide whether a graph claim is authoritative or merely useful.

AI RESPONSIBILITIES AND BOUNDARIES

AI may extract entities, propose links, reconcile records, and generate graph queries. It must preserve evidence and uncertainty and must not create canonical identity or relationship, infer permission, or publish claims without authority.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Analyst, Data Engineer, Ontology Engineer, Privacy Reviewer, Security Auditor, Governance Reviewer, Critic, Tester, Verifier, Red Team, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

1. Define graph purpose, scope, users, subjects, and prohibited traversal.
2. Establish vocabulary, identity, relationship, temporal, provenance, and access rules.
3. Ingest and map sources with lineage, quality, conflict, and deletion handling.
4. Separate asserted, derived, inferred, uncertain, and generated edges.
5. Evaluate identity collision, link quality, leakage, stale data, and query behavior.
6. Validate graph answers with domain owners and affected parties.
7. Monitor updates, inference drift, access, correction, and retirement.

HUMAN-AI INTERACTION REQUIREMENTS

Show nodes, edges, source, timestamp, provenance, confidence, inference path, permissions, and reviewer. Users must challenge or correct links and see source evidence. Graph traversal cannot bypass access or disclosure controls.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version ontology, identifiers, nodes, edges, assertions, inferences, source mappings, policies, indexes, queries, and snapshots. Enforce provenance and permissions at traversal and export boundaries. Preserve temporal and deletion semantics.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show graph version, source, lineage, identity decisions, inference, conflicts, access, freshness, and queries. Allow authorized approve, correct, merge, split, restrict, quarantine, withdraw, rebuild, and retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Graph charter; entity and relationship model; vocabulary and identity policy; source and lineage map; inference register; query and access contract; quality report; privacy and security review; correction and deletion plan; and handoff.

EVIDENCE REQUIREMENTS

Each graph claim identifies node and edge versions, source, assertion or inference type, method, timestamp, confidence, permission, reviewer, and limitation. A connected path is not proof that its inferred meaning is true.

VERIFICATION AND ACCEPTANCE TESTS

Test identity collision, duplicate, stale, conflicting, unauthorized, deleted, inferred, cyclic, and poisoned relationships; verify provenance, query permissions, temporal behavior, correction, export, and independent reproduction.

FAILURE STATES AND ANTI-PATTERNS

Identity collapse; inferred edge treated as fact; source laundering; unrestricted traversal; stale relationships; hidden conflict; graph completeness claims from one source; and deletion that leaves derived links active.

RECOVERY AND RECONSIDERATION

Quarantine affected nodes or edges on provenance, identity, access, or quality failure. Preserve prior snapshot, notify owners, correct or rebuild, review downstream answers, and reopen after vocabulary, source, identity, or policy change.

COMPLETION CRITERIA

Purpose, graph model, source, identity, provenance, inference, access, quality, lifecycle, verification, recovery, and authority are explicit. Status remains CONTENT-COMPLETE pending acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve graph IDs, snapshots, mappings, ontology versions, queries, evidence, decisions, corrections, incidents, dissent, and gaps. Hand off to Ontology, Semantic Systems, Data, Search, AI, Evaluation, Privacy, Security, and Operations.

RELATED CODEX SECTIONS

030 Information Architecture; 031 Data Architecture; 055 Knowledge Architecture; 154 Data Modeling; 155 Data Governance; 176 Ontology Engineering; 177 Semantic Systems Engineering; 180 Model Evaluation.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Information and Data Architecture, and evidence lexicon. Actual vocabularies, identity authorities, source rights, inference policy, privacy rules, and deletion obligations remain unresolved.

176

SECTION 176 - ONTOLOGY ENGINEERING

CODEX/03_SECTIONS/176_ONTOLOGY_ENGINEERING.txt

SHA-256 918a7f495fa22097ebb24d12d07cb6337373d46e736b7367d739c3751db87016

6,081 BYTES / PRESERVED SOURCE IDENTITY

SECTION 176 — Ontology Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed design and maintenance of formal concepts, classes, properties, constraints, axioms, mappings, and terminology that establish shared meaning across domains and systems.

REQUIRED OUTCOME

- An ontology scope, competency questions, concepts, relationships, constraints, mappings, owners, and version policy.
- Evidence that definitions fit domain usage, stakeholder needs, data, information, and system behavior.
- Controlled change, translation, conflict, deprecation, and adoption paths.

WHY THIS MATTERS

An ontology can encode one group's assumptions as universal truth. Misclassified concepts and silent version drift propagate through data, search, analytics, models, and decisions.

APPLICABILITY

Applies when shared formal meaning, reasoning, mapping, semantic validation, cross-domain exchange, or ontology-based AI is required. Depth increases with legal, safety, identity, public, or automated decision impact.

NON-APPLICABILITY CONDITIONS

May be non-applicable where a bounded local vocabulary has no cross-system, inferential, decision, or lifecycle effect. Record scope, authority, and reopening trigger; a glossary is not automatically an ontology.

PREREQUISITES AND DEPENDENCIES

Requires domain authorities, information and data models, competency questions, stakeholders, translation needs, governance, and versioning. Depends on Sections 007, 016, 030, 031, 055, 175, and 177.

REQUIRED INPUTS

Purpose; domain; users; competency questions; terms; sources; definitions; relations; constraints; mappings; jurisdictions; languages; inference needs; access; lifecycle; and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Domain owners approve meaning, scope, axioms, contested terms, mappings, deprecations, and exceptions. Ontologists facilitate and document; they do not impose a definition over legitimate domain authority or lived experience.

AI RESPONSIBILITIES AND BOUNDARIES

AI may extract terms, propose hierarchies and mappings, and test consistency. It must preserve source and dissent and must not declare a canonical concept, resolve contested meaning, or infer policy from similarity.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Domain Analyst, Ontology Engineer, Information Architect, Data Governance Reviewer, Critic, Verifier, Tester, Documentation Writer, Memory Curator, Governance Reviewer, and Accessibility Reviewer.

REQUIRED ACTIVITIES

1. Define scope, competency questions, users, jurisdictions, languages, and non-goals.
2. Inventory source vocabularies, conflicts, terms, and authoritative owners.
3. Model concepts, properties, constraints, axioms, mappings, and provenance.
4. Test competency questions, consistency, completeness, ambiguity, and unintended inference.
5. Review accessibility, cultural, legal, privacy, and stakeholder effects.
6. Version, publish, deprecate, migrate, and communicate changes.
7. Record dissent, unresolved meaning, and owner acceptance.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, definition, rationale, scope, confidence, mapping, inference, reviewer, and version. Humans must edit and contest concepts without erasing history. Generated mappings remain proposals and cannot grant access or authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version concepts, relations, axioms, constraints, mappings, translations, namespaces, reasoners, and applications. Separate semantic definition from implementation schema. Preserve provenance, compatibility, migration, deprecation, and inference limits.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show ontology version, owner, scope, term, definition, source, mappings, usages, inferences, conflicts, and lifecycle. Allow authorized approve, edit, contest, deprecate, map, restrict, supersede, and retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Ontology charter; competency-question register; concept and relation model; axioms and constraints; source and mapping register; glossary; version and deprecation policy; validation report; decisions, dissent, and handoff.

EVIDENCE REQUIREMENTS

Each concept and axiom identifies authority, source, scope, rationale, version, reviewer, applicability, and limitation. A lexical similarity, reasoner result, or mapping does not prove domain truth or acceptance.

VERIFICATION AND ACCEPTANCE TESTS

Run competency questions; test consistency, ambiguity, unintended entailment, mapping collisions, version compatibility, access implications, multilingual terms, and deprecation migration. Independently verify representative definitions.

FAILURE STATES AND ANTI-PATTERNS

Universal vocabulary without authority; ontology by majority vote; hidden axioms; semantic overreach; mapping by string match; contested terms erased; version drift; and inferred policy presented as definition.

RECOVERY AND RECONSIDERATION

Quarantine affected inferences on semantic conflict, invalid axiom, mapping error, or authority dispute. Preserve prior version and evidence, restrict downstream use, consult domain owners, migrate or supersede deliberately, and reopen.

COMPLETION CRITERIA

Scope, terms, relations, axioms, sources, owners, mappings, tests, version, conflicts, lifecycle, and authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve ontology IDs, namespaces, versions, definitions, mappings, tests, decisions, dissent, usages, and gaps. Hand off to Information, Data, Knowledge Graph, Semantic Systems, Search, AI, and Governance.

RELATED CODEX SECTIONS

007 Glossary; 016 Domain Analysis; 030 Information Architecture; 055 Knowledge Architecture; 175 Knowledge Graph Engineering; 177 Semantic Systems Engineering; 342 Knowledge Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance. Actual domain authorities, contested meanings, legal definitions, languages, reasoning requirements, mappings, and acceptance criteria remain unresolved.

177

SECTION 177 - SEMANTIC SYSTEMS ENGINEERING

CODEX/03_SECTIONS/177_SEMANTIC_SYSTEMS_ENGINEERING.txt

SHA-256 310ab92e49c846200b837450a46bdf392bb56c6747cd4311135f936e22fcfe7b

6,043 BYTES / PRESERVED SOURCE IDENTITY

SECTION 177 — Semantic Systems Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define engineering of systems that interpret, map, reason over, validate, or exchange meaning across data, language, domains, and applications with explicit provenance and authority.

REQUIRED OUTCOME

- A semantic contract connecting ontology, data, information, queries, inference, users, and decisions.
- Verified mappings, reasoning boundaries, provenance, access, contradiction handling, and lifecycle controls.
- A clear distinction between semantic consistency and domain correctness.

WHY THIS MATTERS

Semantic systems can be internally consistent while consistently representing the wrong meaning. Hidden mappings and inference can affect access, ranking, decisions, and generated answers without visible accountability.

APPLICABILITY

Applies to semantic APIs, knowledge representations, reasoning engines, federated meaning, ontology services, semantic search, validation, and AI grounding. Depth increases with automated inference, cross-domain exchange, or external consequence.

NON-APPLICABILITY CONDITIONS

May be non-applicable to local syntax or direct lookup with no meaning mapping, inference, or cross-boundary interpretation. Record the exact boundary and reopening condition.

PREREQUISITES AND DEPENDENCIES

Requires Information, Data, Knowledge, Ontology, API, Identity, Privacy, Security, and Governance authorities. Depends on Sections 030, 031, 042, 055, 175, and 176.

REQUIRED INPUTS

Semantic purpose; ontologies; vocabularies; mappings; data; context; inference rules; query contract; provenance; identity; permissions; contradiction policy; performance; and owner acceptance.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve meaning, mappings, inference limits, contradictions, access, publication, and downstream use. Semantic engineers cannot resolve domain disputes or turn logical entailment into policy authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may propose mappings, embeddings, queries, and explanations. It must preserve source and uncertainty and must not silently infer meaning, permission, identity, or obligation.

APPLICABLE EXISTING AGENT ROLES

Architect, Ontology Engineer, Information Architect, Data Engineer, Researcher, Analyst, Governance Reviewer, Privacy Reviewer, Security Auditor, Critic, Tester, Verifier, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

1. Define semantic boundary, users, domains, contexts, and prohibited inference.
2. Inventory concepts, mappings, data, sources, and authority.
3. Specify semantic contracts, query behavior, provenance, access, and contradiction policy.
4. Implement and test mapping, validation, reasoning, retrieval, and explanation.
5. Evaluate ambiguity, drift, identity collision, leakage, and unintended entailment.
6. Validate with domain owners and affected users.
7. Version, monitor, deprecate, migrate, and hand off.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, semantic path, mapping, inference, confidence, context, permissions, and reviewer. Users must inspect and challenge reasoning. Generated semantic links remain proposals and cannot alter records or policy without authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version semantic contracts, ontologies, mappings, reasoners, embeddings, indexes, queries, provenance, policies, and outputs. Enforce context and access at query and inference boundaries. Preserve contradictions and temporal validity.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show semantic baseline, source, concept, mapping, inference, query, conflict, access, freshness, and downstream uses. Provide authorized correct, restrict, quarantine, rerun, supersede, withdraw, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Semantic charter; ontology and mapping register; contract; reasoning and contradiction policy; provenance and access design; query and explanation examples; validation report; decision record; and continuity handoff.

EVIDENCE REQUIREMENTS

Claims identify semantic version, source, mapping, inference rule, context, query, result, method, reviewer, and limitation. Logical consistency, embedding similarity, or query success does not prove domain validity.

VERIFICATION AND ACCEPTANCE TESTS

Test mappings, contradictions, ambiguity, temporal changes, unauthorized inference, stale indexes, identity collisions, version compatibility, query reproducibility, deletion, and explanation. Independently reproduce representative results.

FAILURE STATES AND ANTI-PATTERNS

Semantic drift; hidden mapping; inference treated as source; closed-world assumption unnoticed; embedding similarity as truth; contradiction suppression; unauthorized traversal; and schema compatibility mistaken for meaning compatibility.

RECOVERY AND RECONSIDERATION

Restrict affected reasoning on semantic conflict, mapping failure, access loss, or provenance gap. Preserve prior version, identify downstream decisions, correct or rebuild, obtain domain review, and reopen after ontology or data change.

COMPLETION CRITERIA

Meaning, mappings, inference, provenance, access, contradictions, tests, lifecycle, evidence, and authority are explicit. Status remains CONTENT-COMPLETE pending source reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve contracts, ontologies, mappings, reasoners, queries, results, decisions, dissent, evidence, and gaps. Hand off to Knowledge Graph, Ontology, Information, Data, API, Search, AI, Evaluation, and Governance.

RELATED CODEX SECTIONS

030 Information Architecture; 031 Data Architecture; 042 API Architecture; 055 Knowledge Architecture; 175 Knowledge Graph Engineering; 176 Ontology Engineering; 180 Model Evaluation; 183 AI Governance.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Information, Data, Knowledge, and Ontology sections. Actual domain semantics, inference policy, mapping ownership, privacy, legal definitions, and acceptance remain unresolved.

178

SECTION 178 - AGENT SYSTEMS ENGINEERING

CODEX/03_SECTIONS/178_AGENT_SYSTEMS_ENGINEERING.txt

SHA-256 70458e618d777e1870a58c01790cd30dadad8398c2bde0d871eea16d63d25dab

6,429 BYTES / PRESERVED SOURCE IDENTITY

SECTION 178 — Agent Systems Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define engineering of systems that perceive context, plan, reason, call tools, coordinate, learn, or act toward goals while preserving explicit identity, authority, state, human oversight, evidence, and compensation.

REQUIRED OUTCOME

- A bounded agent contract for goal, capabilities, tools, data, permissions, state, planning, action, review, and termination.
- Reproducible evaluation of planning, tool use, delegation, refusal, recovery, security, safety, and user control.
- Immutable decisions and append-only evidence for consequential action.

WHY THIS MATTERS

Agents combine model uncertainty with state, tools, and external effects. A persuasive plan or completed tool call is not authorization, correctness, or safe recovery.

APPLICABILITY

Applies to autonomous, semi-autonomous, tool-using, multi-agent, workflow, planning, and long-running agents. Depth increases with external mutation, delegation, sensitive data, autonomy, persistence, or multi-party impact.

NON-APPLICABILITY CONDITIONS

May be non-applicable to stateless generation with no tool, state, delegation, or external effect. Record boundary, owner, evidence date, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires requirements, process, identity, authorization, data, tool contracts, policy evaluation, observability, evaluation, safety, security, and HITL authority. Depends on Sections 169, 170, 180-183, 185, 186, and 188.

REQUIRED INPUTS

Goal; user; agent identity; capability token; tools; data; policies; state; memory; delegation; approval; mutation class; stop conditions; trace; evaluation; compensation; and retirement rules.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans define goal, scope, capability, approval, protected outcomes, exceptions, risk, and final decisions. They may stop, revoke, narrow, compensate, or retire. Agents never derive authority from a goal or model confidence.

AI RESPONSIBILITIES AND BOUNDARIES

Agents may plan, recommend, and invoke authorized tools within a complete governance binding. They must verify state and scope, expose uncertainty, and stop on expired capability, policy mismatch, missing quorum, or unresolved ambiguity.

APPLICABLE EXISTING AGENT ROLES

Architect, Implementer, Planner, Coordinator, Governance Reviewer, Security Auditor, Safety Auditor, Researcher, Analyst, Tester, Verifier, Critic, Red Team, Documentation Writer, Memory Curator, and Human-in-the-Loop Operator.

REQUIRED ACTIVITIES

1. Define goal, actor, resources, authority, mutation class, and prohibited action.
2. Model planning, state, memory, tools, delegation, approval, and termination.
3. Bind each mutation to trace, policy evaluation, capability, decision, and audit records.
4. Test prompt injection, tool misuse, delegation, replay, concurrency, failure, and recovery.
5. Validate human cockpit, stop, revoke, appeal, explanation, and compensation.
6. Monitor traces, policy outcomes, costs, drift, and incidents.
7. Baseline, hand off, and retire without deleting prior decisions.

HUMAN-AI INTERACTION REQUIREMENTS

Show goal, plan, state, identity, capability, policy result, tool, data, mutation, decision, and trace. Require approval or quorum when applicable. Preserve terminal stream semantics, visible stop, correction, appeal, and compensation paths.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent RunRequest, ToolInvocationRequest, PolicyEvaluationRequest and Result, GovernanceEvaluationBinding, MutationGovernanceBinding, DecisionRecord, AuditDecision, trace, artifact, and compensation links as immutable or append-only records. Enforce default-deny and expiry at tool boundaries.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show active runs, goal, state, plan, pending actions, authority, capability scope, policy, decisions, traces, artifacts, cost, and incidents. Provide authorized pause, deny, revoke, quarantine, compensate, resume, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Agent charter; capability and tool catalog; state and delegation model; policy and governance binding; decision and audit records; evaluation suite; trace and artifact package; runbook; incident and compensation plan; and handoff.

EVIDENCE REQUIREMENTS

Every consequential action identifies trace, request, agent identity, capability, policy result, decision record, tool, resource, input, output, reviewer, and result. A plan, tool response, or successful run is not proof of authorization or outcome.

VERIFICATION AND ACCEPTANCE TESTS

Expire or revoke capability; alter policy; remove quorum; inject conflicting instructions; replay requests; duplicate tools; corrupt state; lose provider; interrupt stream; and verify default-deny, audit, terminal event, compensation, and recovery.

FAILURE STATES AND ANTI-PATTERNS

Goal treated as permission; prompt as policy; missing capability binding; local-only policy check; hidden delegation; unbounded loops; silent tool mutation; lost terminal event; and destructive rollback.

RECOVERY AND RECONSIDERATION

Stop affected execution on authority, trace, policy, state, tool, or safety failure. Preserve immutable records and artifacts, quarantine outputs, compensate only through linked transactions, review impact, and resume only after fresh evaluation and authority.

COMPLETION CRITERIA

Goal, boundary, identity, capabilities, policy, state, tools, decisions, audit, trace, evaluation, stop, compensation, and owner authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve run IDs, decisions, bindings, traces, tool calls, artifacts, state, incidents, dissent, and gaps. Hand off to AI Engineering, Governance, Safety, Security, MLOps, Model Operations, Prompt Engineering, and HITL Operations.

RELATED CODEX SECTIONS

048 Governance Architecture; 056 AI Architecture; 169 AI Engineering; 180 Model Evaluation; 182 AI Safety; 183 AI Governance; 185 Model Operations; 186 Prompt Engineering; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, AI Architecture, Owner Cockpit Requirements, and Status and Evidence Lexicon. Actual capability tokens, policy endpoints, quorum rules, tool registry, mutation classes, and compensation contracts remain unresolved until supplied.

179

SECTION 179 - MODEL TRAINING

CODEX/03_SECTIONS/179_MODEL_TRAINING.txt

SHA-256 0cc7b0805cfb63a7fb9a410e7254de48b457df06b1ed8725be6333b971ebde46

6,044 BYTES / PRESERVED SOURCE IDENTITY

SECTION 179 — Model Training

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed preparation, training, fine-tuning, preference optimization, checkpointing, and selection of model artifacts with explicit data, objective, compute, security, evaluation, and release boundaries.

REQUIRED OUTCOME

- A training plan connecting authorized purpose, data, objective, configuration, compute, artifacts, and evaluation.
- Reproducible runs with lineage, resource, contamination, privacy, security, and failure evidence.
- Controlled checkpoint selection and handoff; training is not deployment approval.

WHY THIS MATTERS

Training can encode data defects, leakage, unsafe capability, bias, or hidden memorization. A lower loss or preferred output does not prove the model serves its authorized purpose safely.

APPLICABILITY

Applies to pretraining, fine-tuning, transfer learning, reinforcement or preference optimization, distillation, quantization-aware training, and retraining. Depth increases with sensitive data, scale, autonomy, or external use.

NON-APPLICABILITY CONDITIONS

May be non-applicable to using an unchanged external model with no training or artifact change, or conceptual discussion with no run claim. Record scope and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires data rights and quality, objective authority, compute and supply-chain controls, evaluation, safety, privacy, security, and model-operations gates. Depends on Sections 167-169, 180-184.

REQUIRED INPUTS

Training objective; data and license; population; labels or preference source; architecture; configuration; compute; budget; split; contamination controls; safety constraints; evaluator; checkpoint policy; and release authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve purpose, data use, objective, resource budget, safety limits, evaluation gate, checkpoint selection, publication, and residual risk. Training staff cannot expand use or declare alignment.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft datasets, code, configurations, schedules, and diagnostics. It must preserve run identity and cannot invent metrics, hide failed runs, or select a checkpoint for deployment without authorized review.

APPLICABLE EXISTING AGENT ROLES

Architect, Researcher, Data Scientist, Implementer, Optimizer, Security Auditor, Privacy Reviewer, Safety Auditor, Tester, Verifier, Critic, Red Team, Governance Reviewer, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

1. Define purpose, objective, prohibited capability, data, and release boundary.
2. Validate data lineage, rights, quality, representation, contamination, and split.
3. Record code, configuration, hardware, software, seeds, budget, and dependencies.
4. Run training with checkpoint, failure, resource, and access controls.
5. Evaluate generalization, robustness, subgroup, privacy, security, safety, and memorization.
6. Compare checkpoints against baseline and document selection rationale.
7. Hand off only with evidence and explicit next approval.

HUMAN-AI INTERACTION REQUIREMENTS

Expose run, data, objective, configuration, checkpoint, metrics, failures, cost, uncertainty, and reviewer. AI-generated training changes remain proposals. Preserve pause, cancel, evidence, access, and safe checkpoint handling.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version datasets, loaders, code, libraries, hardware, configuration, seeds, checkpoints, logs, metrics, evaluators, and policies. Protect artifacts and secrets; enforce lineage, retention, access, and immutable run records.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show training state, data, objective, resources, checkpoints, metrics, failures, risks, authority, and evidence. Provide authorized pause, cancel, quarantine, revoke, compare, select, archive, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Training charter; data and rights register; objective and configuration; reproducible environment; run and checkpoint register; evaluation report; safety, privacy, and security review; selection decision; and handoff.

EVIDENCE REQUIREMENTS

Each claim identifies run, data, split, objective, configuration, hardware, checkpoint, metric, evaluator, result, uncertainty, and limitation. Loss, reward, or preference score is not deployment or safety evidence by itself.

VERIFICATION AND ACCEPTANCE TESTS

Reproduce a run; verify data split and contamination; test checkpoint integrity, dependency changes, memorization, robustness, subgroups, privacy, security, safety, and failed-run visibility; independently review selection.

FAILURE STATES AND ANTI-PATTERNS

Untracked data; test leakage; hidden failed runs; metric gaming; checkpoint confusion; unsafe data reuse; unauthorized compute; silent retraining; and training success treated as production readiness.

RECOVERY AND RECONSIDERATION

Freeze affected runs or checkpoints on data, artifact, compute, security, or safety failure. Preserve evidence, use last qualified artifact only if still valid, investigate, re-evaluate, and reopen when data, objective, or population changes.

COMPLETION CRITERIA

Purpose, data, objective, run, artifact, resources, evaluation, risks, selection, recovery, evidence, and authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve runs, checkpoints, data, code, environments, metrics, decisions, failed attempts, dissent, incidents, and gaps. Hand off to ML Engineering, Deep Learning, Evaluation, Alignment, MLOps, Safety, and Model Operations.

RELATED CODEX SECTIONS

164 Data Science; 167 Machine Learning Engineering; 168 Deep Learning Engineering; 180 Model Evaluation; 181 Model Alignment; 182 AI Safety; 184 MLOps; 253 Model Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance. Actual training objectives, data rights, compute authority, safety gates, evaluator independence, and deployment criteria remain unresolved until supplied.

180

SECTION 180 - MODEL EVALUATION

CODEX/03_SECTIONS/180_MODEL_EVALUATION.txt

SHA-256 8f529e974a1d369e0600440babffc480d42d4fa49da3ebb47f749486dd0b4909

6,036 BYTES / PRESERVED SOURCE IDENTITY

SECTION 180 — Model Evaluation

STATUS: CONTENT-COMPLETE

PURPOSE

Define independent, reproducible evaluation of models and AI systems against authorized tasks, protected properties, risks, users, environments, and operational outcomes.

REQUIRED OUTCOME

- An evaluation charter with claims, datasets, scenarios, metrics, thresholds, uncertainty, and decision authority.
- Evidence covering expected, boundary, adversarial, subgroup, safety, privacy, security, accessibility, robustness, and operational behavior.
- A decision that states what the evidence supports and what remains unknown.

WHY THIS MATTERS

Evaluation is often reduced to a score selected after the fact. Governed evaluation prevents benchmark success, model preference, or a green dashboard from becoming an unsupported safety or readiness claim.

APPLICABILITY

Applies to model selection, training gates, release, monitoring, incident review, retraining, and claims about capability, safety, fairness, robustness, or alignment. Depth increases with consequence and autonomy.

NON-APPLICABILITY CONDITIONS

May be non-applicable to a purely deterministic non-model change or an exploratory test with no acceptance or operational claim. Record scope, claim boundary, owner, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires model and system contract, intended use, evaluation data, baseline, risk and impact context, test authority, and independent review. Depends on Sections 167-170, 179, 181-183, 253, and 254.

REQUIRED INPUTS

Claim; model and system version; task; population; scenarios; protected properties; test data; baseline; metrics; thresholds; uncertainty; evaluator; independence; environment; and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve claims, metrics, thresholds, exclusions, evaluator independence, release decision, exceptions, and residual risk. Evaluators report honestly and cannot convert a result into permission without owner action.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate tests, score outputs, summarize results, and identify gaps. It must preserve exact inputs and grading rules and must not grade its own success as independent proof or hide failures.

APPLICABLE EXISTING AGENT ROLES

Tester, Verifier, Analyst, Researcher, Critic, Red Team, Safety Auditor, Security Auditor, Privacy Reviewer, Accessibility Reviewer, Governance Reviewer, Documentation Writer, Memory Curator, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

1. Define claim, use, population, risk, metric, threshold, and decision.
2. Build or select representative and adversarial evaluation sets.
3. Predeclare procedures, scoring, exclusions, and uncertainty where possible.
4. Evaluate baseline, subgroups, boundary, robustness, safety, privacy, security, and operations.
5. Run independent review, sensitivity, replication, and red-team checks.
6. Record failures, abstentions, disagreement, and evidence limits.
7. Issue a bounded decision and monitoring or reopening conditions.

HUMAN-AI INTERACTION REQUIREMENTS

Show claim, model, data, grader, rubric, result, uncertainty, failures, reviewer, and status. Humans must inspect representative failures and can reject or narrow claims. Evaluation output cannot silently grant authority.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version evaluation sets, prompts, graders, rubrics, models, environments, seeds, results, traces, and decisions. Separate development, validation, and acceptance data. Preserve reproducibility, evaluator access, and evidence immutability.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show evaluation baseline, claim, coverage, scores, uncertainty, failures, subgroups, grader, independence, decision, and freshness. Allow authorized rerun, challenge, narrow, reject, accept, publish, withdraw, and reopen.

REQUIRED OUTPUTS AND ARTIFACTS

Evaluation charter; dataset and scenario register; rubric and grader record; baseline comparison; results and uncertainty; failure catalog; red-team report; decision record; limitations; monitoring gates; and handoff.

EVIDENCE REQUIREMENTS

Each result identifies claim, model, version, data, scenario, rubric, grader, environment, metric, uncertainty, result, reviewer, and limitation. Aggregate scores must not hide failures or unsupported populations.

VERIFICATION AND ACCEPTANCE TESTS

Reproduce results; vary prompts, graders, seeds, data, thresholds, and environments; inspect false positives and negatives; test leakage, gaming, subgroup and adversarial cases; and verify the decision does not exceed coverage.

FAILURE STATES AND ANTI-PATTERNS

Benchmark overreach; grader bias; self-evaluation treated as independence; cherry-picked samples; metric gaming; threshold drift; hidden abstentions; pass/fail without uncertainty; and evaluation performed after deployment only.

RECOVERY AND RECONSIDERATION

Invalidate affected claims on grader, data, model, environment, or procedure change. Preserve prior evidence, notify owners, narrow or withdraw release, rerun independently, review downstream decisions, and reopen.

COMPLETION CRITERIA

Claim, data, procedure, grader, coverage, results, failures, uncertainty, decision, validity, and reopen conditions are explicit. Status remains CONTENT-COMPLETE until reconciliation and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

CODEX/03_SECTIONS/180_MODEL_EVALUATION.txt

Preserve evaluation IDs, data, rubrics, graders, prompts, outputs, traces, results, decisions, dissent, and gaps. Hand off to Training, Alignment, Safety, MLOps, Model Operations, Testing, Governance, and HITL.

RELATED CODEX SECTIONS

164 Data Science; 167 Machine Learning Engineering; 179 Model Training; 181 Model Alignment; 182 AI Safety; 184 MLOps; 185 Model Operations; 253 Model Testing; 254 Data Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Testing and AI sections, and evidence lexicon. Actual claims, populations, thresholds, evaluator independence, risk class, and acceptance authority remain unresolved.

181

SECTION 181 - MODEL ALIGNMENT

CODEX/03_SECTIONS/181_MODEL_ALIGNMENT.txt

SHA-256 cad01bcab95bbee192743b37c1533a8e30145b60dd5f42e967b4662c22ece337

6,134 BYTES / PRESERVED SOURCE IDENTITY

SECTION 181 — Model Alignment

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed work to shape model behavior toward authorized objectives, policies, values, user needs, safety constraints, and truthful uncertainty without claiming that training preferences establish universal morality or authority.

REQUIRED OUTCOME

- Explicit behavior objectives, prohibited behaviors, value conflicts, feedback sources, and decision authority.
- Evidence of helpfulness, honesty, refusal, robustness, bias, manipulation resistance, and human-control behavior.
- Transparent limitations and residual disagreement.

WHY THIS MATTERS

Alignment targets are choices made under incomplete and conflicting evidence. Rewarding apparent compliance can produce sycophancy, concealment, over-refusal, or proxy gaming rather than trustworthy behavior.

APPLICABILITY

Applies to instruction tuning, preference optimization, constitutional or policy training, refusal behavior, value shaping, reward modeling, and behavior modification. Depth increases with autonomy, power, user impact, or contested values.

NON-APPLICABILITY CONDITIONS

May be non-applicable to unchanged model use or technical training with no behavior, policy, preference, or objective change. Record boundary, authority, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires explicit use, policies, values, affected parties, evaluation, safety, governance, data rights, and independent review. Depends on Sections 179, 180, 182, 183, 186, and 188.

REQUIRED INPUTS

Behavior objective; policies; values; prohibited actions; feedback; preference data; reward or rubric; disagreement; threat model; target users; evaluation; refusal and appeal; and owner authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans define and approve values, policies, tradeoffs, protected groups, refusal boundaries, exceptions, and residual risk. No alignment artifact can authorize a decision or substitute for legitimate governance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft rubrics, scenarios, critiques, and behavior tests. It must preserve disagreement and cannot claim moral correctness, fabricate feedback, or infer that reward means alignment.

APPLICABLE EXISTING AGENT ROLES

Researcher, Analyst, Safety Auditor, Governance Reviewer, Domain Expert, Critic, Red Team, Tester, Verifier, Privacy Reviewer, Security Auditor, Documentation Writer, Memory Curator, and Decision Arbiter Advisory.

REQUIRED ACTIVITIES

1. State behaviors, values, policies, prohibited use, and decision authority.
2. Audit preference and feedback sources, representation, incentives, and conflicts.
3. Define reward, rubric, refusal, uncertainty, and appeal contracts.
4. Train or tune with reproducible data, configuration, and checkpoints.
5. Evaluate helpfulness, honesty, refusal, robustness, manipulation, bias, and safety.
6. Review failure cases and value disagreement with independent humans.
7. Record residual uncertainty, acceptance, monitoring, and reopening triggers.

HUMAN-AI INTERACTION REQUIREMENTS

Expose policy, rubric, feedback provenance, model version, refusal reason, uncertainty, and review. Users must challenge outputs and appeal decisions. Alignment must not remove correction, explanation, human service, or safe refusal.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version objectives, policies, rubrics, preference data, reward models, training runs, checkpoints, evaluations, prompts, and deployment. Preserve behavior provenance and prevent policy changes from silently invalidating prior claims.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show alignment target, policy version, feedback, disagreements, evaluations, refusals, incidents, and authority. Allow authorized narrow, pause, revoke, compare, retrain, withdraw, compensate, and retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Alignment charter; policy and value register; feedback and preference provenance; rubric; training record; behavior and red-team evaluation; disagreement log; decision record; limitations; monitoring and handoff package.

EVIDENCE REQUIREMENTS

Claims identify behavior target, policy, feedback, model, training, scenario, evaluator, result, disagreement, uncertainty, and limitation. Preference, reward, or refusal rate does not prove universal alignment or safe action.

VERIFICATION AND ACCEPTANCE TESTS

Test sycophancy, deception, reward hacking, jailbreak, manipulation, over-refusal, under-refusal, ambiguity, subgroup behavior, policy conflict, prompt injection, and correction. Verify independent human review and reproducible results.

FAILURE STATES AND ANTI-PATTERNS

Reward equals values; majority feedback equals legitimacy; policy hidden in prompts; disagreement erased; obedience rewarded over truth; safety benchmark overreach; and aligned behavior treated as authorization.

RECOVERY AND RECONSIDERATION

Suspend affected behavior on unsafe drift, policy conflict, feedback poisoning, evaluator failure, or rights impact. Preserve artifacts and dissent, use prior qualified behavior or human fallback, review impacts, and reopen after value or policy change.

COMPLETION CRITERIA

Targets, values, policies, data, training, evaluation, disagreements, limits, recovery, evidence, and authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve behavior targets, policies, feedback, checkpoints, evaluations, failures, decisions, dissent, incidents, and gaps. Hand off to Training, Evaluation, Safety, Governance, Prompt Engineering, MLOps, Model Operations, and HITL.

RELATED CODEX SECTIONS

004 Evidence Verification and Truth; 018 Risk Management; 170 Generative AI Engineering; 179 Model Training; 180 Model Evaluation; 182 AI Safety; 183 AI Governance; 186 Prompt Engineering; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract and shared governance. Actual values, policies, feedback populations, cultural authorities, evaluation standards, safety thresholds, and acceptance authority remain unresolved.

182

SECTION 182 - AI SAFETY

CODEX/03_SECTIONS/182_AI_SAFETY.txt

SHA-256 b4fcbd9deae03b04d01797665d01cf6036f429ffa2f5fd4f9efe5d87a8400469

6,171 BYTES / PRESERVED SOURCE IDENTITY

SECTION 182 — AI Safety

STATUS: CONTENT-COMPLETE

PURPOSE

Define the identification, prevention, evaluation, monitoring, containment, and recovery of AI harms, including unsafe capability, misuse, failure, loss of control, human impact, and system interaction risk.

REQUIRED OUTCOME

- A safety case with hazards, affected parties, severity, likelihood, controls, evidence, residual risk, and authority.
- Tested safe states, refusal, interruption, fallback, incident, communication, and compensation paths.
- No claim of safety beyond declared scope and evidence.

WHY THIS MATTERS

AI failures can scale, persist, deceive, discriminate, or interact with tools and institutions. Safety is a system property requiring human authority and recovery, not a model score or warning label.

APPLICABILITY

Applies to any AI system with material user, public, physical, environmental, financial, privacy, security, rights, or autonomy consequence. Depth is proportional to hazard and reversibility.

NON-APPLICABILITY CONDITIONS

May be non-applicable to bounded non-AI work or analysis with no AI claim or consequence. Record exact scope and reopen if capability, use, population, or integration changes.

PREREQUISITES AND DEPENDENCIES

Requires system boundary, hazard analysis, domain and user evidence, threat model, evaluation, operating concept, incident authority, and owner acceptance. Depends on Sections 050, 167-170, 180, 181, 183-185, and 188.

REQUIRED INPUTS

Use; hazards; affected parties; model and system; tools; data; autonomy; operating environment; constraints; safety requirements; tests; monitoring; stop and fallback; communications; and compensation authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans own safety decisions, hazard acceptance, release, emergency action, notification, and retirement. Independent safety specialists may block unsafe work. AI cannot self-certify safety or waive a hazard.

AI RESPONSIBILITIES AND BOUNDARIES

AI may identify hazards, generate scenarios, simulate failures, and monitor bounded signals. It must expose uncertainty and escalate. It must not suppress safety findings, redefine hazards, or continue after revoked authority.

APPLICABLE EXISTING AGENT ROLES

Safety Auditor, Architect, Researcher, Analyst, Critic, Red Team, Tester, Verifier, Governance Reviewer, Security Auditor, Privacy Reviewer, UX Reviewer, Documentation Writer, Memory Curator, and Human-in-the-Loop Operator.

REQUIRED ACTIVITIES

1. Define safety scope, hazards, affected parties, severity, and authority.
2. Build safety case and map hazards to requirements, controls, tests, and evidence.
3. Assess misuse, emergent interaction, autonomy, provider, data, and human factors.
4. Test normal, boundary, adversarial, degraded, loss-of-control, and recovery cases.
5. Validate stop, isolation, refusal, fallback, communication, appeal, and compensation.
6. Monitor incidents, near misses, drift, controls, and evidence freshness.
7. Review residual risk and reopen on material change.

HUMAN-AI INTERACTION REQUIREMENTS

Show hazard, control, state, authority, uncertainty, stop condition, reviewer, and evidence. Humans must have effective interruption and appeal. Safety information cannot be hidden behind model confidence, automation, or a conversational interface.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent hazards, controls, safety requirements, states, interlocks, evidence, incidents, decisions, and compensations as versioned or immutable records. Enforce safety at system boundaries and preserve independent stop and audit paths.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show safety case, open hazards, control status, evidence, incidents, authority, current mode, and safe actions. Provide authorized stop, isolate, revoke, restrict, notify, compensate, restore, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Safety charter; hazard log; safety case; control and interlock design; evaluation and red-team report; incident and near-miss register; emergency runbook; decision and residual-risk record; and handoff.

EVIDENCE REQUIREMENTS

Each safety claim identifies hazard, scope, system version, scenario, control, test, result, uncertainty, reviewer, date, and limitation. A test suite, refusal rate, or compliance statement is not universal safety evidence.

VERIFICATION AND ACCEPTANCE TESTS

Exercise unsafe input, tool misuse, prompt injection, loss of monitor, authority expiry, provider loss, human absence, sensor failure, physical stop, fallback, communication, compensation, and recovery. Verify controls cannot be bypassed through alternate paths.

FAILURE STATES AND ANTI-PATTERNS

Safety as documentation only; hazard omission; model-only boundary; no independent stop; warning as control; risk accepted by default; hidden near misses; and recovery dependent on the failed AI system.

RECOVERY AND RECONSIDERATION

Contain and stop affected behavior on a credible hazard or control failure. Preserve immutable decisions and evidence, protect affected parties, communicate, compensate where authorized, investigate independently, and require fresh safety review before resuming.

COMPLETION CRITERIA

Hazards, controls, tests, evidence, residual risk, authority, stop, recovery, communication, and reopening are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve safety case, hazards, controls, tests, incidents, decisions, traces, compensations, dissent, and gaps. Hand off to Governance, Security, Privacy, Reliability, Operations, Evaluation, MLOps, Model Operations, and HITL.

RELATED CODEX SECTIONS

018 Risk Management; 050 Safety Architecture; 182 AI Safety; 183 AI Governance; 214 Incident Response; 244 Resilience Testing; 252 Safety Testing; 253 Model Testing; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Safety Architecture, AI Engineering, and evidence lexicon. Actual hazards, safety standards, affected parties, control authority, emergency powers, and residual-risk criteria remain unresolved.

183

SECTION 183 - AI GOVERNANCE

CODEX/03_SECTIONS/183_AI_GOVERNANCE.txt

SHA-256 c94ea530614d7adc337b874a9ec3f6bc0383a1f6f8a2509a7c80340cd7943d80

6,514 BYTES / PRESERVED SOURCE IDENTITY

SECTION 183 — AI Governance

STATUS: CONTENT-COMPLETE

PURPOSE

Define the authority, policy, accountability, evidence, approval, monitoring, and lifecycle controls for AI systems and AI-enabled mutations.

REQUIRED OUTCOME

- A governed AI inventory with purpose, owner, impact, risk, capability, data, provider, lifecycle, and decision status.
- Policy evaluation linked to immutable DecisionRecord and AuditDecision artifacts.
- Complete mutation bindings for consequential model, prompt, tool, data, deployment, and policy changes.

WHY THIS MATTERS

Governance fails when a local check is mistaken for policy evaluation, a model capability becomes authority, or an approval cannot be replayed. Explicit records make decisions accountable, reviewable, and reversible only through compensation.

APPLICABILITY

Applies to AI design, procurement, training, evaluation, deployment, operation, tool use, data use, prompt changes, policy changes, and retirement. It applies more deeply with regulated use, autonomy, sensitive data, external effect, or mutation.

NON-APPLICABILITY CONDITIONS

May be non-applicable to non-AI work or read-only analysis with no AI claim, policy effect, or mutation. Record scope, controlling policy, owner, evidence date, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires authority model, policy sources, impact assessment, AI inventory, data and system context, evaluation, safety, privacy, security, and audit capability. Depends on Sections 048, 056, 180, 182, 184, 185, and 188.

REQUIRED INPUTS

Intent; actor; AI system; purpose; data; model; provider; capability; resource; mutation class; traceId; capability token; GovernanceContext; PolicyEvaluationRequest and Result; quorum; resolution policy; and compensation path.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans define policy, delegate authority, approve exceptions, provide quorum where required, accept risk, review evidence, and authorize or deny mutations. No operator override may erase a prior immutable decision or bypass policy evaluation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may prepare requests, summarize policy, identify gaps, and recommend decisions. It must not treat local validation as governance evaluation, self-issue capability, approve itself, or mutate without a complete binding and valid decision.

APPLICABLE EXISTING AGENT ROLES

Governance Reviewer, Decision Arbiter Advisory, Architect, Security Auditor, Privacy Reviewer, Safety Auditor, Researcher, Analyst, Critic, Verifier, Documentation Writer, Memory Curator, and Human-in-the-Loop Operator.

REQUIRED ACTIVITIES

1. Register AI system, purpose, owner, impact, capabilities, dependencies, and lifecycle.
2. Establish GovernanceContext and PolicyEvaluationRequest for each governed action.
3. Obtain PolicyEvaluationResult, capability scope, quorum proof, and resolution policy.
4. Create immutable DecisionRecord and AuditDecision linked to trace and evidence.
5. Execute only the authorized mutation and emit append-only observability events.
6. Monitor drift, violations, incidents, appeals, and policy changes.
7. Compensate linked prior mutations and preserve replayable history.

HUMAN-AI INTERACTION REQUIREMENTS

Show policy, evaluation, capability, decision, quorum, trace, resource, mutation, and outcome. Humans can deny, narrow, pause, revoke, appeal, or request reconsideration. Stream completion must include a terminal event and preserve audit linkage.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use immutable DecisionRecord, GovernanceEvaluationBinding, MutationGovernanceBinding, PolicyEvaluationRequest and Result, ApprovalDecisionRequest, AuditDecision, trace, artifact, and compensation records. Enforce default-deny, precedence, expiry, and append-only audit at service boundaries.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show inventory, impact, policy result, binding completeness, decision, quorum, authority, traces, evidence, mutations, incidents, and compensations. Provide authorized approve, deny, revoke, pause, quarantine, reconsider, and compensate controls.

REQUIRED OUTPUTS AND ARTIFACTS

AI inventory; impact record; policy catalog; evaluation requests and results; capability records; decision and audit records; mutation bindings; quorum proofs; trace and artifact links; exception register; incident record; and compensation plan.

EVIDENCE REQUIREMENTS

Every governance claim identifies policy, evaluation ID and result, traceId, actor, capability scope, resource, action, mutation class, decision, quorum, evidence, and validity. Local tests, approvals, or logs alone do not prove governed authorization.

VERIFICATION AND ACCEPTANCE TESTS

Attempt missing trace, expired token, mismatched resource, local-only check, absent evaluation, insufficient quorum, conflicting policy, lifecycle regression, missing terminal event, and destructive rollback; verify default-deny and immutable audit.

FAILURE STATES AND ANTI-PATTERNS

Rubber-stamp approval; policy copied into prompt; local-only enforcement; self-approval; incomplete binding; hidden override; mutable decision; audit replacement; quorum bypass; and destructive reversal.

RECOVERY AND RECONSIDERATION

Deny or stop on binding, evaluation, capability, quorum, trace, or policy failure. Preserve records, identify safe read-only work, obtain fresh evaluation and approval, and use compensation transactions linked to prior decisions rather than destructive reversal.

COMPLETION CRITERIA

Authority, policy, evaluation, binding, decision, audit, trace, quorum, mutation, observability, compensation, and lifecycle are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve immutable IDs, policy versions, evaluations, decisions, bindings, quorum proofs, traces, artifacts, events, incidents, appeals, and gaps. Hand off to Safety, Security, Privacy, MLOps, Operations, Audit, and HITL.

RELATED CODEX SECTIONS

003 Authority Scope and Delegation; 004 Evidence Verification and Truth; 048 Governance Architecture; 056 AI Architecture; 180 Model Evaluation; 182 AI Safety; 184 MLOps; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Owner Cockpit Requirements, Status and Evidence Lexicon, and governing decision-binding requirements. Concrete policy registries, token issuer, quorum rules, retention, and authority assignments remain unresolved.

184

SECTION 184 - MLOPS

CODEX/03_SECTIONS/184_MLOPS.txt

SHA-256 73017a5c74f0a44811d36a6d65c3d8357f5ed5a0dab640d8144b047f79081c59

5,979 BYTES / PRESERVED SOURCE IDENTITY

SECTION 184 — MLOps

STATUS: CONTENT-COMPLETE

PURPOSE

Define the governed delivery and lifecycle practice for data, models, code, evaluation, deployment, monitoring, incident response, retraining, and retirement.

REQUIRED OUTCOME

- Reproducible, traceable promotion of ML artifacts through controlled environments.
- Gates for data, evaluation, security, safety, policy, approval, observability, rollback, and compensation.
- Continuous evidence that distinguishes pipeline health from model or outcome validity.

WHY THIS MATTERS

Automation can accelerate an unapproved or invalid model into production. MLOps must preserve evidence and authority across frequent data, code, model, and infrastructure changes.

APPLICABILITY

Applies to ML build, registry, validation, release, serving, monitoring, retraining, promotion, rollback, incident, and retirement workflows. Depth increases with autonomy, continuous learning, sensitive data, or external effect.

NON-APPLICABILITY CONDITIONS

May be non-applicable to static non-ML software or an offline experiment with no pipeline, promotion, or operational claim. Record exact boundary and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires data and model contracts, evaluation, governance, security, privacy, safety, observability, operations, and owner authority. Depends on Sections 167, 179, 180, 183, 185, 253, and 266-270.

REQUIRED INPUTS

Pipeline; artifacts; data; registry; environments; tests; evaluation gates; policy result; capability; trace; deployment strategy; monitoring; incident; rollback or compensation; and retirement conditions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve promotion gates, exceptions, retraining, production changes, risk, and retirement. Operators may stop pipelines and revoke authority; automation cannot waive failed gates.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate pipeline code, diagnose failures, and recommend promotion. It must preserve logs and evidence and cannot approve its own output, suppress failures, or mutate environments without a complete binding.

APPLICABLE EXISTING AGENT ROLES

Architect, Implementer, DevOps Engineer, Tester, Verifier, Governance Reviewer, Security Auditor, Privacy Reviewer, Safety Auditor, Critic, Optimizer, Planner, Documentation Writer, Memory Curator, and HITL Operator.

REQUIRED ACTIVITIES

1. Define artifact, environment, promotion, and evidence contracts.
2. Version source, data, features, model, code, image, configuration, and dependencies.
3. Enforce quality, evaluation, security, safety, policy, and approval gates.
4. Promote through reproducible stages with immutable records and traces.
5. Monitor data, model, service, policy, cost, and outcome drift.
6. Exercise incident, pause, rollback configuration, fallback, compensation, and retirement.
7. Review pipeline changes and reopen invalidated claims.

HUMAN-AI INTERACTION REQUIREMENTS

Show pipeline state, artifact lineage, gate result, policy evaluation, decision, authority, trace, and environment. Human approval is explicit for consequential promotion. Preserve stop, evidence, and manual recovery if automation fails.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use immutable artifact and decision identities, append-only events, reproducible environments, segregated credentials, signed or equivalent provenance, and default-deny promotion. Keep rollback distinct from compensation for external effects.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show pipeline, model, data, gates, risks, drift, environment, approvals, traces, incidents, and evidence freshness. Provide authorized pause, deny, revoke, quarantine, promote, fallback, compensate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

MLOps charter; pipeline and environment map; artifact registry; gate and policy records; deployment and promotion logs; monitoring and alert plan; incident and compensation runbook; decision and handoff package.

EVIDENCE REQUIREMENTS

Each promotion claim identifies source, data, model, code, environment, gates, policy result, decision, trace, actor, result, and limitation. Green pipelines prove execution, not validity, safety, or accepted outcome.

VERIFICATION AND ACCEPTANCE TESTS

Break data, tests, evaluator, registry, credentials, policy, capability, deployment, monitor, provider, and stream termination; verify promotion denies, evidence remains, alerts fire, and recovery follows approved controls.

FAILURE STATES AND ANTI-PATTERNS

Auto-promotion on weak metrics; mutable registry; missing lineage; silent retraining; test bypass; shared credentials; green pipeline equals model quality; and destructive rollback of governed history.

RECOVERY AND RECONSIDERATION

Pause affected pipeline or deployment on gate, artifact, policy, drift, or authority failure. Preserve records, use prior qualified or manual mode, compensate external effects, re-evaluate, and resume only with fresh authorization.

COMPLETION CRITERIA

Artifacts, environments, gates, policy, decisions, traces, monitoring, incident, recovery, compensation, and retirement are explicit. Status remains CONTENT-COMPLETE pending reconciliation and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve pipeline IDs, artifacts, environments, logs, decisions, bindings, traces, incidents, drift, dissent, and gaps. Hand off to Model Operations, Training, Evaluation, Security, Governance, Operations, and Retirement.

RELATED CODEX SECTIONS

167 Machine Learning Engineering; 179 Model Training; 180 Model Evaluation; 183 AI Governance; 185 Model Operations; 258 Configuration Management; 266 Continuous Integration; 267 Continuous Delivery; 270 DevSecOps.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, architecture, testing, and delivery sections. Actual registries, gate thresholds, signing, environments, approval authority, service levels, and compensation contracts remain unresolved.

185

SECTION 185 - MODEL OPERATIONS

CODEX/03_SECTIONS/185_MODEL_OPERATIONS.txt

SHA-256 572a7e0d309ecc0cdff2fa6f0fb1c40ae0d1a64e75404c7969b651b3900135a4

5,896 BYTES / PRESERVED SOURCE IDENTITY

SECTION 185 — Model Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Define operation of models in live and controlled environments, including serving, routing, versioning, monitoring, drift, incidents, access, cost, human review, fallback, and retirement.

REQUIRED OUTCOME

- An operational model inventory and serving contract with owner, version, scope, authority, dependencies, and lifecycle.
- Reliable observability of model, service, data, policy, user, and outcome behavior.
- Safe stop, fallback, incident, compensation, and retirement paths.

WHY THIS MATTERS

Operational model behavior changes with traffic, data, providers, prompts, tools, and policy. Availability alone cannot show whether the model remains valid, safe, or authorized.

APPLICABILITY

Applies to deployed, hosted, embedded, batch, edge, shadow, or human-reviewed model use. Depth increases with live learning, external mutation, sensitive data, and high consequence.

NON-APPLICABILITY CONDITIONS

May be non-applicable to an unserved research artifact with no operational claim. Record artifact scope, owner, evidence date, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires model contract, deployment identity, evaluation, monitoring, access, incident, safety, privacy, security, operations, and governance authority. Depends on Sections 167, 180, 182-184, 301-318, and 188.

REQUIRED INPUTS

Model and endpoint; version; data; provider; prompts; tools; users; load; SLOs; quality and drift metrics; policy; capability; traces; incidents; fallback; compensation; and retirement plan.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans own operating mode, traffic, release, thresholds, incidents, communications, risk, and retirement. Operators may pause, route away, revoke, or compensate but cannot erase operational evidence.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize telemetry, detect anomalies, and recommend actions. It must distinguish signal from inference and cannot suppress alerts, change routing, close incidents, or accept risk without authority.

APPLICABLE EXISTING AGENT ROLES

Operations Engineer, Architect, Analyst, Observer, Security Auditor, Privacy Reviewer, Safety Auditor, Governance Reviewer, Tester, Verifier, Critic, Planner, Documentation Writer, Memory Curator, and HITL Operator.

REQUIRED ACTIVITIES

1. Register model, endpoint, owner, scope, version, dependencies, and authority.
2. Define serving, routing, latency, capacity, quality, cost, and evidence contracts.
3. Instrument requests, outputs, errors, policy, tools, users, drift, and incidents.
4. Monitor quality, safety, privacy, security, availability, cost, and outcome signals.
5. Exercise provider loss, model mismatch, data shift, authority expiry, and fallback.
6. Manage incidents, communication, compensation, requalification, and retirement.

HUMAN-AI INTERACTION REQUIREMENTS

Show model, provider, request, output, policy, authority, review, confidence limits, trace, and operational status. Require human approval for material routing or model changes and preserve manual service and stop paths.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version endpoints, weights, runtime, prompts, policies, data, routing, thresholds, telemetry, and decisions. Enforce identity, capability, rate, data, and output controls at serving boundaries. Preserve append-only traces and compensation links.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show live model state, traffic, drift, quality, incidents, dependencies, policy, cost, authority, and evidence. Provide authorized pause, drain, route, revoke, fallback, quarantine, compensate, restore, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Model operations charter; inventory; serving and SLO contract; dashboards and alert rules; drift and incident runbook; provider and dependency map; access and policy records; decision, audit, trace, and retirement package.

EVIDENCE REQUIREMENTS

Claims identify model, endpoint, version, environment, traffic, data, metric, observation period, method, actor, result, and limitation. Uptime, latency, or output samples do not prove model correctness or safety.

VERIFICATION AND ACCEPTANCE TESTS

Test version mismatch, stale data, drift, provider loss, load, malformed input, unauthorized access, policy conflict, monitor failure, alert noise, stop, fallback, compensation, and retirement. Reproduce operational claims.

FAILURE STATES AND ANTI-PATTERNS

Uptime equals quality; unowned drift; alert suppression; silent routing; untracked provider change; model-output logs without privacy control; no manual path; and incident closure without evidence.

RECOVERY AND RECONSIDERATION

Restrict or stop affected model behavior on drift, incident, policy, security, privacy, safety, or evidence failure. Preserve state, notify owners, use qualified fallback, compensate impacts, re-evaluate, and resume only with authority.

COMPLETION CRITERIA

Inventory, serving, authority, observability, quality, drift, incidents, recovery, compensation, evidence, and lifecycle are explicit. Status remains CONTENT-COMPLETE pending reconciliation and acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve endpoints, versions, telemetry, alerts, decisions, traces, incidents, compensations, dissent, and gaps. Hand off to MLOps, Operations, Evaluation, Safety, Security, Privacy, Governance, and HITL.

RELATED CODEX SECTIONS

029 Operational Architecture; 167 Machine Learning Engineering; 180 Model Evaluation; 182 AI Safety; 183 AI Governance; 184 MLOps; 301 Site Reliability Engineering; 313 Observability Engineering; 318 Alerting.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Operational Architecture, and evidence lexicon. Actual SLOs, drift thresholds, provider terms, incident authority, data rights, and retirement requirements remain unresolved.

186

SECTION 186 - PROMPT ENGINEERING

CODEX/03_SECTIONS/186_PROMPT_ENGINEERING.txt

SHA-256 91fc117189f832c88e123d1c19e4f9c7b6fa212935df23fdd50c8b793a6d61c7

5,897 BYTES / PRESERVED SOURCE IDENTITY

SECTION 186 — Prompt Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed design, testing, versioning, and operation of prompts and instructions as behavior-shaping inputs, without treating them as security, policy, authorization, or evidence substitutes.

REQUIRED OUTCOME

- A prompt contract with purpose, audience, model, context, tools, output, policy, failure, and review boundaries.
- Reproducible prompt versions, evaluations, injection tests, provenance, and release decisions.
- Explicit limits on what prompt changes can and cannot authorize.

WHY THIS MATTERS

Prompts influence behavior but are mutable, context-sensitive, and exposed to injection and ambiguity. A well-written prompt cannot replace access controls, policy enforcement, validation, or human authority.

APPLICABILITY

Applies to system instructions, templates, few-shot examples, routing prompts, tool instructions, evaluator prompts, retrieval context, and prompt optimization. Depth increases with tools, sensitive data, external action, or policy consequence.

NON-APPLICABILITY CONDITIONS

May be non-applicable to static text with no model interaction, behavior claim, or downstream action. Record scope, model boundary, owner, evidence date, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires AI system contract, model, data, tools, policy, evaluation, security, privacy, safety, and human-review authority. Depends on Sections 169, 170, 180-183, and 188.

REQUIRED INPUTS

Intent; user; model; prompt layers; context; data; tools; output schema; policy; capability; threat model; examples; evaluation; version; retention; and owner decision.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve purpose, instructions, policy interpretation, tool scope, user notice, release, and risk. Prompt engineers cannot grant access, authority, or exceptions by wording.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft, compare, optimize, and test prompts. It must preserve versions and test context and must not hide injection, fabricate evaluation, or claim prompt-only protection.

APPLICABLE EXISTING AGENT ROLES

Prompt Engineer, Architect, Researcher, Analyst, Implementer, Tester, Verifier, Security Auditor, Privacy Reviewer, Safety Auditor, Critic, Red Team, Governance Reviewer, Documentation Writer, and HITL Operator.

REQUIRED ACTIVITIES

1. Define prompt purpose, layer, model, users, context, tools, and prohibited use.
2. Separate instructions from policy, permissions, data, and validation controls.
3. Version prompts, examples, retrieval context, output schema, and configuration.
4. Test instruction conflict, injection, leakage, ambiguity, long context, and model change.
5. Evaluate quality, truthfulness, refusal, safety, accessibility, cost, and robustness.
6. Require review for tool, policy, output, or authority effects.
7. Monitor drift and preserve rollback configuration and compensation paths.

HUMAN-AI INTERACTION REQUIREMENTS

Show prompt version, model, context, tools, policy, output, evaluator, and review. Users must see consequential limitations and can correct or stop. Prompt changes cannot bypass policy evaluation or capability binding.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Store prompts, layers, examples, context, model, tools, policies, output schemas, tests, evaluations, and decisions as versioned artifacts. Enforce authorization, validation, sandboxing, and audit outside prompt text.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show active prompt, model, context, tools, policy, evaluation, traces, failures, authority, and evidence. Provide authorized compare, stage, approve, reject, pause, revoke, fallback, withdraw, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Prompt charter; prompt registry; model and context record; tool and output contract; injection and safety suite; evaluation report; release decision; trace and incident record; and handoff.

EVIDENCE REQUIREMENTS

Claims identify prompt, model, context, tools, policy, data, scenario, evaluator, result, and limitation. A single response or prompt comparison does not prove general behavior, security, or authorization.

VERIFICATION AND ACCEPTANCE TESTS

Test prompt injection, instruction hierarchy, context poisoning, leakage, tool escalation, malformed output, model substitution, long context, ambiguity, refusal, correction, and release rollback. Reproduce accepted results.

FAILURE STATES AND ANTI-PATTERNS

Prompt as access control; hidden system instruction; unversioned examples; overfitting to tests; injection suppression; model confidence as authorization; secrets in prompts; and prompt change without evaluation or audit.

RECOVERY AND RECONSIDERATION

Withdraw or disable affected prompt on injection, leakage, unsafe output, model change, policy conflict, or evidence failure. Preserve prior version and traces, use qualified fallback, review impacts, and re-evaluate before release.

COMPLETION CRITERIA

Purpose, prompt layers, model, context, tools, policy, tests, evaluation, authority, evidence, recovery, and lifecycle are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve prompt IDs, versions, contexts, model, tools, tests, outputs, traces, decisions, incidents, dissent, and gaps. Hand off to Generative AI, Agent Systems, Evaluation, Safety, Governance, MLOps, and HITL.

RELATED CODEX SECTIONS

056 AI Architecture; 089 Conversational Design; 169 AI Engineering; 170 Generative AI Engineering; 178 Agent Systems Engineering; 180 Model Evaluation; 182 AI Safety; 183 AI Governance.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, AI Architecture, and evidence lexicon. Actual prompt owners, models, tools, policy layers, injection threats, user notices, and release authority remain unresolved.

187

SECTION 187 - SYNTHETIC DATA ENGINEERING

CODEX/03_SECTIONS/187_SYNTHETIC_DATA_ENGINEERING.txt

SHA-256 885ca3bf8e7706a0bd1841f72c2f1917c6ba3f8a81f43e98a45e1963dd255482

6,443 BYTES / PRESERVED SOURCE IDENTITY

SECTION 187 — Synthetic Data Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define governed creation, validation, use, disclosure, retention, and retirement of data generated to resemble or simulate real data for development, testing, analysis, training, evaluation, or operations.

REQUIRED OUTCOME

- A synthetic-data purpose, generator, source basis, population, privacy, quality, utility, and limitation record.
- Evidence that synthetic data is labeled, fit for its bounded use, not mistaken for observed fact, and not a hidden disclosure channel.
- Controlled generation, access, correction, deletion, and retirement.

WHY THIS MATTERS

Synthetic data can reduce exposure but can also memorize, reproduce, amplify bias, create unrealistic correlations, or be treated as real evidence. Its value and limits must be measured for the intended use.

APPLICABILITY

Applies to generated records, simulations, augmented samples, masked or perturbed data, test fixtures, synthetic images, text, audio, features, labels, and training or evaluation data. Depth increases with sensitive source data or consequential models.

NON-APPLICABILITY CONDITIONS

May be non-applicable to hand-written non-sensitive test constants with no claim of realism, privacy, or population representation. Record scope, owner, source status, and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires source-data authority, purpose, privacy and security review, generator identity, quality and utility criteria, downstream use, and retention. Depends on Sections 031, 156, 164, 167, 179, 180, and 183.

REQUIRED INPUTS

Purpose; source data; lawful basis; generator; configuration; population; sensitive attributes; privacy target; utility metrics; bias plan; disclosure; labeling; downstream model or test; and deletion requirements.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans approve source use, privacy target, generator, disclosure, realism claim, downstream use, and retention. Data owners and privacy authorities decide whether synthetic output remains sensitive or linkable.

AI RESPONSIBILITIES AND BOUNDARIES

AI may generate, profile, and test synthetic data within authorization. It must preserve synthetic status, provenance, and uncertainty and must not claim privacy, representativeness, or independence without evidence.

APPLICABLE EXISTING AGENT ROLES

Data Engineer, Data Scientist, Researcher, Analyst, Privacy Reviewer, Security Auditor, Data Governance Reviewer, Tester, Verifier, Critic, Red Team, Documentation Writer, Memory Curator, and Governance Reviewer. Existing role contracts remain controlling.

REQUIRED ACTIVITIES

1. Define purpose, population, source, sensitivity, use, and prohibited inference.
2. Assess source rights, memorization, linkage, re-identification, and disclosure risk.
3. Select generator and record configuration, seed, transformations, and output label.
4. Test utility, fidelity, coverage, bias, rare cases, privacy, and leakage.
5. Validate downstream use and prevent synthetic data from masquerading as observed data.
6. Control access, retention, correction, deletion, and retirement.
7. Record failures, limitations, decisions, and reopening conditions.

HUMAN-AI INTERACTION REQUIREMENTS

Show synthetic status, source basis, generator, configuration, privacy result, utility, intended use, and limitations. Humans approve disclosure and downstream use. Users must distinguish generated records from observed records and retain correction and deletion paths.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Version sources, generators, seeds, configurations, outputs, labels, privacy tests, utility tests, and downstream artifacts. Enforce synthetic labeling and provenance in storage, APIs, exports, prompts, logs, and model pipelines. Preserve deletion and lineage.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show dataset identity, synthetic status, source, generator, privacy, utility, bias, users, downstream uses, retention, and evidence. Allow authorized restrict, quarantine, relabel, regenerate, withdraw, delete, and retire actions.

REQUIRED OUTPUTS AND ARTIFACTS

Synthetic-data charter; source and rights record; generator and configuration; privacy and utility report; bias and rare-case analysis; labeling and access contract; downstream impact record; decision; and handoff.

EVIDENCE REQUIREMENTS

Claims identify source, generator, version, seed, population, configuration, privacy method, utility metric, bias result, reviewer, use, and limitation. Similarity or privacy scores do not prove absence of disclosure or real-world validity.

VERIFICATION AND ACCEPTANCE TESTS

Test memorization, nearest-neighbor disclosure, linkage, rare records, attribute inference, bias, distribution shift, schema, labels, access, deletion, downstream model behavior, and independent regeneration. Verify synthetic status survives all interfaces.

FAILURE STATES AND ANTI-PATTERNS

Synthetic labeled as real; source leakage; privacy score overreach; rare-case omission; bias amplification; generator drift; hidden synthetic data in training or evaluation; and deletion that leaves derived copies.

RECOVERY AND RECONSIDERATION

Quarantine synthetic data on disclosure, provenance, labeling, quality, or downstream failure. Preserve evidence, restrict access, notify owners, delete or regenerate under authority, review affected models and decisions, and reopen.

COMPLETION CRITERIA

Purpose, source, generator, privacy, utility, bias, label, downstream use, controls, evidence, recovery, and authority are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve dataset IDs, sources, generators, seeds, configs, tests, labels, decisions, downstream links, incidents, dissent, and gaps. Hand off to Data Governance, Data Quality, ML, Training, Evaluation, Privacy, Security, and Operations.

RELATED CODEX SECTIONS

031 Data Architecture; 155 Data Governance; 156 Data Quality Engineering; 164 Data Science; 167 Machine Learning Engineering; 179 Model Training; 180 Model Evaluation; 254 Data Testing; 188 Human-in-the-Loop Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, Data Architecture, AI, and evidence sections. Actual source rights, privacy thresholds, utility criteria, downstream use, rare populations, and retention authority remain unresolved.

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SECTION 188 - HUMAN IN THE LOOP OPERATIONS

CODEX/03_SECTIONS/188_HUMAN_IN_THE_LOOP_OPERATIONS.txt

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6,774 BYTES / PRESERVED SOURCE IDENTITY

SECTION 188 — Human-in-the-Loop Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Define how humans supervise, review, approve, correct, challenge, interrupt, appeal, and recover AI or automated work in live operations without reducing human involvement to decorative confirmation.

REQUIRED OUTCOME

- A human-control model with role, authority, workload, timing, information, training, escalation, and decision boundaries.
- Review that is meaningful, accessible, attributable, and able to change the result before consequence.
- Durable records linking AI output, human action, policy evaluation, decision, trace, and outcome.

WHY THIS MATTERS

Human review can fail through overload, automation bias, missing context, unclear authority, poor timing, inaccessible interfaces, or retaliation for dissent. A human in a workflow is not automatically human control.

APPLICABILITY

Applies when humans review, approve, monitor, label, correct, appeal, override, supervise, or recover AI and automated outputs. Depth increases with consequence, scale, speed, vulnerability, autonomy, or irreversible action.

NON-APPLICABILITY CONDITIONS

May be non-applicable where no human-control claim exists and the system is bounded non-AI automation, or where a read-only analysis has no operational consequence. Record the exact boundary and reopening trigger.

PREREQUISITES AND DEPENDENCIES

Requires AI and process contract, authority, policy evaluation, capability, decision, audit, trace, accessibility, staffing, training, privacy, security, safety, and incident context. Depends on Sections 006, 048, 182, 183, 185, and 214.

REQUIRED INPUTS

Decision or action; AI output; model and policy; evidence; human role; authority; capability token; workload; timing; review rubric; context; escalation; stop; appeal; compensation; and staffing and accessibility constraints.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans exercise actual judgment, approve or deny, record rationale, challenge evidence, escalate, stop, revoke, and accept only delegated responsibility. Owners must staff, train, protect, and support reviewers and preserve independent appeal.

AI RESPONSIBILITIES AND BOUNDARIES

AI must present relevant evidence, uncertainty, alternatives, provenance, and limits; wait for required approval; and expose review state. It must not pressure, manipulate, conceal dissent, pre-fill approval deceptively, or treat silence as consent.

APPLICABLE EXISTING AGENT ROLES

Human-in-the-Loop Operator, Governance Reviewer, Decision Arbiter Advisory, Safety Auditor, Security Auditor, Privacy Reviewer, User Experience Reviewer, Accessibility Reviewer, Analyst, Observer, Critic, Verifier, Documentation Writer, and Memory Curator. Existing role contracts remain controlling.

REQUIRED ACTIVITIES

1. Define human role, authority, consequence, review point, and prohibited delegation.
2. Provide context, evidence, uncertainty, alternatives, time, and accessible controls.
3. Review, approve, deny, correct, escalate, appeal, or stop according to policy.
4. Record human identity, rationale, decision, dissent, and trace linkage.
5. Monitor workload, speed, disagreement, override, error, fatigue, and retaliation risk.
6. Test model, interface, staffing, accessibility, provider, and communication failure.
7. Use compensation and reconsideration for affected prior actions.

HUMAN-AI INTERACTION REQUIREMENTS

The reviewer must know the AI identity, output status, evidence, uncertainty, policy result, capability, resource, and consequence. Review must occur before the relevant mutation and be able to change it. The system must expose stop, appeal, correction, and terminal state.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent human assignment, role, authority, capability token, review request, evidence, PolicyEvaluationResult, DecisionRecord, AuditDecision, trace, mutation binding, outcome, and compensation link. Enforce quorum and expiry; append audit rather than overwrite.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show queue, urgency, workload, context, AI output, evidence, policy, authority, reviewer, decision, dissent, trace, and unresolved state. Provide authorized approve, deny, defer, escalate, pause, revoke, correct, appeal, and compensate controls.

REQUIRED OUTPUTS AND ARTIFACTS

HITL operating model; role and authority matrix; review rubric; queue and workload design; accessibility and training plan; decision and audit records; trace package; escalation and appeal runbook; incident and compensation plan; and handoff.

EVIDENCE REQUIREMENTS

Each review identifies request, AI and policy versions, evidence, reviewer, authority, capability, timing, decision, rationale, dissent, trace, outcome, and limitation. A click, presence of a reviewer, or approval rate does not prove meaningful oversight.

VERIFICATION AND ACCEPTANCE TESTS

Test missing context, time pressure, overload, disagreement, inaccessible control, stale evidence, expired authority, policy mismatch, missing quorum, deceptive prefill, provider loss, interruption, appeal, and compensation. Verify humans can change outcomes.

FAILURE STATES AND ANTI-PATTERNS

Rubber-stamping; silence as approval; reviewer overload; automation bias; inaccessible review; hidden evidence; punitive dissent; approval after mutation; no appeal; and human review used to legitimize an ungoverned system.

RECOVERY AND RECONSIDERATION

Pause or narrow actions when meaningful review is unavailable, authority expires, evidence is missing, workload is unsafe, or a control fails. Preserve decisions and traces, notify owners, provide appeal and compensation, staff qualified review, and resume only after fresh evaluation.

COMPLETION CRITERIA

Role, authority, evidence, timing, workload, accessibility, review, decision, audit, trace, stop, appeal, compensation, and lifecycle are explicit. Status remains CONTENT-COMPLETE pending reconciliation and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve review IDs, assignments, evidence, outputs, decisions, dissent, workload, incidents, appeals, traces, and gaps. Hand off to AI Governance, Safety, Security, Privacy, Model Operations, Incident Management, Training, and Audit.

RELATED CODEX SECTIONS

001 Human-AI Interaction Requirements; 006 Owner Cockpit Requirements; 048 Governance Architecture; 182 AI Safety; 183 AI Governance; 185 Model Operations; 214 Incident Response; 332 Command and Control.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance files, Owner Cockpit Requirements, Status and Evidence Lexicon, AI Safety, and AI Governance. Actual staffing, authority, quorum, accessibility, workload, appeal, and compensation rules remain unresolved until supplied.